

GMP3

İçindekiler

1	Redmine Introduction	4
2	ECR – External System Connection Types	8
2.1	iWE280 Base – RS232 / USB / ETHERNET(TCP) Connections	8
2.2	iWE280 Base – USB Cable Connection	8
2.3	iWE280 Base – ETHERNET (TCP) Cable Connection	8
2.4	iDE280 – RS232 Connection	9
2.5	iDE280 – ETHERNET Cable Connection	9
2.6	MOVE5000F Base– RS232 Connection	10
2.7	MOVE5000F USB – Standart USB Connection	10
2.8	MOVE5000F BASE - ETHERNET (TCP) Cable Connection	10
2.9	Flow Control, BUSY, PING & ECHO	11
2.10	Unique ID	13
2.10.1	Creation Of Unique Id on ECR	13
2.10.2	Creation Of Unique Id on TSM	13
3	Transacion Flows and Daigrams	14
3.1	Synchronous (Single Command) Mode	14
3.2	Asynchronous (Batch Command) Mode	15
4	FUNCTIONS	16
4.1	FP3 Functions For Synchronous Mode	16
4.1.1	FP3_Start	16
4.1.2	FP3_OptionFlags	16
4.1.3	FP3_TicketHeader	16
4.1.4	FP3_ItemSale	17
4.1.5	FP3_Payment	18
4.1.6	FP3_PrintTotalsAndPayments	19
4.1.7	FP3_PrintUserMessage	19
4.1.8	FP3_PrintBeforeMF	19
4.1.9	FP3_PrintMF	19
4.1.10	FP3_Close	19
4.2	Additional Functions in Transactions	19
4.2.1	FP3_GetPaymentApplicationInfo	19
4.2.2	FP3_VoidItem	19
4.2.3	FP3_VoidPayment	19
4.2.4	FP3_VoidAll	20
4.2.5	FP3_ReversePayment	20
4.2.6	FP3_Plus, FP3_Minus, FP3_Inc, FP3_Dec	20
4.2.7	FP3_SetOnlineInvoice	20
4.2.8	FP3_KasaAvans	21
4.2.9	FP3_CustomerAvans (Müşteri Avans)	21
4.2.10	FP3_KasaPayment	21
4.2.11	FP3_FunctionBankingRefundExt	21
4.2.12	FP3_GetTaxRates	22
4.2.13	FP3_GetDepartments/FP3_SetDepartments	22
4.2.14	FP3_GetTicket	22
4.3	Fiscal Printer Functions For Asynchronous Mode	23
4.3.1	MultipleCommand	23
4.3.2	Prepare_Start	24
4.3.3	Prepare_TicketHeader	24
4.3.4	Prepare_ItemSale	24
4.3.5	Prepare_Payment	24
4.3.6	Prepare_PrintTotalsAndPayments	24
4.3.7	Prepare_PrintBeforeMF	25
4.3.8	Prepare_PrintMF	25
4.3.9	Prepare_Close	25
4.3.10	Prepare_PrintUserMessage	25
4.3.11	Prepare_GetPLU	25
4.3.12	Prepare_ReversePayment	26
4.3.13	Prepare_SetInvoice	26

4.3.14	Prepare_SetParkingTicket	26
4.3.15	Prepare_KasaAvans	26
4.3.16	Prepare_KasaPayment	27
4.3.17	Prepare_Pretotal	27
4.3.18	Prepare_DisplayPaymentSummary	27
4.3.19	Prepare_Plus	27
4.3.20	Prepare_Minus	27
4.3.21	Prepare_Inc	28
4.3.22	Prepare_Dec	28
4.3.23	Prepare_VoidPayment	28
4.3.24	Prepare_VoidItem	28
4.3.25	Prepare_SendFrontStationPrint	29
4.3.26	Prepare_GetTicket	29
5	Transaction Flows	30
5.1	Sale Ticket Flow	30
5.2	Invoice Sale Ticket	31
5.3	Online Invoice Sale Ticket	32
5.4	Avans Ticket	33
5.5	Yemek Ceki Ticket	34
5.6	Tax Free Ticket	35
5.7	Refund Ticket	36
5.8	Parking Entry Ticket	37
5.9	Paper Detection	38
5.10	OKC Reset	39
5.11	Convert To Invoice	40
5.12	Cancel Payment	41
5.13	Bank Payment/Refund Type Selection	43
5.13.1	Credit Card Payment	43
6	Transaction Handle ID	46
6.1	APP_ERR_GMP3_INVALID_HANDLE	46
6.2	APP_ERR_GMP3_NO_HANDLE	46
	Using Customizable Currency In Cash Payments	48
	Using Food Voucher in Fiscal Receipt	50
	Vat Control Process	51
7	GMP3 – Error Handling Guide	52
7.1	Error Codes and Possible Actions	52
7.1.1	TRAN_RESULT_NO_PAPER = 0x0020	52
7.1.2	APP_ERR_FISCAL_INVALID_ENTRY = 2009	52
7.1.3	APP_ERR_APL_COMPLETE_PAYMENT = 2021	52
7.1.4	APP_ERR_APL_CREDIT_CANNOT_BIGGER_THAN_REMAIN_AMOUNT = 2022	52
7.1.5	APP_ERR_CASHIER_ENTRY_REQUIRED = 2053	52
7.1.6	APP_ERR_NOT_ALLOWED = 2064	52
7.1.7	APP_ERR_PAYMENT_FOUND = 2069	53
7.1.8	APP_ERR_ALREADY_DONE = 2080	53
7.1.9	APP_ERR_PAYMENT_NOT_SUCCESSFUL_AND_NO_MORE_ERROR_CODE = 2085	53
7.1.10	APP_ERR_PAYMENT_NOT_SUCCESSFUL_AND_MORE_ERROR_CODE = 2086	53
7.1.11	APP_ERR_VAS_NOT_AVAILABLE = 2088	53
7.1.12	APP_ERR_NOT_ALLOWED_AS_GMP3_TRANSACTION_IS_PENDING = 2097	53
7.1.13	APP_ERR_YEMEKCEKI_PAYMENT = 2119	53
7.1.14	APP_ERR_GMP3_INVALID_DATE_TIME = 2310	53
7.1.15	APP_ERR_GMP3_INVALID_HANDLE = 2317	54
7.1.16	APP_ERR_GMP3_APP_CHECKSUM_MISMATCH = 2338	54
7.1.17	APP_ERR_GMP3_NO_HANDLE = 2341	54
7.1.18	APP_ERR_GMP3_PAYMENT_CAN_NOT_VOID = 2357	54
7.1.19	APP_ERR_GMP3_Z_REQUIRED = 2417	54
7.1.20	APP_ERR_GMP3_APPLICATION_NOT_FOUND = 2440	54
7.1.21	DLL_RETCODE_PORT_NOT_OPEN = 0xF000	54
7.1.22	DLL_RETCODE_ACK_NOT_RECEIVED = 0xF01B	54

7.1.23	DLL_RETCODE_TIMEOUT= 0xF003	54
7.1.24	DLL_RETCODE_DATA_RECV_ERR = 0xF007	55
7.1.25	DLL_RETCODE_RECV_BUSY = 0xF01C	55
7.1.26	DLL_RETCODE_PAIRING_REQUIRED = 0xF020	55
7.1.27	APP_ERR_FIS_LIMIT_ASILAMAZ=2067	55
7.2	Thoughts On Error Handling	55
7.3	Other Errors	55
8	GDP_v15r32	65
8.1	GMP3 Simulator	65
8.2	GMP.XML	66
8.3	Using GMP3 Simulator	67
8.3.1	Sales Ticket	69
8.3.2	Invoice Sales Ticket	70
8.3.3	Use of Dynamic Vat Table	73
8.3.4	Disconnection and Out of Paper Scenarios	73

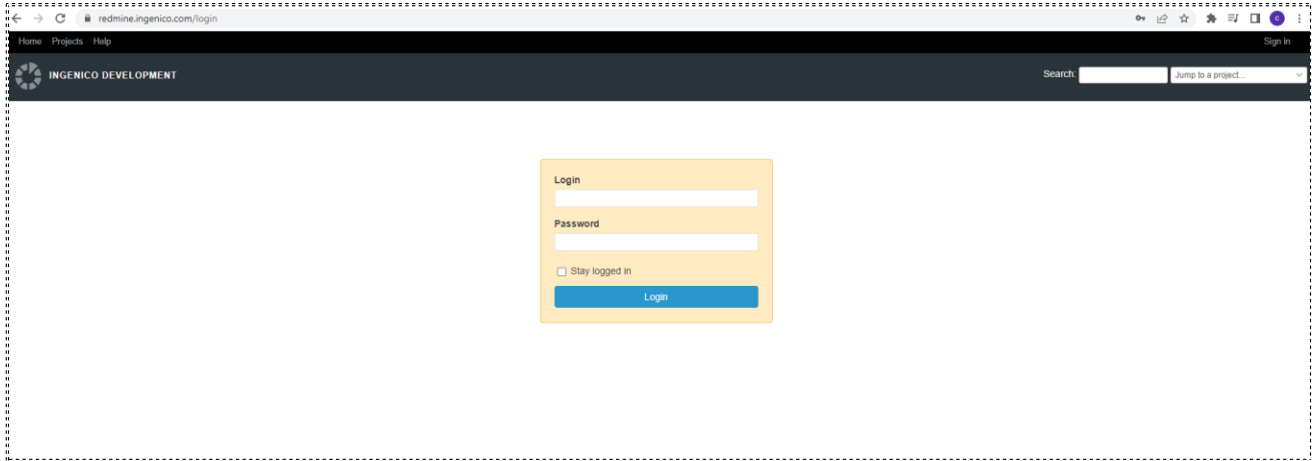
1 Redmine Introduction

Redmine is a web-based project management and bug tracking system software widely used all over the world.

As Worldline group, we use Redmine to support you for your requests and errors which you encounter during intagratation of your external software with our ECR devices.

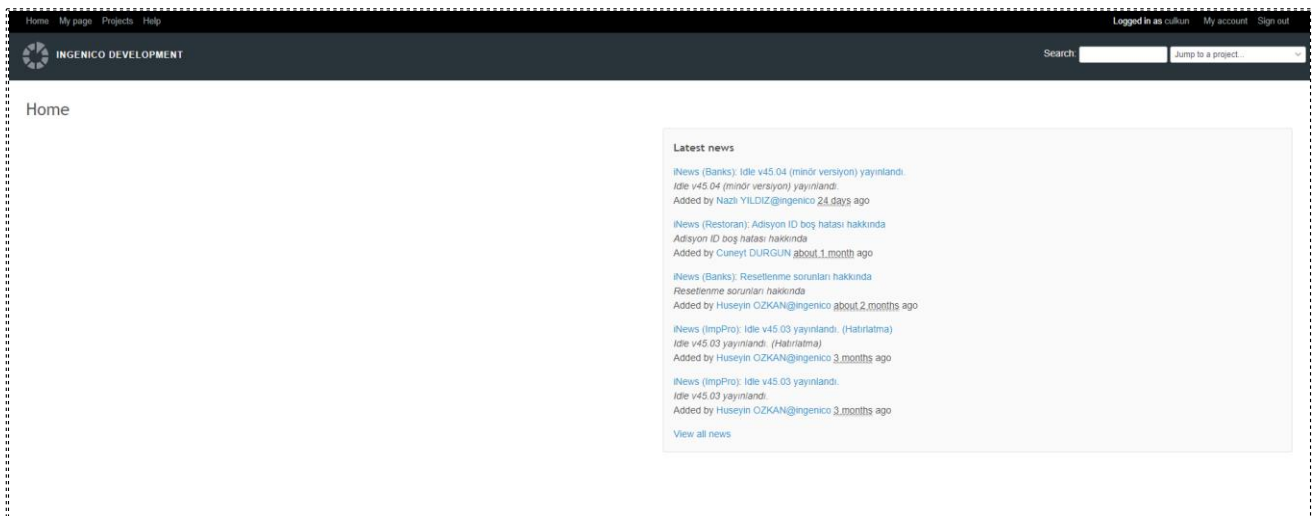
We just want to give you some briefing about Redmine as how 0to use it.

Login panel is below.

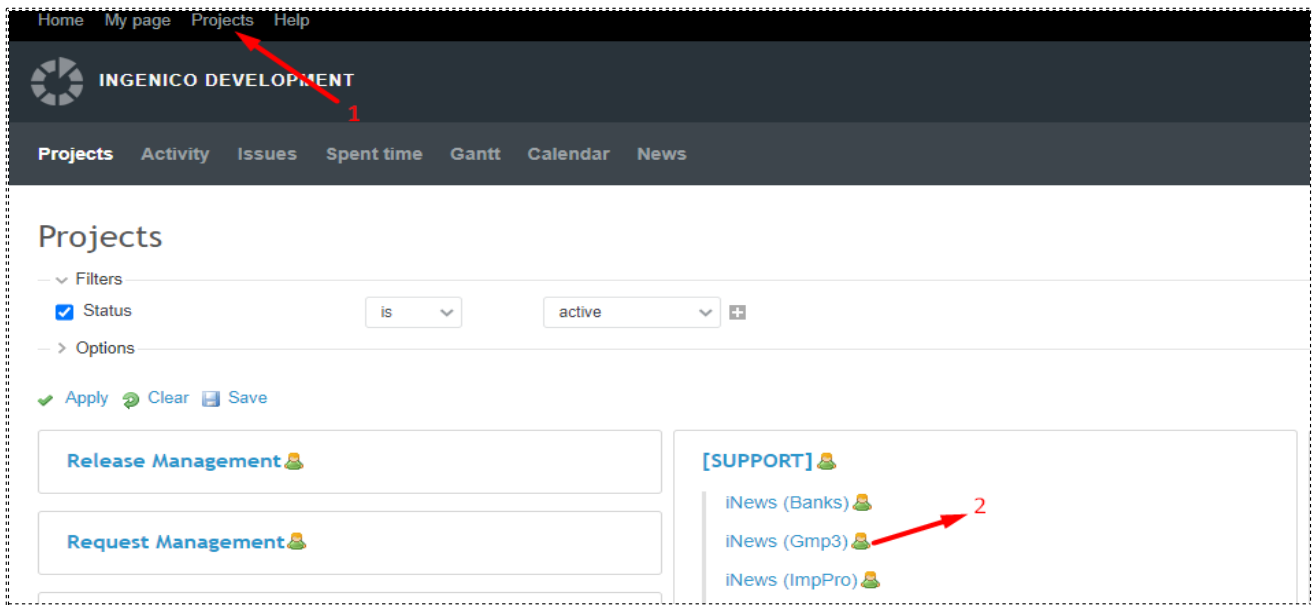
A screenshot of the Redmine login page. The browser address bar shows 'redmine.ingenico.com/login'. The page has a dark header with 'INGENICO DEVELOPMENT' on the left and a search bar on the right. In the center, there is a light orange login box with fields for 'Login' and 'Password', a 'Stay logged in' checkbox, and a blue 'Login' button.

After login, you are going to see the main page.

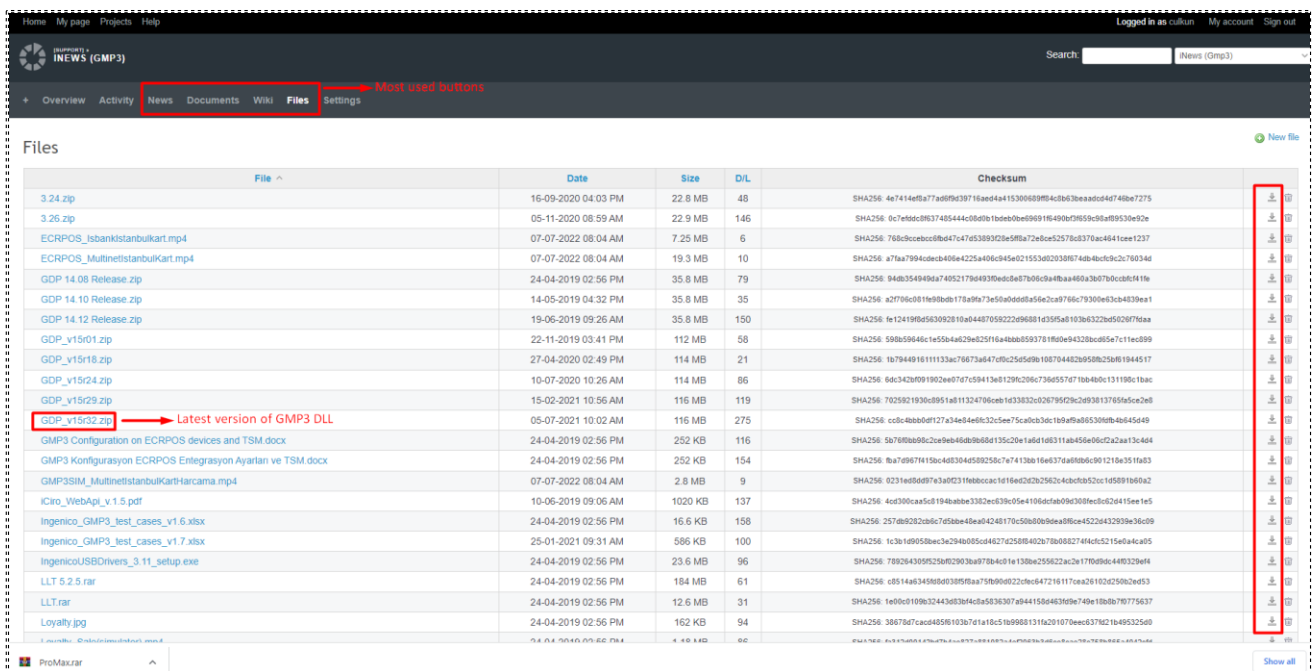
In addition, latest news/changes will be seen about our all projects which we published recently

A screenshot of the Redmine main page after login. The header shows 'Logged in as calkan' and 'My account Sign out'. The main content area has a 'Home' link on the left and a 'Latest news' section on the right. The 'Latest news' section lists several updates with links to view them, such as 'iNews (Banks): Idle v45.04 (minder versiyon) yayinlandi.' and 'iNews (ImpPro): Idle v45.03 yayinlandi. (Hastirlama)'.

When click Project+iNews (Gmp3) as below can be downloaded all documents and files of GMP3 DLL.



In this section there are some buttons direct to download documents and files of GMP3 DLL.



ECR

ECR devices(Electronic Cash Register) are designed, produced, certified and operated by the Government technical requirements. ECRPOS is a version of ECR which has an embeded EFT-POS functionality on it. INGENICO has two different models of ECRPOS, namely IWE280 and IDE280. Both devices are certified by Turkish Government and used in the field with success.

ECR (and ECRPOS) are designed to be used in different use cases like "stand-alone", "Cable Connection" and "TSM Connection".

GMPSmartDll is intended to use in "GMP3 Cable Connection" and "TSM Connection" modes to connect INGENICO IWE280 and IDE280 Model EFT-POS Embeded ECR devices. It can not be used with any other ECR or EFT-POS (even if it is INGENICO product).

GMP3

GMP3 stands for "Gelirler idaresi Mesaging Protocol 3" which is released by Government. GMP3 is a subset of GMP protocol which covers secure connectivity of external system to ECRs by cables. This document mostly covers GMP3 connectivity methods for INGENICO ECRs.

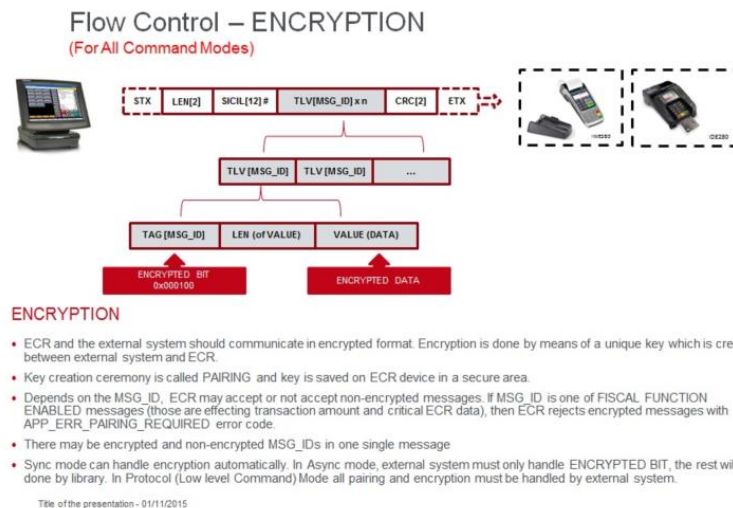
TSM

TSM is "Trusted Service Manager" which is defined and certified by Turkish Government Ministry of Finance. It's set of servers which is used to control ECRs. There are some mandatory services on it like connection to Government Servers, Bank Servers, ext. And there are also some value added services that ECRs and the merchants can benefit. This document also describes some methods which enables ECRs to connect Value Added Servers.

Pairing

PAIRING is the security procedure applied between External System and the ECR as defined in GMP3 Cable Connection. After pairing, the physical link becomes fully encrypted and secure.

Encryption



External System

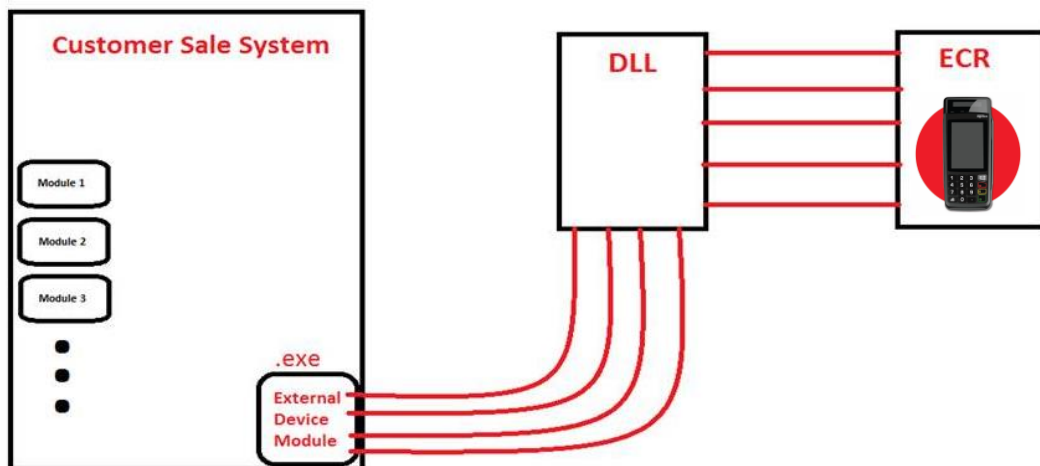
External System is any kind of external application (some time hardware + application) which connects to ECR via one of the connection methods defined in this document. It may also be called as "User Application" or "External Application".

HASH of External System

As GMP3 rules, INGENICO is obliged to control any External System which is connected to INGENICO ECRs by means of GMP3. It is the INGENICO responsibility to control and validate the external system in terms of fiscal regulation. INGENICO is performing a HASH value control of the external application to ensure that the external application is not changed without control and validation. it works like that;

- TSM is able to set a set of HASH values per ECR in the field. So, each ECR can be remotely configured for valid HASH values;
- ECR download TSM parameters to update valid HASH table;
- GmpSmartDll is able to get HASH of the external application that it is called;
- GmpSmartDll is sending the HASH code to ECR during the Pairing operation;
- ECR checks and validate HASH value during the pairing;

Now, you could set your environment like picture in below.



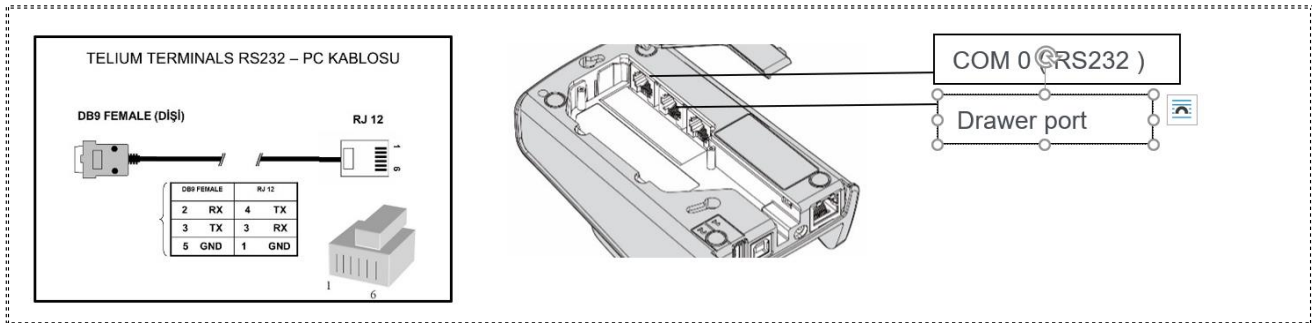
If you call DLL function via ExternalDeviceModule.exe, hash value wont change and once you set a hash value, it will be available a long time. Otherwise, if your big system(customer sale system) invokes DLL functions, when you change even a button location, hash value will change and you have to send your application to test and INGENICO has to define a new hash value for your external system application.

2 ECR – External System Connection Types

.According to the external system connection type, the wiring diagrams and the ports to be connected are given below.

- USB
- RS232
- TCP/IP (Ethernet)

2.1 iWE280 Base – RS232 / USB / ETHERNET(TCP) Connections



.The DB9 end of the cable should be connected to the RS232 port of the EXTERNAL SYSTEM, and the RJ12 end of the cable should be connected to the COM 0 terminal under the BASE of the cash register.

2.2 iWE280 Base – USB Cable Connection

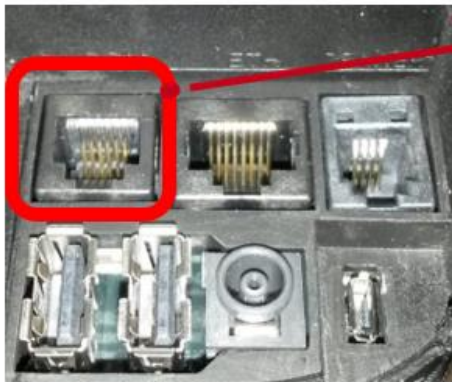


The standard USB end of the cable should be connected to the standard USB port of the EXTERNAL SYSTEM, and the USB -B end of the cable should be connected to the BASE -USB B female end of the cash register.

2.3 iWE280 Base – ETHERNET (TCP) Cable Connection

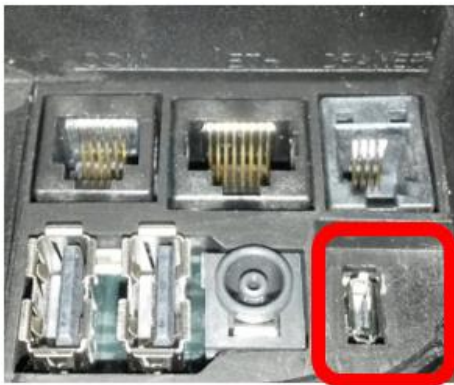


2.4 iDE280 – RS232 Connection



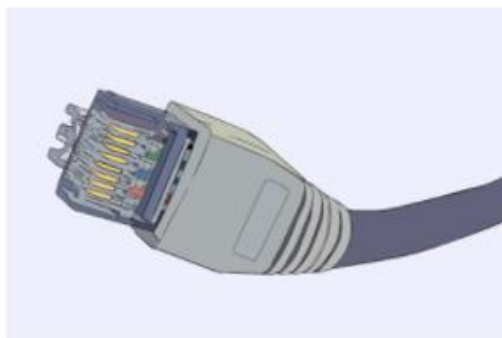
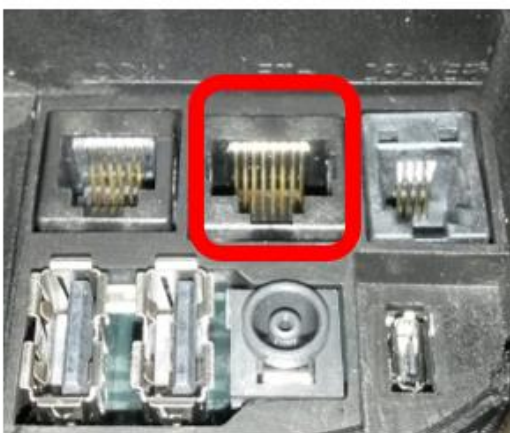
iDE280 - COM 0 (RS232)

iDE280 Micro Base – Standart USB Connection

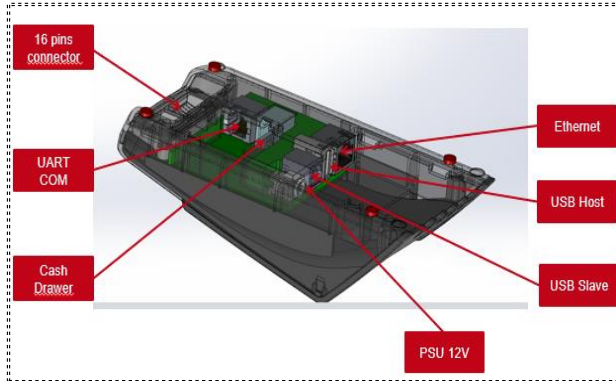


The Standard USB end of the cable should be connected to the standard USB port of the EXTERNAL SYSTEM, and the Micro USB end of the cable should be connected to the Micro USB port of the iDE280 cash register.

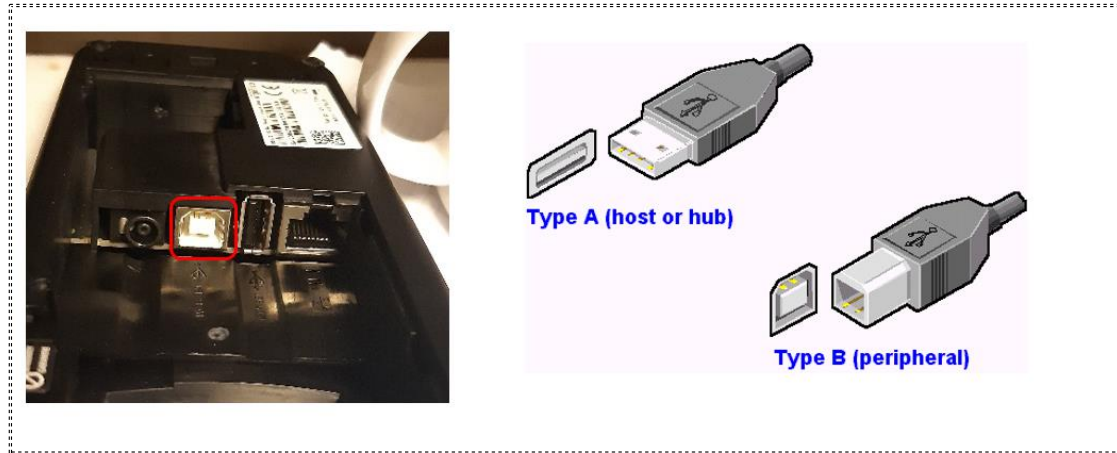
2.5 iDE280 – ETHERNET Cable Connection



2.6 MOVE5000F Base– RS232 Connection



2.7 MOVE5000F USB – Standart USB Connection



2.8 MOVE5000F BASE - ETHERNET (TCP) Cable Connection



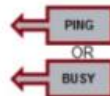
2.9 Flow Control, BUSY, PING & ECHO

There are 3 methods to check connection availability of ECR.

These are;

- FP3_PING
- FP3_BUSY
- FP3_ECHO

Flow Control – PING (For All Command Modes)



PING

- PING is one byte control character (0x57) sent by External System and replied (echo) by ECR. By using PING, external system may control the presence of ECR and its availability.
- There are 3 options for an external system to receive after sending PING;
 - 1)Receiving PING; ECR is online and available
 - 2)Receiving BUSY; ECR is online but BUSY
 - 3)Timeout (1 sec); ECR is not online
- It's the fastest way to control the ECR availability and can be used in external system idle times to know the status of ECR (Available, Offline or Busy).
- Even if it can be used on both TCP/IP & RS232/USB communication links, it's mostly designed for RS232/USB links.

Flow Control – BUSY (For All Command Modes)

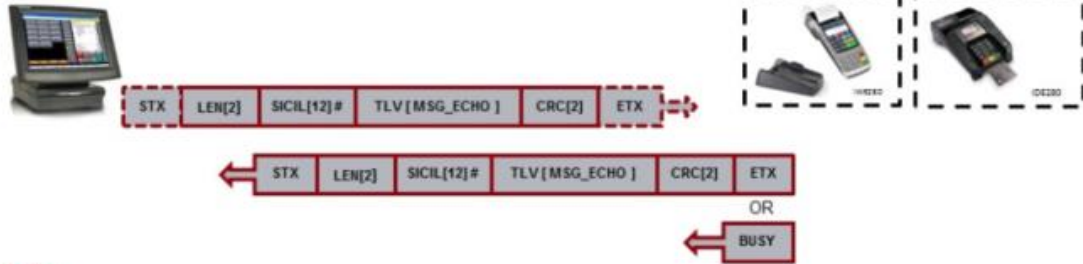


BUSY (APP_ERR_BUSY)

- ECR sends BUSY to external system for the one of the following conditions;
 - 1)ECR is performing a background task (visible on display or not visible)
 - 2)ECR is serving to an other port
 - 3)ECR is not on the idle (main) display
- ECR GMP3 driver waits 1 second before sending BUSY to external system. If ECR ends the BUSY reason, message is processed normally and the external system does not receive BUSY message
- ECR does not save any data from the coming Request message after sending BUSY. It is fully external system responsibility to resend the message or cancel it.
- ECR may send BUSY between multiple command set. The control is on the communication layer and high level function status does not effect this behavior.
- BUSY message (0x11) is same for all modes and communication interfaces.
- BUSY messages are handled by Sync and Async modes automatically which generates error code of APP_ERR_BUSY

Flow Control – ECHO

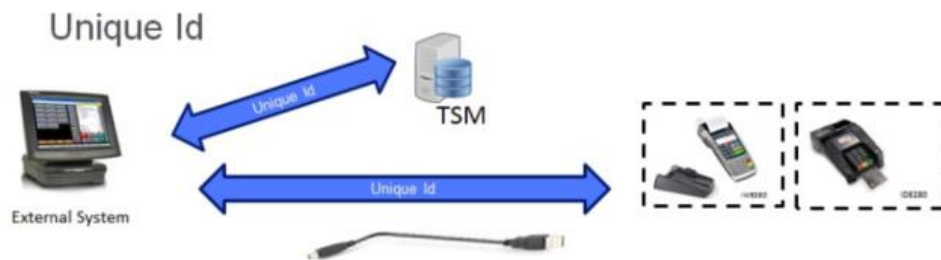
(For All Command Modes)



ECHO

- ECHO is a type GMP3 package sent by external systems and replied by ECR. By using ECHO, external system may control the presence of ECR, its availability and some generic data (Z No, F No, Casheir, ext).
- There are 3 options for an external system to receive after sending ECHO command;
 - 1)Receiving ECHO package; ECR is online and available
 - 2)Receiving BUSY; ECR is online but BUSY
 - 3)Timeout (3 sec); ECR is not online
- It's a way to control the ECR availability before some major functions (before pairing or transactions,ext) but it is not the fastest way for external system to know the status of ECR (Available, Offline or Busy) on idle times.
- It can be used on both TCP/IP & RS232/USB communication links.

2.10 Unique ID



Unique Id

- It is a Unique string which provides a trustable track of ECR transaction driven by External Systems. The format of Unique id is Vendor depended. Ingenico is using 24 bytes Unique id defined by Ingenico itself.
- External systems must get an Unique Id at the beginning of each transaction. And then this unique id will be verified by ECR to continue the transaction.
- There are two different source for Unique Id:
 - 1) ECR: Each cable connected ECR can produce it's set of Unique id and provide it to external system before the transaction start. This Unique Id is called «Local Unique Id» and can not be used other than this parent ECR!
 - 2) TSM: TSM can also produce a Unique Id by signed TSM keys. External System can connect to TSM and ask for a Unique Id + Signed value of it. Each Ingenico ECR is able to verify TSM Unique Id by using Signed value of it.
- So Ingenico Offer you different use cases of Unique Id:
 - 1) Local Unique Id Created by ECR – Verified by the same ECR
 - 2) TSM Unique Id Created by TSM – Verified by any Cable Connected ECR which should belong to the same merchant !
 - 3) TSM Unique Id Created by TSM – Verified by any TSM Connected ECR which should belong to the same merchant !

There is only one function to manage (create & verify) the unique id in Ingenico ECRs.

For Synchronous and Asynchronous Mode users it is FP3_Start function.

For Protocol (Low level command) mode users it is

subCommand_GMP3_SUBCOMMAND_FISCALPRINTER_START

2.10.1 Creation Of Unique Id on ECR

The related function is called with Unique Id value of all 0x00's. This is replied by the ECR by creating a new Unique Id on its internal resources.

•External system may continue the transaction with Ticket Header and other fiscal functions. In this case external system does not need to re-use Unique id again. When the transaction ends, Unique id will be printed automatically at the end of the ticket. So the creation and verification of the Unique id is done on the same session.

•Alternatively, external system may call FP3_Close (or subCommand_GMP3_SUBCOMMAND_FISCALPRINTER_CLOSE for low level implementation) to close the session. In this case External system must save the Unique to use later during the fiscal ticket printing phase.

2.10.2 Creation Of Unique Id on TSM

There is a webService for this functionality. Please contact to Ingenico for details...

Verification (Using) Of Unique Id on ECR

The related function (FP3_Start or subCommand_GMP3_SUBCOMMAND_FISCALPRINTER_START) is called with an Unique Id (either ECR Local Unique Id or TSM based Unique id with signature value). Function verifies the Unique Id and continue the transaction.

3 Transaction Flows and Daigrams

There are 2 types of implemantation mode between ECR and External System.

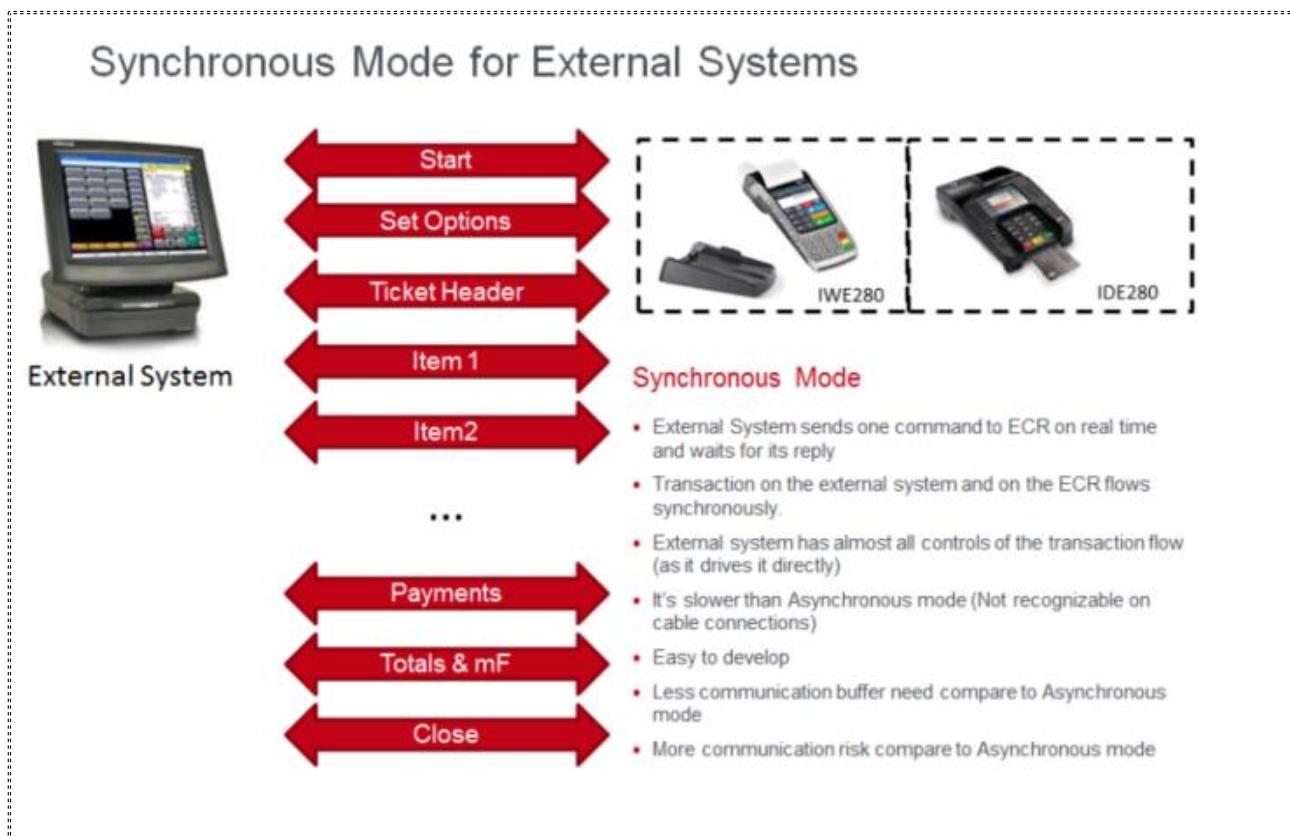
These are;

3.1 Synchronous (Single Command) Mode

Synchronous (Single Command) Mode is the basic implemation method that can be used between ECR and the External System. It is simple a command sent by external system and a reply from ECR device for each action (like start of ticket, item print, ext).

If the connection type is GMP3 Cable Connection, then Pairing must be completed before any other command sent from External System.

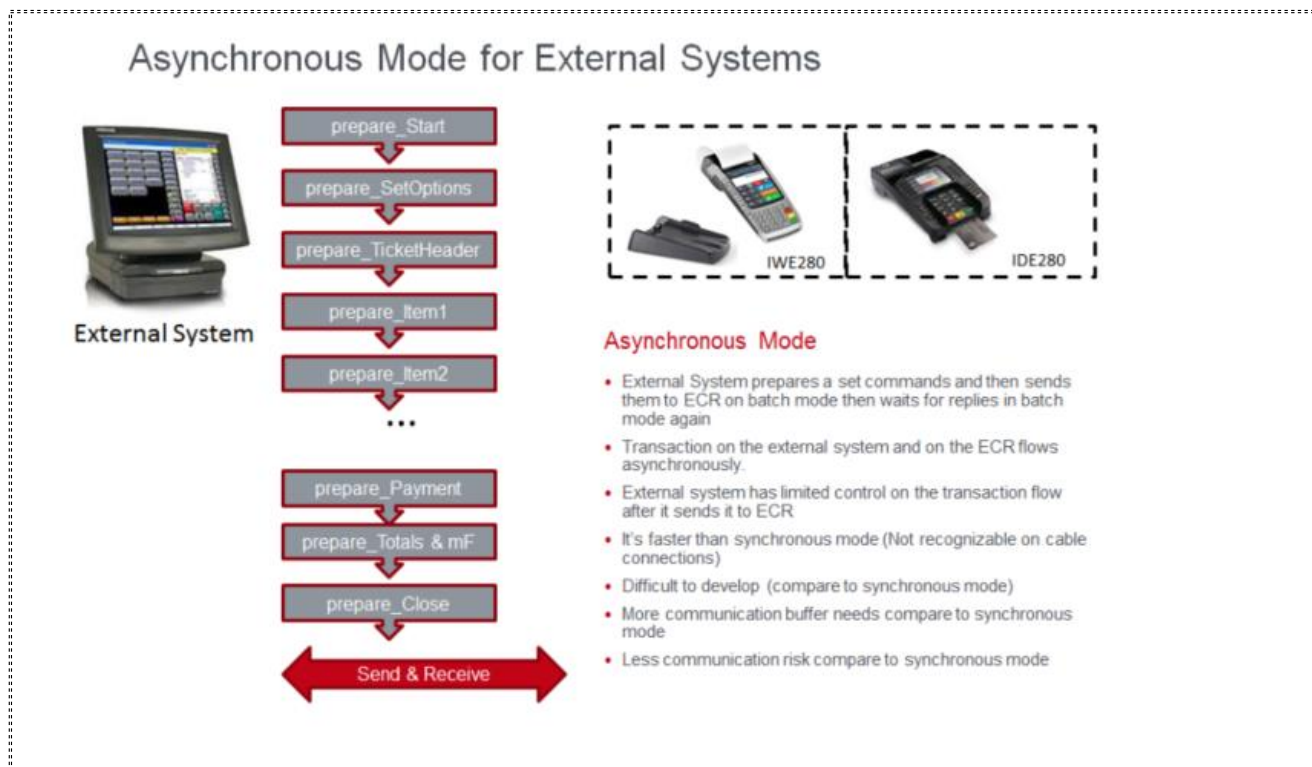
It is advised to call FP3_Echo before any other Synchronous command on GMP3 Cable Connection. This way, external application can be sure about the physical connection and the availability of the ECR.



3.2 Asynchronous (Batch Command) Mode

Asynchronous (Batch Command) Mode is used to send a set of commands in batch mode. ECR will process the commands one by one starting from the first one. If one of the commands are failed during execution, ECR returns with the result code of previously executed commands and the last one. It's external system responsibility to decode the error codes and decides new commands if needed.

If the connection type is GMP3 Cable Connection, then Pairing must be completed before any other command sent from External System.



4 FUNCTIONS

ECR and External System communication is provided by GMP3 functions. Duties of these functions' are same, but names are a little bit different depending on the mode which is used.

FP3 Functions For Synchronous Mode

FP3_Start
FP3_Close
FP3_TicketHeader
FP3_ItemSale
FP3_PrintTotalsAndPayments
FP3_PrintBeforeME
FP3_PrintME
FP3_PrintUserMessage
FP3_PrintUserMessage_Ex (For bigger size data, such as QRCode)
FP3_GetTicket
FP3_Payment
FP3_ReversePayment
FP3_SetInvoice
FP3_SetParkingTicket
FP3_KasaAvans
FP3_CustomerAvans
FP3_KasaPayment
FP3_Pretotal
FP3_DisplayPaymentSummary
FP3_Plus
FP3_Minus
FP3_Inc
FP3_Dec
FP3_VoidPayment
FP3_VoidAll
FP3_VoidItem

4.1 FP3 Functions For Synchronous Mode

4.1.1 FP3_Start

- First function to start a transaction.
- Return value is not OK(it actually returns 0 for OK) when an another transaction is still active on Ingenico device.
- Parameter: CurrentInterface, represents current options for external application – ECR Pos communication.
- Parameter: TranHandle, for only this function, it'll be sent as reference with value of 0, and it'll return transaction handle information, which represents that particular transaction.

4.1.2 FP3_OptionFlags

- This function is used for setting some options for that particular transaction.
- Parameter: TranHandle, this is created by DLL on FP3_Start method as explained above. This value now must be passed as argument.
- Parameter: activeFlags, this returns currently active flags.
- Parameter: FlagsToBeSet, external application may set desired options with this parameter.

4.1.3 FP3_TicketHeader

- This function trigger header print from Ingenico device.
- Parameter: ticketType([TTicketType](#)), it basically determines what kind of receipt header will be printed from Ingenico device. E.g fiscal receipt, invoice sale, taxfree sale.

Values of Ticket Types in GMPSmartDLL as below.

```

TTasnifDisi = 0,
TProcessSale = 1,      //Fiscal Ticket
TZReport = 2,
TXReport = 3,
TEJReport = 4,
TFiscal2Z = 5,
TFiscal2T = 6,
TFiscalCumm = 7,
TAvans = 8,           //Non_Fiscal Ticket
TPayment = 9,         //Non_Fiscal Ticket
TZDonemReport = 10,
TXDonemReport = 11,
TXPluSale = 12,
TInvoice = 13,        //Non_Fiscal Ticket
TVoidSale = 14,       //Non_Fiscal Ticket
TRefund = 15,         //Non_Fiscal Ticket
TYemekceki = 16,     //Non_Fiscal Ticket
TOtopark = 17,        //Non_Fiscal Ticket
TZReportForce = 18,
TInfo = 19,           //Non_Fiscal Ticket
TTaxFree = 20,        //Fiscal Ticket
TDailyMemory = 21,
TKasaAvans = 22,      //Non_Fiscal Ticket
TCariHesap,           //Non_Fiscal Ticket
TUniqueId = 127,
TLAST                // Bu satir hep sonda kalmali

```

4.1.4 FP3_ItemSale

- This function is used for adding a new item for current transaction. E.g shoes, bread, towels, vegetables. Obviously can be called multiple times.
- This function gets two different parameters as reference. One stItem([ST_ITEM](#)) and the other called m_stTicket([ST_TICKET](#)).
- m_stTicket object is used for returning information back to external system. These values can be found as needed but it's out of scope so it'll not be analysed in depth. Check "ST_SALEINFO" object under m_stTicket object for more information.
- stItem object contains all the information about an item, explained below;
- Parameter: type, this must be set as 1 most of the times. There are other possibilities, but it's not widely used. For other options, check out "[ITEM_TYPE_](#)" prefixed constant in Defines class.
- Parameter: subType, this is not currently in use, can be set as 0.
- Parameter: deptIndex: represents index of related department on Ingenico device. For example, it must be 0 if you want to use first department for this item.
- Parameter: amount, price of that item, should be multiplied by 100. For example, item's price is 175 TL, amount must be set as 17500.
- Parameter: currency, represents currency of an item's price. Must be set to 949 for Turkish Lira.
- Parameter: count, represents how many of an item is wanted to be added. Note: amount represent only one item's price, calculation takes place on Ingenico device.
- Parameter: unitType, represents type of count should be used for an item. [1\(EltemUnitTypes.ITEM_NUMBER\)](#) means countable, this one is widely used. There is also kilogram, litre etc.
- Parameter: name, represents name of that item, this value will be printed through Ingenico device.
- Parameter: barcode, if barcode information is used on external system, this can be set to this parameter.
- Parameter: flag, some specific attributes can be added to this parameter like specifying an item will be tax free.

4.1.5 FP3_Payment

- This function is used for payment process. It can be called multiple times as needed. For example, transaction total amount is 500 TL and Payment function can be called with 300 TL credit card and for the rest, 200 TL it can be called as cash or any other supported types.
- Payment function accepts two argument as objects. One is stPaymentRequest(ST_PAYMENT_REQUEST) and the other is m_stTicket(ST_TICKET)
- stPaymentRequest object contains all details about a payment request. Essential parameters is analysed below;
- Parameter: BankPaymentUniqueld, set by external system, must be unique, simulator code contains a method to create one.
- Parameter: typeOfPayment, represents main payment method, e.g credit card, cash.
- Parameter: subtypeOfPayment, determines what kind of payment will take place, such as regular sale, instalment sale.
- Parameter: payAmount, paying amount, for example, must be set to 15000 for 150 Turkish Lira.
- Parameter: payAmountBonus, this must be set for loyalty sales. Note: it might seem, partial payment(regular sale + loyalty sale) at a time, it's not supported.
- Parameter: payAmountCurrencyCode, currency code of payment. 949 for Turkish Lira.
- Parameter: bankBkmId, Can be set to related bank bkm id, in case of a specific bank usage is wanted to be forced. Otherwise, must be set to 0, and Ingenico device will decide which bank application to proceed.
- Parameter: transactionFlag, can be used for setting some options. Usage can be found in the simulator code. Deatils are out of the scope of this document.
- m_stTicket object contains important information about payment response. Please see parameters with explanation when necessary, below.
- This information can be found under ST_TICKET -> ST_PAYMENT -> ST_BANK_PAYMENT_INFO typed object.
- Parameter: batchNo, related bank application batch number. Usually represent an end of day for a bank application. Automatically incremented by bank application on each end of day.
- Parameter: stan, related bank application transaction's number. Incremented on each transaction by bank application.
- Parameter: bankBkmId, determine related bank application. If bankBkmId is sent as 0 on payment request, this value can be used to decide which bank application is used for a transaction.
- Parameter: authorizeCode, authorization code of a transaction.
- Parameter: terminalId: Unique id for banking application installed on a particular Ingenico device. It's very important in case of an issue about a banking application, with this value bank host is able know details about that application.
- Parameter: merchantId: Unique id for a bank's customer.
- Parameter: rrn, stands for receiver registration number, provided by bank host, can be empty for offline banking payments.
- Parameter: bankName, obviously related bank name in string.
- Parameter: numberOfInstallments, Instalment count, can behave differently bank to bank, some banks cumulate extra instalment, some don't.
- Parameter: stCard, contains card pan and card input type such as chip, contactless.
- Parameter: ST_PaymentErrMsg, consists of error messages if any.

Values of Payment Types in GMPSmartDLL as below.

```
PAYMENT_ALL = 0x000FFFFF,  
PAYMENT_CASH_TL = 0x00000001,  
PAYMENT_CASH_CURRENCY = 0x00000002,  
PAYMENT_BANK_CARD = 0x00000004,  
PAYMENT_YEMEKCEKI = 0x00000008,  
PAYMENT_MOBILE = 0x00000010,  
PAYMENT_HEDIYE_CEKI = 0x00000020,  
PAYMENT_IKRAM = 0x00000040,  
PAYMENT_ODEMESIZ = 0x00000080,  
PAYMENT_KAPORA = 0x00000100,  
PAYMENT_PUAN = 0x00000200,  
PAYMENT_GIDER_PUSULASI = 0x00000400,  
PAYMENT_BANKA_TRANSFERRI = 0x00000800,  
PAYMENT_CEK = 0x00001000,  
PAYMENT_ACIK_HESAP = 0x00002000,  
PAYMENT_DIGER = 0x00004000, //      xx
```

4.1.6 FP3_PrintTotalsAndPayments

- This function is used for printing total line, after this function call, any item or payment cannot be cancelled but whole receipt still can be cancelled, takes only standard arguments(handle, interface, timeout)and return code is enough to understand if it's successful or not.

4.1.7 FP3_PrintUserMessage

- This function is used for printing custom messages, logos, barcodes and QR codes. It can be used throughout any kind of transaction.
- Logo files must be under certain properties, monochrome, depth:1, filename length max:12, Width under 384 px.
- This function accepts two main objects, m_stTicket(ST_TICKET) and stUserMessage(ST_USER_MESSAGE), m_stTicket does not take an important responsibility. Following parameters are set in stUserMessage object.
- Parameter: len, length of message.
- Parameter: message, message itself.
- Parameter: flag, determines lots of details, bitmap, barcode, text, centered etc.

4.1.8 FP3_PrintBeforeMF

- This function is used for saving a fiscal or non-fiscal receipt on to EJU memory, only takes standard arguments. Once this function is called, there is no coming back, that transaction must be completed.

4.1.9 FP3_PrintMF

- This function is used for printing fiscal symbol on slip, only takes standard arguments. This function also triggers banking slip printing.

4.1.10 FP3_Close

- This function is aimed to be used as transaction closer, only takes standard arguments.It should be called after all transaction's end.

4.2 Additional Functions in Transactions

These functions may or may not be used in any transaction, depending on implementation needs.

4.2.1 FP3_GetPaymentApplicationInfo

- This function returns bank applications installed on an Ingenico device.
- It takes an argument called StPaymentApplicationInfo(ST_PAYMENT_APPLICATION_INFO) and it'll be an empty object during request. DLL fill this object with proper data as a collection, each member represents a bank application. Analysed below.
- Parameter: u16BKMIId, represents a bank's unique EFT code.
- Parameter: u16ApplId, represents bank's unique id, determined on Ingenico side.

4.2.2 FP3_VoidItem

- This function is used for cancelling one or more item included before. It also prints out necessary information about that cancelling.
- Parameter: ItemIndex, order of unwanted item, starts from zero.
- Parameter: ItemCount, determines how many item will be cancelled, for example, there is a particular item, added three times before, only one of these can also be cancelled.

4.2.3 FP3_VoidPayment

- This function is used for voiding previous successful payments when necessary. Notice, this can't be used after total amount printed out with "FP3_PrintTotalsAndPayments".
- Parameter: PaymentIndex, order of unwanted payment, starts from zero. If the payment was in credit card, Ingenico device will take of any other details.
- m_stTicket(ST_TICKET) is also sent as an argument but it does not have a big responsibility, can be used for checking current situation of that particular transaction.

4.2.4 FP3_VoidAll

Function sends a command to ECR to void ALL Ticket immediately.

Parameters

[in] **hInt**: Interface Handle

[in] **hTrx**: Transaction Handle

[out] **pstTicket**: Pointer to ST_TICKET structure that will keep the result of the operation

[in] **timeoutInMilliseconds**: Timeout value in ms that the driver will wait a reply

4.2.5 FP3_ReversePayment

Function sends a command to ECR to start a reverse payment (negative payments)(REFUND, VOID, ext) operation

Parameters

hInt : Interface Handle

hTrx :Transaction Handle

stPaymentRequest : structure to ST_PAYMENT_REQUEST items which keeps parameters for a payment operation

numberOfPaymentRequests :number of reverse payment requests

[out] **pstTicket** :Pointer to ST_TICKET structure that will keep the result of the operation

timeoutInMilliseconds : Timeout value in ms that the driver will wait a reply

Values of Reverse Payment Types in GMPSmartDLL as below.

```
REVERSE_PAYMENT_CASH = 0x00100000,  
REVERSE_PAYMENT_BANK_CARD_VOID = 0x00200000,  
REVERSE_PAYMENT_BANK_CARD_REFUND = 0x00400000  
REVERSE_PAYMENT_YEMEKCEKI = 0x00800000,  
REVERSE_PAYMENT_MOBILE = 0x01000000,  
REVERSE_PAYMENT_HEDIYE_CEKI = 0x02000000,  
REVERSE_PAYMENT_PUAN = 0x04000000,  
REVERSE_PAYMENT_ACIK_HESAP = 0x08000000,  
REVERSE_PAYMENT_KAPORA = 0x10000000,  
REVERSE_PAYMENT_GIDER_PUSULASI = 0x20000000,  
REVERSE_PAYMENT_BANKA_TRANSFERRI = 0x40000000,
```

4.2.6 FP3_Plus, FP3_Minus, FP3_Inc, FP3_Dec

- FP3_Plus: amount increase.
- FP3_Minus: amount decrease.
- FP3_Inc: percentage increase.
- FP3_Dec: percentage decrease.
- All of these functions can be used for only one item or for total amount.
- Parameter: Rate, amount or percentage, 1000 for 10 TL decrease/increase, 10 for %10 decrease/increase.
- Parameter: itemNo: desired item index, should be set as 0xFFFF for total amount decrease/increase.
- Parameter: szText: message text of a decrease/increase, for example, "father's day campaign".

4.2.7 FP3_SetOnlineInvoice

- This function must be used before FP3_TicketHeader, if external system will use online invoice integration.
- Parameters will be sent in stOnlineInvoiceInfo([ST_ONLINE_INVOICE_INFO](#)) object, listed below.
- Parameters: CustomerName, VKN, HomeAddress, District, City, Country, Mail, WebSite, Phone, TaxOffice, Etn, DespatchNo, Identifier, OrderNo, Type, OrderDate, DespatchDate, SellerIdentifier_OnlineInvoice, SellerIdentifier_OnlineArchive

4.2.8 FP3_KasaAvans

Function sends a command to ECR to SET the AVANS ticket parameters
(AVANS : An amount that is accepted as INCOME without ny item sale. It's like PRE_PAYMENT)

Parameters

[in] **hInt** : Interface Handle
[in] **hTrx** : Transaction Handle
[in] **amount** : Amount to set as AVANS
[out] **pstTicket** : Pointer to ST_TICKET structure that will keep the result of the operation
[in] **timeoutInMilliseconds** : Timeout value in ms that the driver will wait a reply

4.2.9 FP3_CustomerAvans (Müşteri Avans)

Function sends a command to ECR to SET the AVANS ticket parameters
(AVANS : An amount that is accepted as INCOME without ny item sale. It's like PRE_PAYMENT)

Parameters

[in] **hInt**: Interface Handle
[in] **hTrx**: Transaction Handle
[in] **amount**: Amount to set as AVANS
[out] **pstTicket**: Pointer to ST_TICKET structure that will keep the result of the operation
[in] **pCustomerName**: Customer name value for receipt info, it fills in the buffer if not empty
[in] **pTckn Customer** :TCKN value for receipt info, it fills in the buffer if not empty
[in] **pVkn Customer**: VKN value for receipt info, it fills in the buffer while pTckn is NULL and pVkn is not empty
[in] **timeoutInMilliseconds**: Timeout value in ms that the driver will wait a reply

4.2.10 FP3_KasaPayment

Function sends a command to ECR to SET the PAYMENT ticket parameters (his is not the same Payment operation for ordinary tickets)
(PAYMENT : An amount that is paid to some one. It's like giving money to someone)

Parameters

[in] **hInt** : Interface Handle
[in] **hTrx** : Transaction Handle
[in] **amount** : Amount to set as PAYMENT
[out] **pstTicket** : Pointer to ST_TICKET structure that will keep the result of the operation
[in] **timeoutInMilliseconds** :Timeout value in ms that the driver will wait a reply

4.2.11 FP3_FunctionBankingRefundExt

- This function is used for void and refund process.
- Brief background about void and refund. Void takes place when a transaction should be cancelled in the same bank applicaiton batch (after a transaction is done, before banking settlement is done). Refund, on the other hand, takes place when a transaction should be cancelled after that bank settlement or on another Ingenico device.
- Parameter: bankBkmlId, related bank BKM Id, a.k.a Bank EFT Code.
- Parameter: typeOfPayment, void or refund indicator.
- Parameter: payAmount, original transaction's amount.
- Parameter: payAmountCurrencyCode, original transaction's currency code.
- Parameter: subtypeOfPayment, is used for determining different kind of refund Operations.
- Parameter: payAmountBonus, determines bonus amount of original transaction, if there is any
- Parameter: numberOfinstallments, determines intallment count in case of installement sale's refund.

- Parameter: following list of parameters is used for depending on bank application's need, please check out "\GDP 14.xx\ Release\Doc\ECRPOS_ExtDevice_Refund_and_Void_20160824.docx" named document, in order to find out which bank requires which parameters.
- Parameter List: rrn, MerchantId, AuthorizationCode, referenceCodeOfTransaction, TransactionType, TransactionAmount, numberOfInstallments, LoyaltyAmount, TransactionDate, terminalId, BankPaymentUniqueId(will be created at the time of refund)

4.2.12 FP3_GetTaxRates

- This function is used for getting tax rates which is configured on related Ingenico device.
- Tax Rates cannot be set through GMP3 integration, it can be basically considered that on parameter download, Ingenico device will get all of the available tax rates, defined by the government.
- Parameter: stTaxRates, determines an empty ([ST_TAX_RATE](#)) typed object as reference. This function will fill this object with proper tax rates, and these tax rates can be used on departments. Tax index in this object collection should be referred on a department.

4.2.13 FP3_GetDepartments/FP3_SetDepartments

- FP3_GetDepartments and FP3_SetDepartments are analyzed together. Difference is straight-forward. If you call FP3_GetDepartments, it'll return what Ingenico device already has, if you call FP3_SetDepartments it'll change department configuration as you fill stDepartments object.
- This function is used for determining items properties like tax rate, currency, unit type.
- This function accepts ([ST_DEPARTMENT](#)) typed collection object with default values as reference. Function call will fill related values from Ingenico device.
- Let's assume ([ST_DEPARTMENT](#)) typed object named as "stDepartments". Each parameter of "stDepartments" explained below.
- Parameter: szDeptName, determines default name value. It's printed out on slip if the "name" property of [ST_ITEM](#) typed object is not filled.
- Parameter: u8TaxIndex, determines desired tax rate's index.
- Parameter: iCurrencyType, determines currency code. Check out "[ECurrency](#)" enum in simulator code for more information.
- Parameter: iUnitType, determines unit type, e.g countable, gram, liter. Check out "[EItemUnitTypes](#)" enum in simulator code for more information.
- Parameter: u64Limit, this parameter is not widely used, usually set to highest possible value. What it does is that limiting maximum amount of an item with related department. E.g If it set to 100000, then any item more expensive than 1000 TL cannot be added with that department.
- Parameter: u64Price, can be used to set a default price for an item, which is related to this department.
- Parameter: bLuchVoucher, determines whether a food card sale can be used for this department or not. If it sets as 1, then it's tax rate must be set to a tax index that has %8 tax rate defined because of the government regulations.

4.2.14 FP3_GetTicket

-The purpose of this function is to keep the open ticket in the memory during disconnection or receipt expiry scenarios and continue the process when pairing is achieved again

-Function sends a command to ECR to get the details of an open ticket.Function then waits the result of the command for a given timeout value

Parameters

hInt : Interface Handle

hTrx : Transaction Handle

pstTicket : Pointer to ST_TICKET structure that will keep the result of the operation

timeoutInMilliseconds :Timeout value in ms that the driver will wait a reply

4.3 Fiscal Printer Functions For Asynchronous Mode

FP3_MultipleCommand
prepare_Start
prepare_Close
prepare_PrintTotalsAndPayments
prepare_PrintBeforeMF
prepare_PrintMF
prepare_PrintUserMessage
prepare_GetTicket
prepare_GetPLU
prepare_Payment
prepare_ReversePayment
prepare_TicketHeader
prepare_SetInvoice
prepare_SetParkingTicket
prepare_ItemSale
prepare_KasaAvans
prepare_KasaPayment
prepare_Pretotal
prepare_DisplayPaymentSummary
prepare_Plus
prepare_Minus
prepare_Inc
prepare_Dec
prepare_VoidPayment
prepare_VoidAll
prepare_VoidItem
prepare_SendFrontStationPrint

4.3.1 MultipleCommand

Function sends and receive FP3_xxx commands in batch format. Response buffer is parsed into ST_MULTIPLE_RETURN_CODE and ST_TICKET structures until it faces with a problem

Parameters

[in] hInt Interface Handle
[in] hTrx Transaction Handle
[out] returnCodes Array of ST_MULTIPLE_RETURN_CODE struture which will keep the result codes. It's the user responsibility to give enough buffer to receive all return codes
[out] pnumberOfreturnCodes pointer to integer that keeps number of return codes in returnCodes buffer.
[in] sendBuffer Send message buffer (which is prepared with prepare_xxx commands...)
[in] sendBufferLen Send message len
[out] pstTicket Pointer to ST_TICKET structure that will keep the result of the operation
[in] timeoutInMiliseconds Timeout value in ms that the driver will wait a reply

Returns

One of EGlobalReturnCodes
If successful, returns TRAN_RESULT_OK,
else, returns a member from EGlobalReturnCodes

NOTE 1) Full Response Message Format should be like this :
STX(Only Serial Interfaces) - LEN[2] - SICIL[12] - MSGID[3] - LEN[1-2-3] - DATA[n] - CRC[2](Only Serial Interfaces) - ETX(Only Serial Interfaces)
sendBuffer must start with the DATA part of the response package, the rest will be inserted by function & driver

NOTE 2) FP3_xxx commands may be encapsulated as batch commands with functions prepare_xxx. In that case, each prepare_xxx function has its own return code.
This function aims to parse each prepare_xxx function and keep the return code of it in ST_MULTIPLE_RETURN_CODE array.

4.3.2 Prepare_Start

defined(TELIUM2) && !defined(TELIUM3)

Parameters

(uint8 * Buffer,
int MaxSize,
uint8 * uniqueId,
int lengthOfUniqueId,
uint8 * uniqueIdSign,
int lengthOfUniqueIdSign,
uint8 * userData,
int lengthOfUserData)

4.3.3 Prepare_TicketHeader

prepare_TicketHeader (SUBSET of FP3_TicketHeader)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] ticketType one of TTicketType

4.3.4 Prepare_ItemSale

prepare_ItemSale (SUBSET of FP3_ItemSale)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] pstItem pointer to ST_ITEM structure which keeps item details to print on the ticket

4.3.5 Prepare_Payment

prepare_Payment (SUBSET of FP3_Payment)

Parameters

in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] pstPaymentRequest pointer to ST_PAYMENT_REQUEST structure which keeps parameters for a payment operation

4.3.6 Prepare_PrintTotalsAndPayments

prepare_PrintTotalsAndPayments (SUBSET of FP3_PrintTotalsAndPayments)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

4.3.7 Prepare_PrintBeforeMF

prepare_PrintBeforeMF (SUBSET of FP3_PrintBeforeMF)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

Subset Functions do not send the message or do not wait for the reply
Please SEE the explanations of parent function for more details

4.3.8 Prepare_PrintMF

prepare_PrintMF (SUBSET of FP3_PrintMF)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

4.3.9 Prepare_Close

prepare_PrintMF (SUBSET of FP3_Close)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

4.3.10 Prepare_PrintUserMessage

prepare_PrintMF (SUBSET of FP3_PrintUserMessage)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] stUserMessage array of ST_USER_MESSAGE items which keeps parameters for user messages

[in] numberOfMessage number of user messages .

4.3.11 Prepare_GetPLU

prepare_PrintMF (SUBSET of FP3_GetPLU)

Parameters

in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] barcode A NULL terminated string that includes FULL barcode to search in ECR PLU DB

4.3.12 Prepare_ReversePayment

prepare_PrintMF (SUBSET of FP3_ReversePayment)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] pstPaymentRequest structure to ST_PAYMENT_REQUEST items which keeps parameters for a payment operation

[in] numberOfPaymentRequests number of reverse payment requests

4.3.13 Prepare_SetInvoice

prepare_PrintMF (SUBSET of FP3_SetInvoice)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] pstInvoiceInfo pointer to ST_INVIOCE_INFO structure which keeps parameters for an invoice

4.3.14 Prepare_SetParkingTicket

prepare_PrintMF (SUBSET of FP3_SetParkingTicket)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] carIdentification pointer to buffer which keeps parameters for an car parking ticket

4.3.15 Prepare_KasaAvans

prepare_PrintMF (SUBSET of FP3_KasaAvans)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] amount Amount to set as AVANS

[in] pCustomerName Customer name value for receipt info, it fills in the buffer if not empty

[in] pTckn Customer TCKN value for receipt info, it fills in the buffer if not empty

[in] pVkn Customer VKN value for receipt info, it fills in the buffer while pTckn is NULL and pVkn is not empty

4.3.16 Prepare_KasaPayment

prepare_PrintMF (SUBSET of FP3_KasaPayment)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] amount Amount to set as PAYMENT

4.3.17 Prepare_Pretotal

prepare_PrintMF (SUBSET of FP3_Pretotal)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

4.3.18 Prepare_DisplayPaymentSummary

prepare_PrintMF (SUBSET of FP3_DisplayPaymentSummary)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

4.3.19 Prepare_Plus

prepare_PrintMF (SUBSET of FP3_Plus)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

4.3.20 Prepare_Minus

prepare_PrintMF (SUBSET of FP3_Minus)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] Amount Amount to be decreased (it's in the 2 decimal format like (1 TL is represented as 100)

[in] indexOfItem it's the index of the item to to work on. if this index is bigger than item count on the ticket(like 0xFFFF), all ticket amount is effected

4.3.21 Prepare_Inc

prepare_PrintMF (SUBSET of FP3_Inc)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] Rate rate to be increased (it's in the 0 decimal format like (10% is represented as 10)

[in] indexOfItem it's the index of the item to to work on. if this index is bigger than item count on the ticket(like 0xFFFF), all ticket amount is effected

4.3.22 Prepare_Dec

prepare_PrintMF (SUBSET of FP3_Dec)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] Rate rate to be decreased (it's in the 0 decimal format like (10% is represented as 10)

[in] indexOfItem it's the index of the item to to work on. if this index is bigger than item count on the ticket(like 0xFFFF), all ticket amount is effected

4.3.23 Prepare_VoidPayment

prepare_PrintMF (SUBSET of FP3_VoidPayment)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] index it's the index of the payment to void.

4.3.24 Prepare_VoidItem

prepare_PrintMF (SUBSET of FP3_VoidItem)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] index Zero based index of the item to void

[in] ItemCount Count (number) of the items to void. it can be used if the user wants to void a part of the item which is pointed by index. if this value is Zero, then all item count (indexed by index) will be voided.

[in] ItemCountPrecision Precision value of the ItemCount. Exp; It is 2, to represent 2,58 kg for ItemCount = 258

4.3.25 Prepare_SendFrontStationPrint

prepare_PrintMF (SUBSET of FP3_SendFrontStationPrint)

Parameters

[in,out] Buffer buffer to prepare and copy the request message (function does not check if the buffer is big enough or not!)

[in] MaxSize size of buffer

[in] pSendBuffer pointer of data

[in] SendBufferLen size of SendBuffer

[in] MaxLen Receive buffer max len

[in] PrintTimeOut Print Command TimeOut

4.3.26 Prepare_GetTicket

Function sends a command to ECR to get the details of an open ticket

Function then waits the result of the command for a given timeout value

Parameters

[in] hInt Interface Handle

[in] hTrx Transaction Handle

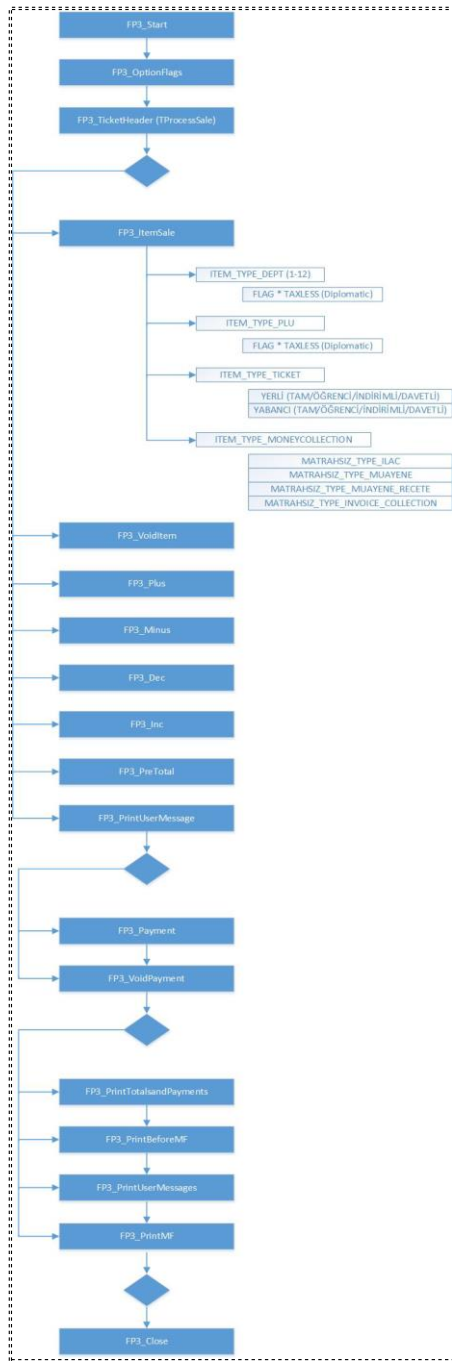
[in,out] pstTicket Pointer to ST_TICKET structure that will keep the result of the operation

[in] timeoutInMilliseconds Timeout value in ms that the driver will wait a reply

5 Transaction Flows

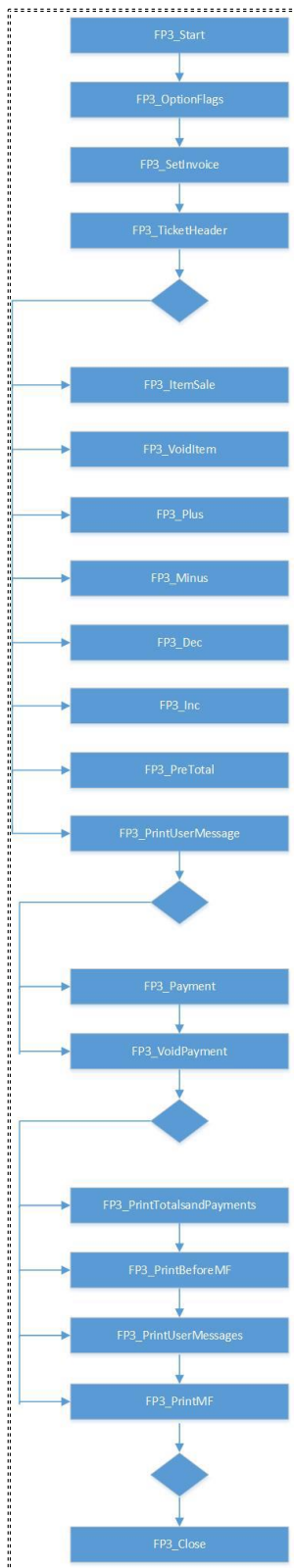
5.1 Sale Ticket Flow

Fiscal Sale Transaction can be done by using the following flows



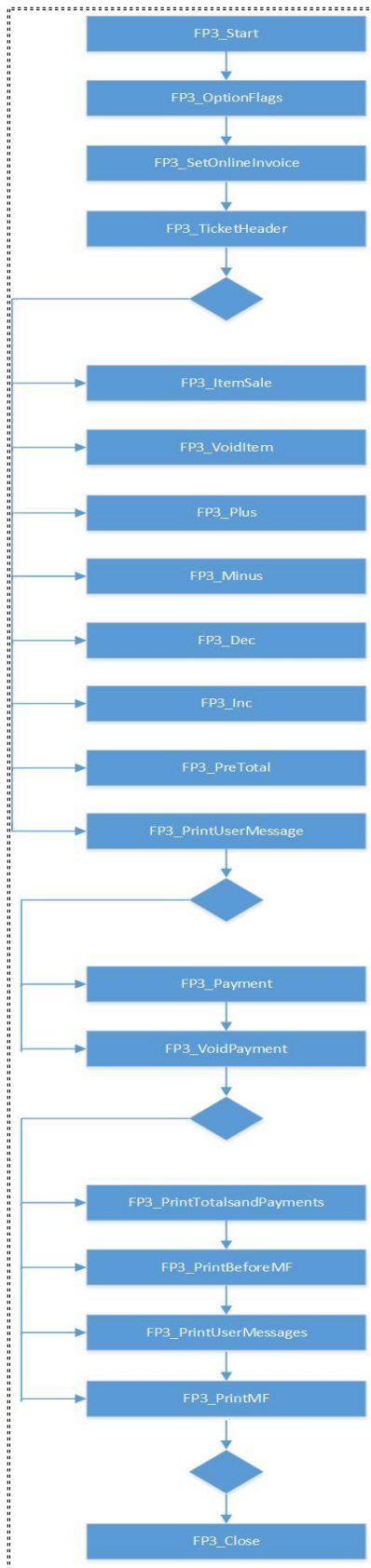
5.2 Invoice Sale Ticket

Invoice Sale Transaction ("Faturali Satis Bilgi Fisi") can be done by using the following flows



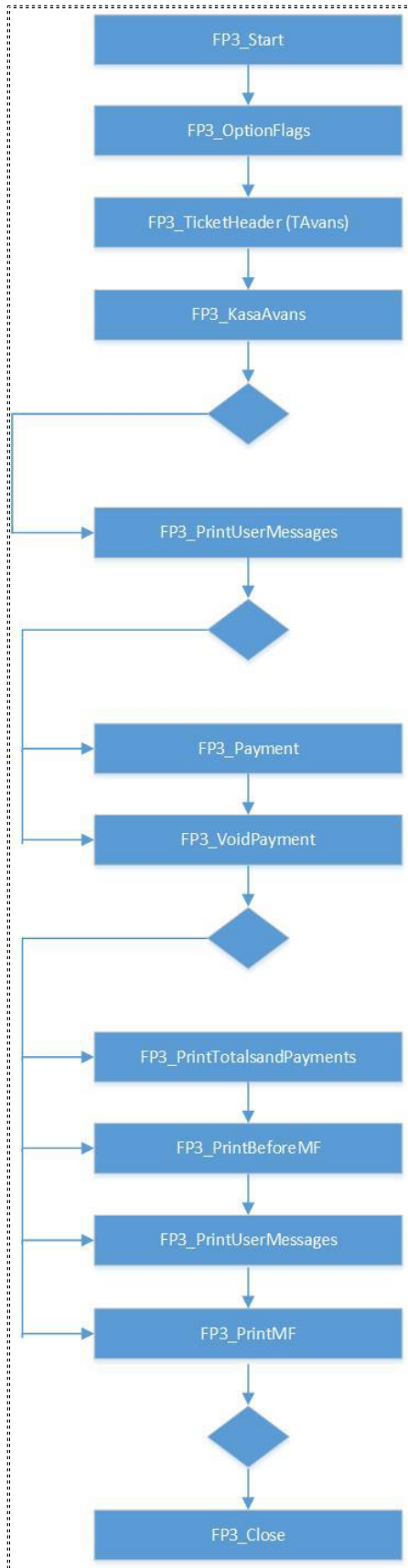
5.3 Online Invoice Sale Ticket

Online Invoice Sale Transaction ("Online Faturali Satis Bilgi Fisi") can be done by using the following flows



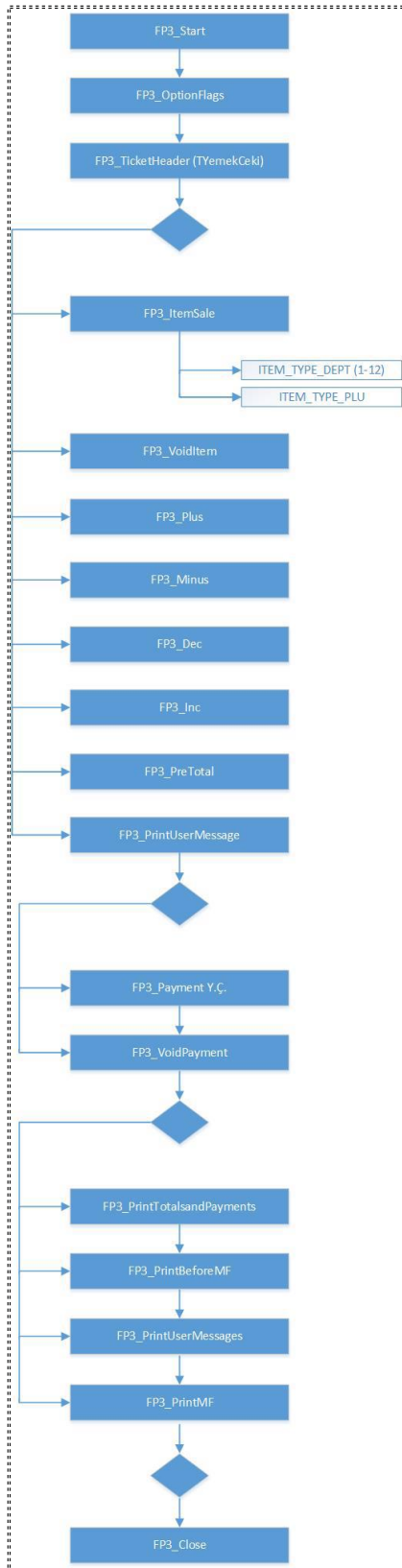
5.4 Avans Ticket

Avans Transaction ("Avans (On Odeme) Bilgi Fisi") can be done by using the following flows



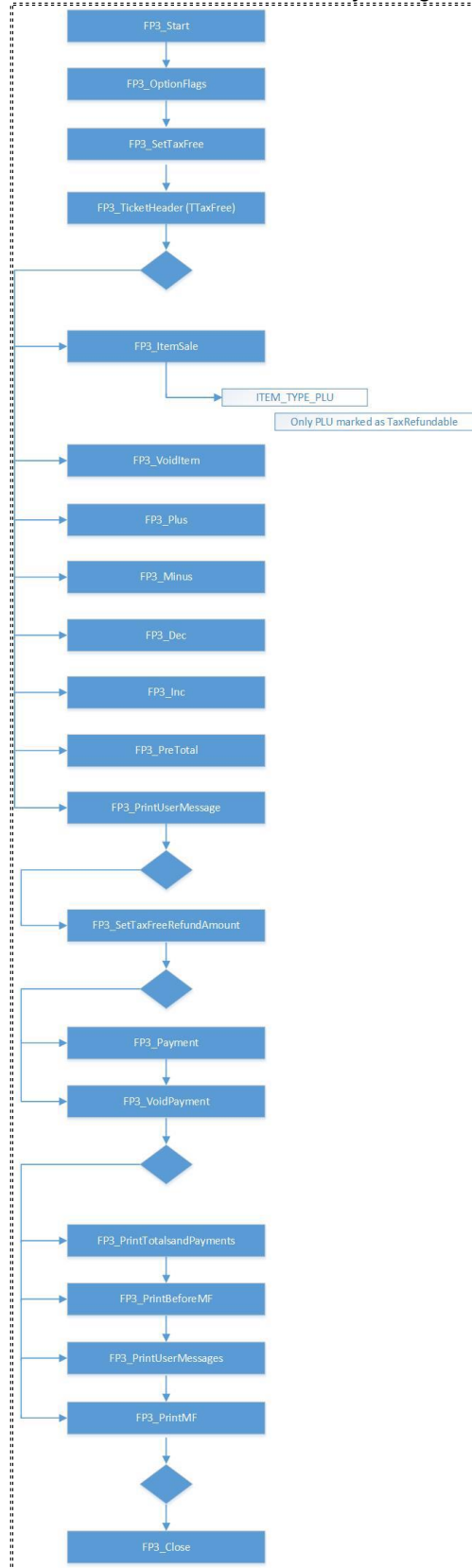
5.5 Yemek Ceki Ticket

YemekCeki Ticket ("Yemek Ceki Bilgi Fisi") can be done by using the following flows



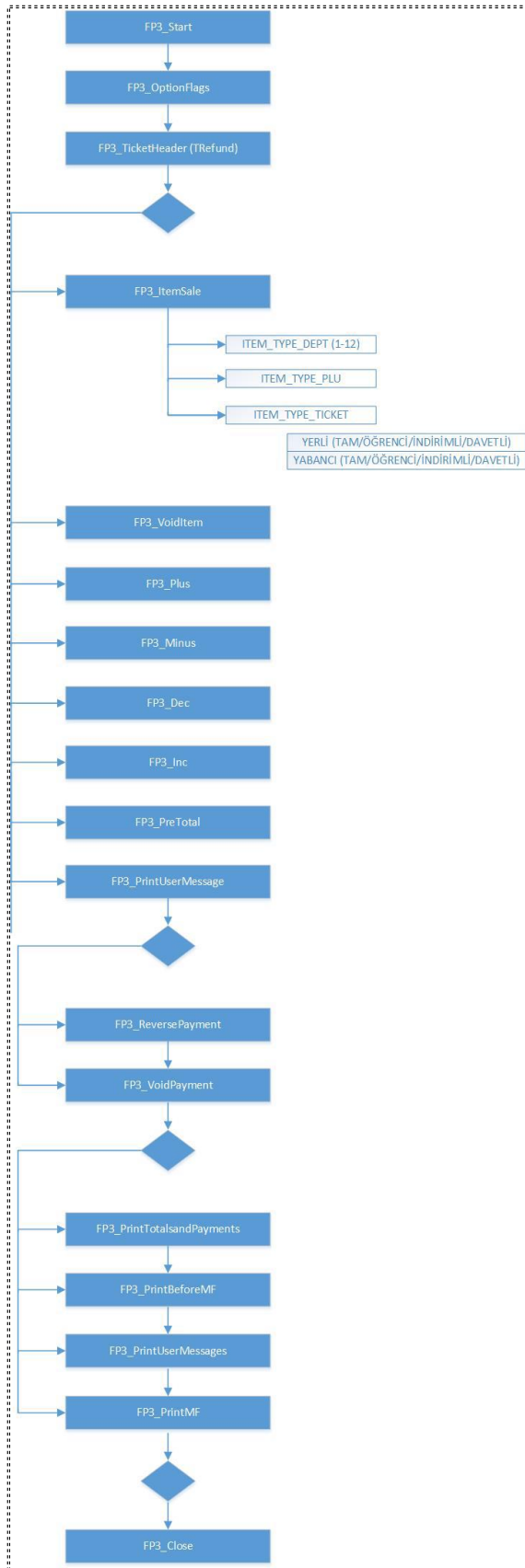
5.6 Tax Free Ticket

Tax Free Ticket can be done by using the following flows



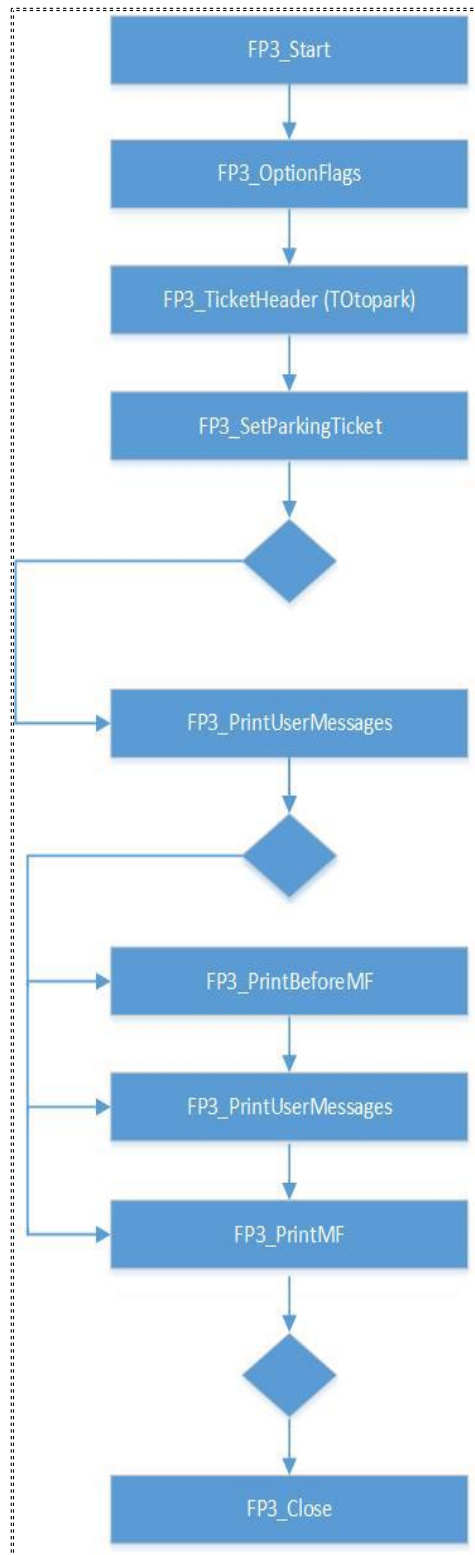
5.7 Refund Ticket

Refund Ticket ("İade Fisi") can be done by using the following flows



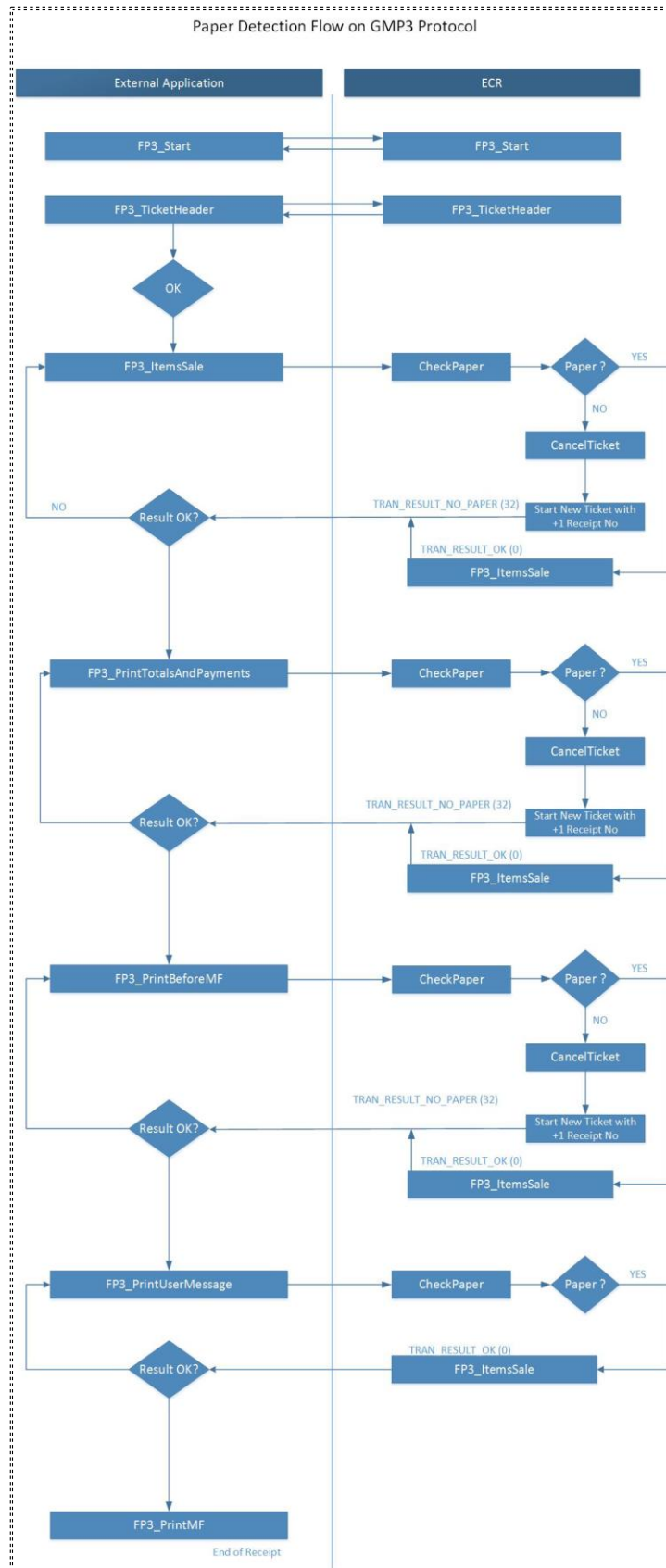
5.8 Parking Entry Ticket

Parking Entry Ticket ("Otopark Giris Bilgi Fisi") can be done by using the following flows



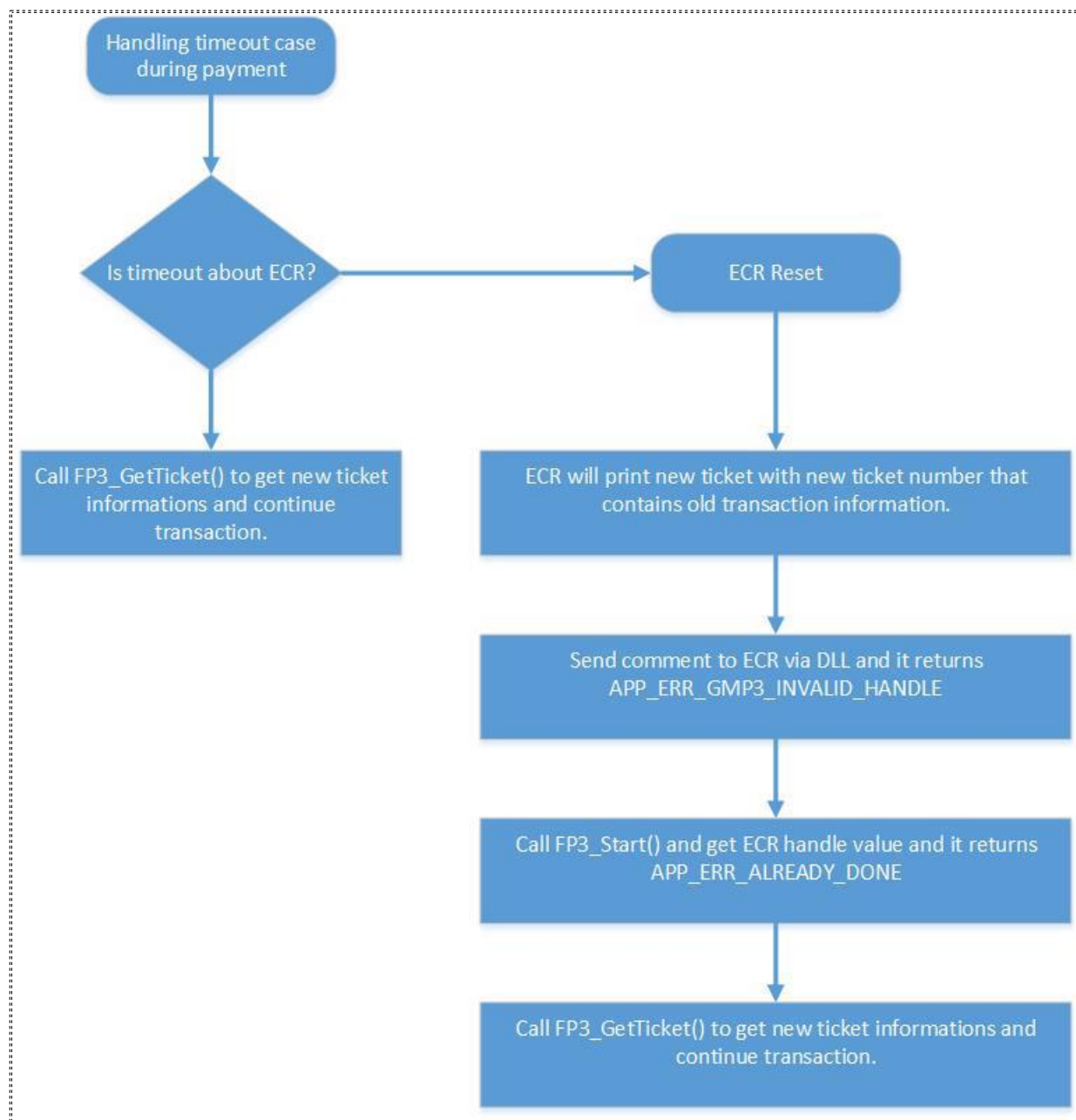
5.9 Paper Detection

Paper Detection can be handled by using the following flows.



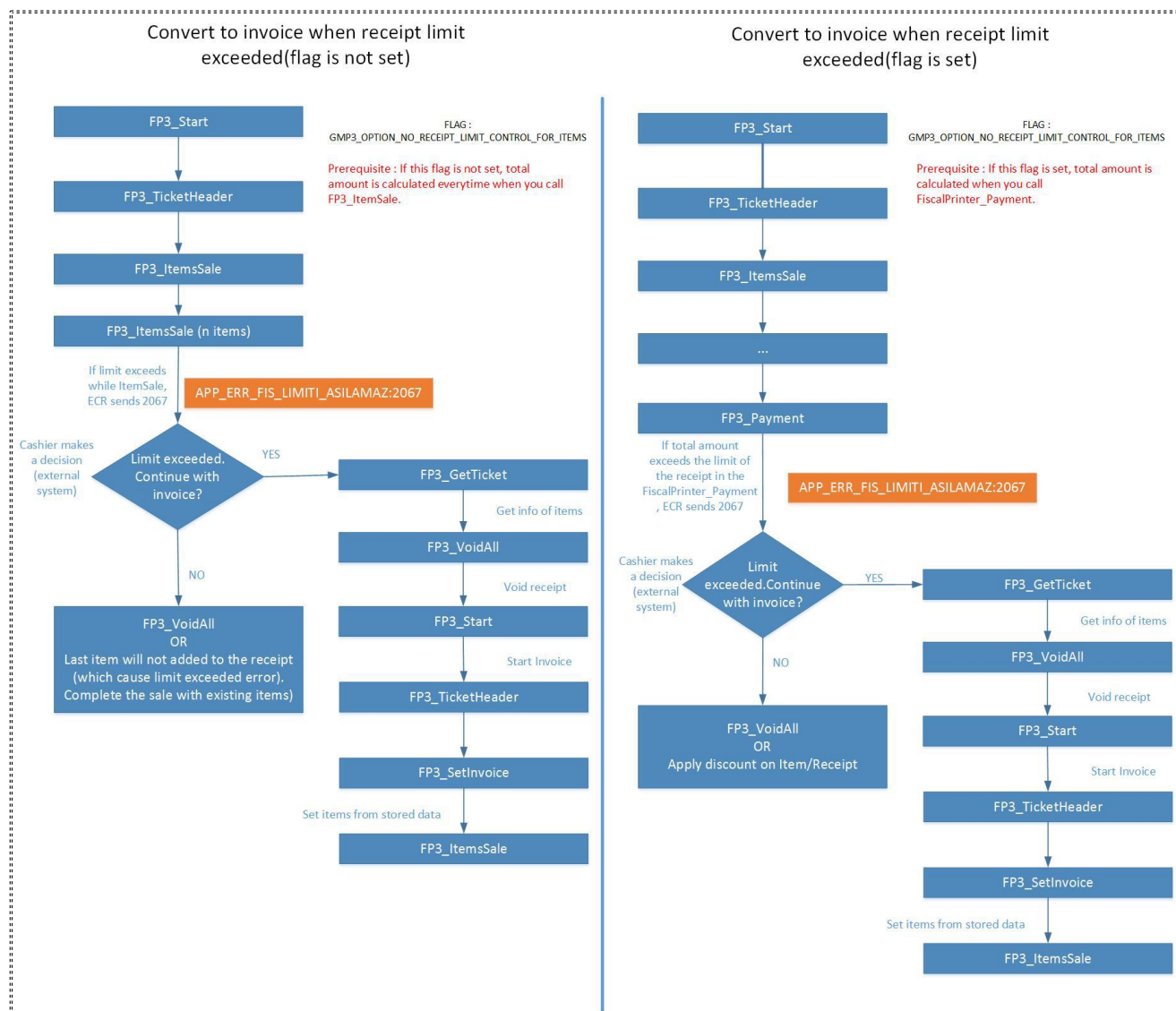
5.10 OKC Reset

OKC Reset can be handled by using the following flows.



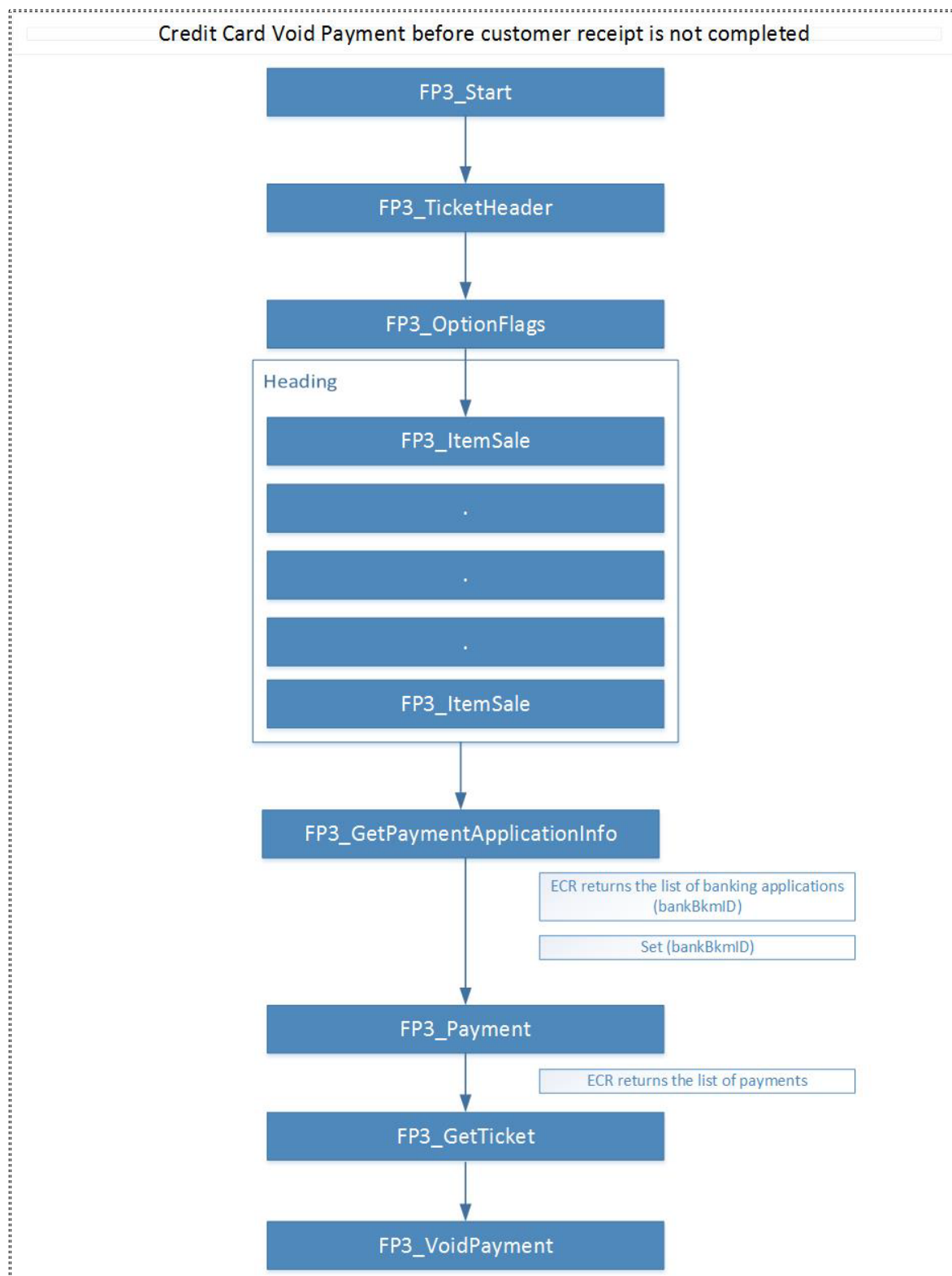
5.11 Convert To Invoice

You may chance the receipt to invoice in case of "receipt limit exceeded".

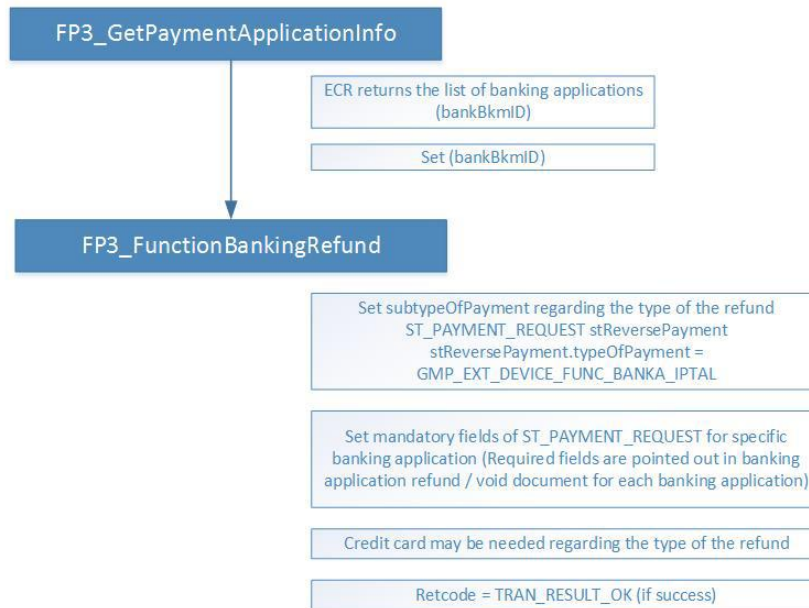


5.12 Cancel Payment

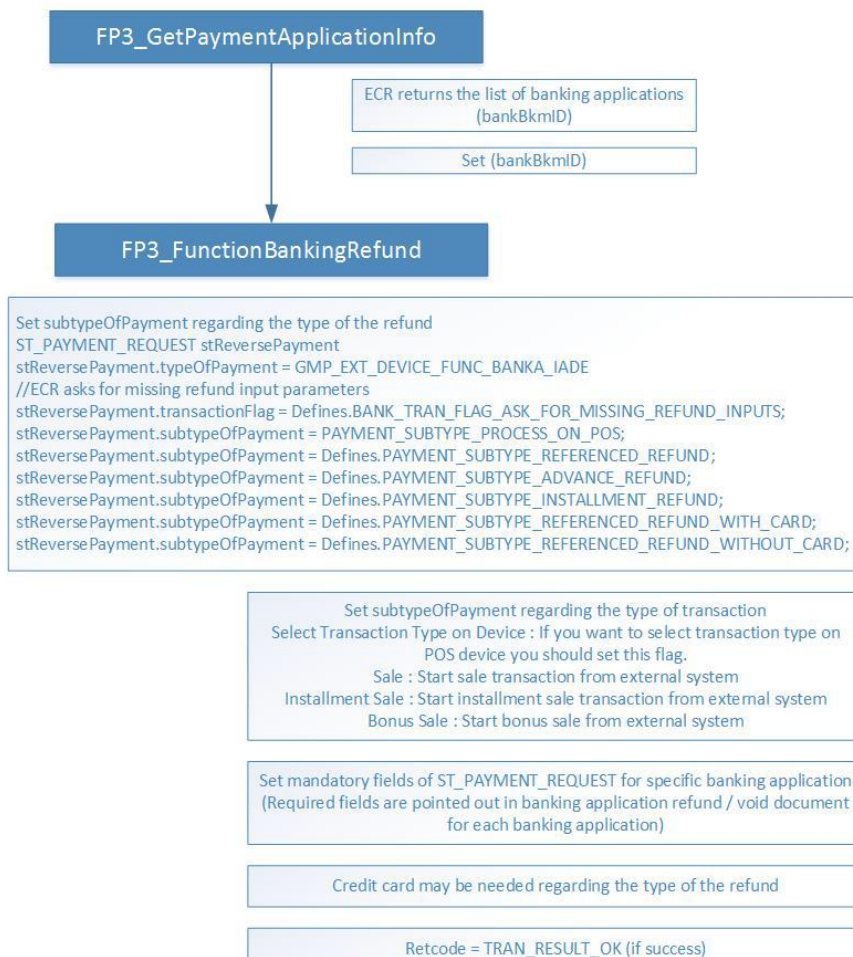
You may cancel or refund customer ticket with given flows.



Credit Card Void Payment after customer receipt is completed and before banking end of day reports



Credit Card Refund Payment after customer receipt is completed and banking end of day reports is completed



5.13 Bank Payment/Refund Type Selection

Sub Payment Type for Bank Transactions

MSG_ID	LENGTH	VALUE
• PAYMENT_SUBTYPE_PROCESS_ON_POS • PAYMENT_SUBTYPE_SALE • PAYMENT_SUBTYPE_INSTALLMENT_SALE • PAYMENT_SUBTYPE_LOYALTY_PUAN • PAYMENT_SUBTYPE_ADVANCE_REFUND • PAYMENT_SUBTYPE_INSTALLMENT_REFUND • PAYMENT_SUBTYPE_REFERENCED_REFUND • PAYMENT_SUBTYPE_REFERENCED_REFUND_WITH_CARD • PAYMENT_SUBTYPE_REFERENCED_REFUND_WITHOUT_CARD	Length Of the VALUE part	• 0x00000000 • 0x00000001 • 0x00000002 • 0x00000003 • 0x00000004 • 0x00000005 • 0x00000006 • 0x00000007 • 0x00000008

External System Transaction Flags of ST_PAYMENT_REQUEST

MSG_ID	LENGTH	VALUE
• BANK_TRAN_FLAG_DO_NOT_ASK_FOR_MISSING_LOYALTY_POINT • BANK_TRAN_FLAG_ALL_INPUT_FROM_EXTERNAL_SYSTEM • BANK_TRAN_FLAG_ASK_FOR_MISSING_REFUND_INPUTS • BANK_TRAN_FLAG_LOYALTY_POINT_NOT_SUPPORTED_FOR_TRANS • Reserved • Reserved • Reserved • BANK_TRAN_FLAG_SALE_WITHOUT_CAMPAIGN	Length Of the VALUE part	• 0x04000000 • 0x08000000 • 0x10000000 • 0x20000000 • 0x40000000 • 0x80000000 • 0x02000000 • 0x01000000

5.13.1 Credit Card Payment

In package 10, payment by using credit card which initilized from external system has been revised.

5.13.1.1 New Payment Types (Credit card)

Existing and new payment types are;

- 1- Select Transaction Type on POS (added)
- 2- Sale (existing)
- 3- Installment Sale (existing)
- 4- Bonus Sale (added)

After selection of banking application on external system, there are couple of options are available in order to identify payment options of the credit card payment model.

Select Transaction Type on POS

If you select select Transaction Type on POS option, payment types are selected from POS device. These payment types may differentiate according the banking applications.

Sale

If you select Sale option, POS device is requested to insert the credit card and no need any other information required regarding payment types.

Installment Sale

If you select Installment Sale option, installment count must be provided to the POS device via external system.

Bonus Sale

If you select Bonus Sale option, bonus amount must be provided to the POS device via external system. For instance, you can use 80 TL of 100 TL (total amount of receipt) by selecting Sale option or Installment Sale option. For the rest of the receipt you can select Bonus Sale option (20 TL in this case), if customer wants to use loyalty points.

Credit Card Refund

In package 10, refund by using credit card which initialized from external system has been revised. Refund process

5.13.1.2 New Refund Types

Existing and new refund types are;

- 1- Referansli iade (Refund with original transaction reference)
- 2- Pesin iade (Cash refund)
- 3- Taksitli iade (Installment card refund)
- 4- Kartsiz Referansli iade - Garanti Bankasi (Refund using reference, credit card no required)
- 5- Kartli Referansli iade - Garanti Bankasi (Refund using reference, credit card required)

Please note that, all banks have their own refund types and different names for common refund types as well. For instance, RRN iade (which is used by Garanti Bank) and E.iade (which is used by YKB-Yapi Kredi Bank) are same. Another important thing is parameters which are requested by banks during refund operation. Not every refund types require all transaction parameters in order the complete refund operation.

Some transaction parameters are taken into account, some are not. Because of this reason, all transaction parameters are requested as an input via external system. And these transaction parameters are used by banking application regarding refund type.

Referansli iade (Refund with original reference number)

In this refund type, transaction parameters are required by banking application that provided via external system.

Pesin iade (Cash refund)

In this refund type, transaction parameters are requested by banking application which provided via external system.

Taksitli iade (Installment card refund)

In this refund type, transaction parameters are required by banking application which provided via external system.

Kartsiz Referansli iade - Garanti Bankasi (Refund using reference, credit card not required)

In this refund type, credit card is not required and transaction parameters are required by banking application which provided via external system (Garanti bankasi).

Kartli Referansli iade - Garanti Bankasi (Refund using reference, credit card required)

In this refund type, credit card is required and transaction parameters are required by banking application which provided via external system (Garanti bankasi).

5.13.1.3 New flags for REFUND (in package 10+)

In this refund type, credit card is required and transaction parameters are required by banking application which provided via external system (Garanti bankasi).

All Input From External System

(BANK_TRAN_FLAG_ALL_INPUT_FROM_EXTERNAL_SYSTEM)

If this flag is not set, ECRPOS asks required data which are not provided from external system. If this flag is set and some of the required data are missing, ECRPOS returns an error (ERROR_MISSING_MANDATORY_FIELD).

Do Not Ask For Missing Loyalty Point

(BANK_TRAN_FLAG_DO_NOT_ASK_FOR_MISSING_LOYALTY_POINT)

If this flag is set, transaction continues without asking the loyalty point. If this flag is not set and loyalty point is not provided from external system, the supported banking applications may ask loyalty point from ECRPOS.

Ask For Missing Refund Inputs

(BANK_TRAN_FLAG_ASK_FOR_MISSING_REFUND_INPUTS)

If this flag is set, banking application requests missing refund parameters which is not provided from external system for refund process. These parameters must be provided manually.

Loyalty Point Not Supported For Transaction

(BANK_TRAN_FLAG_LOYALTY_POINT_NOT_SUPPORTED_FOR_TRANS)

If this flag is set, banking application will not request loyalty point as an input.

Sale Without Campaign

(BANK_TRAN_FLAG_SALE_WITHOUT_CAMPAIGN)

If this flag is set, installment will not apply for the sale.

6 Transaction Handle ID

Transaction handle id is generated by ECRPOS when GMP3 transaction started which is called "Transaction Handle Id".

GMP3 transaction starts when FP3_START function is involved.

There are three method of sending messages to the ECRPOS from the external system.

- 1- Synchronous (Line-by-Line) mode
- 2- Asynchronous (Bulk) mode
- 3- Hybrid (Aynchronous-Asynchronous) mode

1- In the synchronous mode, fiscal messages are sent to the ECRPOS one-by-one by external system. When ECRPOS receives FP3_Start message, It generates transaction handle id and returns the generated value to the external system as a response.

In subsequent messaging of the same transaction, Ingenico DLL uses this handle id within fiscal functions.

2- In the asynchronous mode, fiscal functions are sent to the ECRPOS by using FP3_MultipleCommand function as an one message.

FP3_MultipleCommand function may contains one or more than one fiscal functions that starts from FP3_prepare_Start to FP3_prepare_Close (please refer the transaction flow section for more information).

ECRPOS processes each messages within the MultipleCommand message and returns the response message to the external system which contains all return codes of the FP3_prepare_xyz functions as well as transaction handle id.

3- In the Hybrid mode, keeping transaction flow order in mind some fiscal functions can be send using synchronous mode and some can be send using asynchronous mode.

Transaction handle id is generated when receives FP3_Start message or FP3_prepare_Start message and transaction handle id returns to the external system within response message.

List of return codes which are related with Transaction Handle Id

A- APP_ERR_GMP3_INVALID_HANDLE - 2317

B- APP_ERR_GMP3_NO_HANDLE - 2341

6.1 APP_ERR_GMP3_INVALID_HANDLE

If an active GMP3 transaction exists on the ECRPOS and the transaction handle id of the fiscal function which sent from the external

system differs from the transaction handle id of the current transaction on the ECRPOS, the ECRPOS returns APP_ERR_GMP3_INVALID_HANDLE return code to the external system as response.

When external system receives APP_ERR_GMP3_INVALID_HANDLE, developers of external system should send the Start function to the ECRPOS and retrieve the transaction handle id of the existing transaction.

As a result, ECRPOS returns APP_ERR_ALREADY_DONE to the Start function as well as transaction handle id.

6.2 APP_ERR_GMP3_NO_HANDLE

This return code means, there isn't an ongoing GMP3 transaction on the ECRPOS device. When ECRPOS returns this return code developers may start a GMP3 transaction without any issues.

VAT Calculation

The normal sales amount of the products with the same VAT rate on the receipt is added and the following procedure is performed for each VAT rate.

For a total of 10.00TL and 8% VAT, the internal percentage is calculated as **$((10.00/108) * 8)$** .

Formula => $(\text{Amount}/(100 + \text{VAT})) * \text{VAT}$

If the 3rd row digit after the comma is 5 or higher, the 2nd row digit is increased by 1 (rounded up), otherwise it remains the same (rounded down)

Example: $8.444 = 8.44$ | $8.445 = 8.45$ | $8.446 = 8.45$

VAT values on the Z Report

The VAT account in the Z report is calculated over the sum of the VAT amounts on the receipt.

In the calculation made on the receipt, the VAT amount calculated on the total amount in the Z report and the sums resulting from the collection of VAT on the receipt one by one, may differ when it is made for rounding up and down the penny. Since it is the consumer who pays the VAT financially, the ÖKC takes into account the VAT amount on the receipt given to the consumer when calculating.

Using Customizable Currency In Cash Payments

This document is written for helping those who would like to use different currency cash payments through GMP3. Different currency configuration is managed with an object named “ST_EXCHANGE_PROFILE” this object has another object named “ST_EXCHANGE” as collection. In order to manipulate data as you need, “FP3_SetCurrencyProfile” function can be used with that “ST_EXCHANGE_PROFILE” typed object.

Ingenico Fiscal Devices always have one ready to use exchange profile named “TSM TABLOSU”, and its index is always zero. This profile is not customizable and cannot be deleted. Its currency values is updated from Ingenico TSM Server.

In order to setup your own exchange profile depending on your need, following function must be called;
FP3_SetCurrencyProfile(UInt32 hInt, string supervisorPassword, byte profileIndex, byte ProfileProcessType, ST_EXCHANGE_PROFILE[] pStExchangeProfile)

profileIndex: can be set to 1, since 0 is already filled with “TSM TABLOSU”

ProfileProcessType: 1: update an existing profile, 2: add brand new profile, 3: delete an existing profile.

pStExchangeProfile: represents “ST_EXCHANGE_PROFILE” typed collection of object. Looks like following, assuming you need only dollar and sterlin, simplified for demonstration purposes;

```
{
  "ProfileName": "IngenicoCafe",
  "NumberOfCurrency": 2,
  "ExchangeRecords": [
    {
      "code": 997,
      "prefix": "USD",
      "sign": "$",
      "locationOfSign": 0,
      "tousandSeperator": 46,
      "centSeperator": 44,
      "numberOfCentDigit": 4,
      "rate": 67777
    },
    {
      "code": 826,
      "prefix": "GBP",
      "sign": "£",
      "locationOfSign": 0,
      "tousandSeperator": 46,
      "centSeperator": 44,
      "numberOfCentDigit": 4,
      "rate": 83555
    }
  ]
  //other currencies will be here
}
```

Dynamic Tax Table Integration (TR01)

GIn GMP3 integrations, if the VAT rates change on the device or in any other way, there is a possibility of transacting with incorrect VAT information. In order to avoid such situations, there are some points to be considered in GMP3 integrations. The item coded as TR01 in the approval tests tests this integration.

Situations where VAT control is required

- Immediately after each match,
- every time the ZNo value changes,

If VAT rates are requested to be changed on the device, the ECR will force the Z report, otherwise the VAT rate cannot be changed. For this reason, the correctness of the VAT information should be checked, as there is a risk of VAT change when the ZNo increases..

Using Food Voucher in Fiscal Receipt

- 1) Retail receipt is initialized. (Start, OptionFlags, TicketHeader(TProcessSale))
- 2) Product(s) are entered.
- 3) Payment is made as follows;
The following function is called for food voucher applications.

```
Json_GMPSmartDLL.FP3_GetVasApplicationInfo(CurrentInterface, ref NumberOfTotalRecord, ref  
NumberOfTotalRecordReceived, ref StPaymentApplicationInfo,  
(ushort)EVasType.TLV_OKC_ASSIST_VAS_TYPE_YEMEKCEKI);
```

StPaymentApplicationInfo object is of type ST_PAYMENT_APPLICATION_INFO and contains details of payment applications. The model is as follows.

```
public class ST_PAYMENT_APPLICATION_INFO  
{  
    public byte[] name;  
    public byte index;  
    public UInt16 u16BKmId;  
    public UInt16 Status;  
    public UInt16 Priority;  
    public UInt16 u16AppId;  
    public UInt16 AppType;  
    public UInt16 AppFlag;  
  
    public ST_PAYMENT_APPLICATION_INFO()  
    {  
        name = new byte[20];  
    }  
};
```

The Payment object is created and populated with the relevant data.

```
ST_PAYMENT_REQUEST[] stPaymentRequest = new ST_PAYMENT_REQUEST[1];  
stPaymentRequest[0] = new ST_PAYMENT_REQUEST();
```

```
stPaymentRequest[0].typeOfPayment = (uint)EPaymentTypes.PAYMENT_YEMEKCEKI;
```

```
    stPaymentRequest[0].subtypeOfPayment = 0;  
    stPaymentRequest[0].payAmount = 1000;  
    stPaymentRequest[0].payAmountCurrencyCode = (ushort)ECurrency.CURRENCY_TL;  
    stPaymentRequest[0].bankBKmId = StPaymentApplicationInfo.ElementAt(0).u16AppId; // Sodexo'nun ilk eleman  
    olarak döndüğü varsayılır.  
    stPaymentRequest[0].numberOfInstallments = 0;  
  
    GetPayment(stPaymentRequest, 1);
```

- 4) How to calculate how much can be paid with food checks.

Between the selected departments, payment can be made with meal vouchers equal to the sum of the products whose stDepartments[0].bLuchVoucher parameter is 1. In case of trying to pay more with meal vouchers, an error will be taken.

- 5) How to make Department settings?

bLuchVoucher parameter on the department information must be set to 1 in order to determine the departments where payment can be made with meal checks. The following code can be shaped and used as needed;

```
stDepartments[0].bLuchVoucher = 1;  
retcode = Json_GMPSmartDLL.FP3_SetDepartments(CurrentInterface, ref stDepartments,  
numberOfDepartmentRequested, pf.m_PASS);
```

BKM ID			TeliumAppId
9069	Edenred	ER	22970
9998	Global Y.Ç	LV	52619
9070	Metropol	MR	52634
9068	Multinet	MN	52613
9058	Sodexo	SD	22973
9081	PAYE	PA	52654
9080	Setcard	ST	52649

Vat Control Process

1. While calling the FP3_ItemSale function, there is a parameter named "deptIndex" in the ST_ITEM object sent to Ingenico ECR. This value tells the ECR which department on the device should be used.
2. By calling the FP3_GetDepartments function, the details of the department of the value in deptIndex can be retrieved. The "u8TaxIndex" parameter in these details shows which indexed VAT information the relevant department will use in the tax table.
3. When the FP3_GetTaxRates function is called, the tax table, that is, the VAT information on the ECR is returned. The "taxRate" parameter here displays the VAT rate.

Let's consider an example scenario in the light of the basic information above. In order to reach the "deptIndex" information that we will use by doing the checks, we first find the VAT we are looking for by making the request in step 3, then we call the departments in the 2nd step and find the department with that VAT index, then we check this value in the FP3_ItemSale requests in step 1. In every situation that requires control under heading 1.1, we check the accuracy of the values and make corrections if there are errors.

During the test, when VAT values are manipulated over ECR, the application should be able to handle this situation without any problems.

7 GMP3 – Error Handling Guide

This document is written for helping developers who work on GMP3 integration. If you're not familiar with GMP3 integration at all. I recommend you to check flows and diagrams in "GDP 14.xx Release\Doc\GMPSmartDLL.chm" file and come back later to this document once you understand general concept. Code samples in this document is written in a .NET Environment. Same concept can be applied in all major programming languages accordingly.

7.1 Error Codes and Possible Actions

What you'll get as retCode(result of DLL functions call) can be found in "GDP 14.xx Release\Inc\GMPSmartDLL.h" file. Most of the errors can be found in Define class in simulator source code, if you need to cover more, please copy and paste it from GMPSmartDLL.h file.

7.1.1 TRAN_RESULT_NO_PAPER = 0x0020

Please check "Transaction Flows and Diagrams -> Paper Detection" flows in "GDP 14.xx Release\Doc\GMPSmartDLL.chm" file.

7.1.2 APP_ERR_FISCAL_INVALID_ENTRY = 2009

This error indicates that you filled some parameters incorrectly. This error would not appear, if, in the integration, you prevent cashiers do anything wrong. For example; if you call FP3_VoidPayment method and do not provide one of the available indexes, this error will appear.

7.1.3 APP_ERR_APL_COMPLETE_PAYMENT = 2021

This error warns you that you're not done with payment part and still trying to finish related ticket unexpectedly. In a robust integration, this error should never appear. But If you want to design your application in a defensive way, you can call "FP3_GetTicket" function and take action accordingly.

7.1.4 APP_ERR_APL_CREDIT_CANNOT_BIGGER_THAN_REMAIN_AMOUNT = 2022

This error indicates that external application tries to pay more than remain amount in credit card or voucher card. Paying cash back is not allowed in card payments. It's allowed only in cash payments. This error mostly appears on communication problems in payment process. When an external application call FP3_Payment function, it can be successful on Ingenico device, even though you don't get result properly. In those cases, "FP3_GetTicket" must be called and Payment result must be checked in ST_TICKET typed object returned from "FP3_GetTicket" method.

7.1.5 APP_ERR_CASHIER_ENTRY_REQUIRED = 2053

This error indicates that cashier login did not take place on related device. If you get this error, you can either prompt your user to login on Ingenico device or call "FP3_FunctionCashierLogin_WE" function through GMP3. You can also set auto-login on Ingenico device with following steps below;

F -> YAZARKASAPOS AYARLARI -> 0000 -> KASİYER İŞLEMLERİ -> SÜREKLİ LOGİN(AKTİF)

7.1.6 APP_ERR_NOT_ALLOWED = 2064

This error is usually generated because of lack of permission from TSM. There might be other possibilities but it's hard to explain them all here. If you get this error on test environment, you can contact Ingenico Turkey Support through Redmine.

method. It means that this item cannot be added because of exceeding limit. You have two options here, convert that fiscal ticket into an invoice or end that ticket and start new one for remaining items. For more information, please checkout "Transaction Flows and Diagrams -> Convert to Invoice" in "GMPSmartDLL.chm" document.

7.1.7 APP_ERR_PAYMENT_FOUND = 2069

This error means that there is a successful payment in that ticket, which stops you calling the function you tried to call. For example, in "Sale Ticket" flow, you've got a successful payment but you're trying use FP3_VoidAll method, in order to cancel that ticket. You'll get this error because before calling FP3_VoidAll method, you have make sure that there is no successful payment. You can check current situation with "FP3_GetTicket" and Void successful payments with FP3_VoidPayment then you can call FP3_VoidAll method.

7.1.8 APP_ERR_ALREADY_DONE = 2080

This error is returned when you call FP3_Start function and there is already a ticket open on related Ingenico Device. You can use this error code to check there is a ticket on the device everytime your application starts. If TRAN_RESULT_OK returns instead of this error then FP3_Close function should be called to remove "Total: 0.00 TL" info on the device's screen.

7.1.9 APP_ERR_PAYMENT_NOT_SUCCESSFUL_AND_NO_MORE_ERROR_CODE = 2085

This is returned from FP3_Payment method and indicates that payment was not a success and there's no specific error message from Banking application.

7.1.10 APP_ERR_PAYMENT_NOT_SUCCESSFUL_AND_MORE_ERROR_CODE = 2086

This is returned from FP3_Payment method and indicates that payment was not a success and there's a specific error message from Banking application. For error details, check out "ST_PaymentErrorMessage" class in GMP3 Simulator.

7.1.11 APP_ERR_VAS_NOT_AVAILABLE = 2088

This error indicates that voucher card parameter (BankBkmlId) on FP3_Payment function call is not available on that Ingenico Device or that application is not ready to use, e.g intallation needed.

7.1.12 APP_ERR_NOT_ALLOWED_AS_GMP3_TRANSACTION_IS_PENDING = 2097

If there is a ticket active on Ingenico Device and a non-fiscal function(e.g Z Report) is called, this error will be returned.

7.1.13 APP_ERR_YEMEKCEKI_PAYMENT = 2119

A little bit background about Loucher payment. Loucher payments must've been start as loucher payments and any other payment type was not accepted but depending on customer needs, it changed by the government. Now you can start a regular sale ticket and purchase entire or partial amount from loucher card and the rest can be purchased other types. But you cannot purchase for items, which is not set as "Loucher Available" from it's department(deptIndex), related parameter can be found under "ST_DEPARTMENT" class "bLuchVoucher" property.

Summarize, if total amount to purchase with loucher cards is more than loucher available items' total amount then you'll get this error.

7.1.14 APP_ERR_GMP3_INVALID_DATE_TIME = 2310

This indicates that the date on Ingenico Device is not compatible with external system for some reason. Date time info is normally updated automatically from NPS Server, but it could somehow happen. There is nothing you can about it in terms of software development. Prompting to cashier to fix this might be a good idea. Steps: "F -> YAZARKASAPOS AYARLARI -> 0000 -> TARIH/SAAT"

7.1.15 APP_ERR_GMP3_INVALID_HANDLE = 2317

Functions that related to a ticket accept “handle” as one of their argument. This value represents that particular ticket currently on the device, and this error is returned when this value is wrong. If you get this error, you can basically call “FP3_GetTicket” function and act accordingly.

7.1.16 APP_ERR_GMP3_APP_CHECKSUM_MISMATCH = 2338

This error is returned when you try pairing process. This means there is something wrong about your hash definition on TSM Server or in production, either your application or our DLL has been changed.

7.1.17 APP_ERR_GMP3_NO_HANDLE = 2341

This error shows there is not active transaction on Ingenico Device and your application tries to call some function as if otherwise.

7.1.18 APP_ERR_GMP3_PAYMENT_CAN_NOT_VOID = 2357

In a ticket flow, there’s a certain point that you cannot void that ticket, you’ll just have to finish that ticket. After “FP3_PrintBeforeMF” function called successfully, this ticket is written in fiscal memory and there’s no way to undo. It can happen anytime after you call “FP3_PrintBeforeMF” method, even if you don’t get result properly or get some error like “timeout” or “data receive error”, it still could be successful on Ingenico Device. When you get this error, you can call “FP3_GetTicket” method and check whether that function successful or not with following if statement.

```
if ((pstTicket.TransactionFlags &
(uint)ETransactionFlags.FLG_XTRANS_TICKET_FOOTER_BEFORE_MF_PRINTED) != 0)
```

7.1.19 APP_ERR_GMP3_Z_REQUIRED = 2417

This error basically means that in order to proceed with sales, you must have Z report. You can either trigger it from GMP3 integration or prompt cashier to have it on Ingenico Device.

7.1.20 APP_ERR_GMP3_APPLICATION_NOT_FOUND = 2440

There is a parameter named “BankBkmld” under “ST_PAYMENT_REQUEST” class. If you set this value with an application which is not installed on that Ingenico device. You’ll get this error.

7.1.21 DLL_RETCODE_PORT_NOT_OPEN = 0xF000

7.1.22 DLL_RETCODE_ACK_NOT_RECEIVED = 0xF01B

These errors indicate that you didn’t configure “GDP 14.xx Release\DLL\Windows\x64\GMP.XML” file properly. In general, there are three different aspects to keep in mind, listed below;

- 1) TSM configuration.(You’ll be asked and will be informed about two communication options, two of RS232, USB and TCP/IP
- 2) GMP.XML configuration.
 - a. IsTcpConnection to be TRUE only for TCP/IP.
 - b. USB, PortName can be found on Device Manager.
 - c. USB, Ingenico Driver can be found on Redmine->Files.
- 3) Physical cable setup must be correct obviously.

7.1.23 DLL_RETCODE_TIMEOUT= 0xF003

If ECR can not response during time which is determined in function of gmp3 then this error will be appeared.

7.1.24 DLL_RETCODE_DATA_RECV_ERR = 0xF007

These errors are self-explanatory but the important part is that, getting this error does not mean that function call was not successful. If you assume that it is, it'll definitely cause serious problems. If these errors and other errors you don't cover are returned, you must check current situation with "FP3_GetTicket" functionality and act accordingly.

7.1.25 DLL_RETCODE_RECV_BUSY = 0xF01C

When Ingenico device is busy for doing some task or cashier just get into a menu of Ingenico Device. There is not much to do in terms of GMP3 development. You may want to prompt cashier, getting back to main screen and try again.

7.1.26 DLL_RETCODE_PAIRING_REQUIRED = 0xF020

This error indicates pairing process is requested. If you get this error, you must trigger pairing with "FP3_StartPairingInit" function call, before proceeding any other function.

7.1.27 APP_ERR_FIS_LIMIT_ASILAMAZ=2067

In Republic of Turkey, there is a certain amount that you can't exceed in a fiscal receipt(TTicketType=TProcessSale). This amount is currently 2000 ₺, but it changes every year. The government decides what it will be. We apply this change on TSM side, so, on integration perspective you don't have worry about that. This error is returned from FP3_ItemSale

7.2 Thoughts On Error Handling

Handling errors on GMP3 integration can be very tricky business. Fortunately, we have test case list which we are going to use on accreditation test for creating your production hash value.

Always show return code and corresponding message on your screen. If this error is something that you can handle in your application, then it could be just small warning appearance. This is not essential but would very helpful on production, because it might take time to get related log files from your servers. Taking a photo of the error is much easier.

We may get lots of different errors on our integration other than shared here. These errors are either rare or self-explanatory. If we are not aware of an error, it would be best calling "FP3_GetTicket" function and checking what was going on, "ST_TICKET" typed object will be filled, and this object tells us everything we need to know. You can analyse log files, in order to understand what may go wrong. Entire data flow can be found on log files, name format is "GMPDLL_YYYY_MM_DD_HH:mm:ss.TXT", the file can be found either in the same location as your application executable located or the path where you mention in GMP.XML file "LogPath" tag.

GMP3 Data Flow looks like below essentially;

Structured Data -> GMP3 Tagged Data -> Raw Data ->

Structured Data <- GMP3 Tagged Data <- Raw Data <-

If you want to see summary of function calls, please download notepad++, or any other editor which supports regular expression search. Following search will give you general idea of what was going on, you can delve into detail from there.

7.3 Other Errors

Errors	Codes	Description
TRAN_RESULT_OK	0x0000	TRANSACTION SUCCESFUL,
TRAN_RESULT_NOT_ALLOWED	0x0001	TRANSACTION NOT ALLOWED,
TRAN_RESULT_TIMEOUT	0x0002	TIMEOUT,
TRAN_RESULT_USER_ABORT	0x0004	USER ABORT,
TRAN_RESULT_EKU_PROBLEM	0x0008	EKU PROBLEM,
TRAN_RESULT_CONTINUE	0x0010	TRANSACTION WILL CONTINUE ON ,
TRAN_RESULT_NO_PAPER	0x0020	NO PAPER,

RECEIPT (FIS)CAL APPLICATION USER TYPE ERRORS	Error Codes	Description
APP_ERR_RECEIPT (FIS)CAL_EXCHANGE_RATES_NOT_FOUND	2000	EXCHANGE RATES NOT FOUND,
APP_ERR_RECEIPT (FIS)CAL_ALREADY_CANCELED_ITEM	2001	ALREADY CANCELED ITEM,
APP_ERR_RECEIPT (FIS)CAL_INVALID_DISCOUNT_RATE	2002	INVALID DISCOUNT RATE,
APP_ERR_RECEIPT (FIS)CAL_DISCOUNT_RATE_NOT_SET	2003	DISCOUNT RATE NOT SET,
APP_ERR_RECEIPT (FIS)CAL_INVALID_INCREASE_RATE	2004	INVALID INCREASE RATE,
APP_ERR_RECEIPT (FIS)CAL_INCREASE_RATE_NOT_SET	2005	INCREASE RATE NOT SET,
APP_ERR_RECEIPT (FIS)CAL_DISCOUNT_ALREADY_DONE	2006	DISCOUNT ALREADY DONE,
APP_ERR_RECEIPT (FIS)CAL_INCREASE_ALREADY_DONE	2007	INCREASE ALREADY DONE,
APP_ERR_RECEIPT (FIS)CAL_NO_PRETOTAL	2008	NOT USED,
APP_ERR_RECEIPT (FIS)CAL_INVALID_ENTRY	2009	INVALID ENTRY,
APP_ERR_RECEIPT (FIS)CAL_KDV_RATE_NOT_FOUND	2010	TAX RATE NOT FOUND,
APP_ERR_RECEIPT (FIS)CAL_TICKET_LIMIT_EXCEED	2011	TICKET LIMIT EXCEED,
APP_ERR_RECEIPT (FIS)CAL_SALE_ITEM_LIMIT_EXCEED	2012	SALE ITEM LIMIT EXCEED,
APP_ERR_RECEIPT (FIS)CAL_CASH_ENTRY_LIMIT_EXCED	2013	CASH ENTRY LIMIT EXCED,
APP_ERR_RECEIPT (FIS)CAL_INVALID_CURRENCY	2014	INVALID CURRENCY,
APP_ERR_NOT_ALLOWED_BEFORE_GIB_CONNECTION	2015	NOT ALLOWED BEFORE GIB CONNECTION,
APP_ERR_KISIM_FIYAT_LIMITI_ASILAMAZ	2016	DEPARTMENT AMOUNT LIMIT EXCEED,
APP_ERR_RECEIPT (FIS)CAL_DEPARTMENT_ENTRY_INCOMPLETE	2017	NOT USED,
APP_ERR_RECEIPT (FIS)CAL_EXCHANGE_RATE_NOT_FOUND	2018	EXCHANGE RATE NOT FOUND,
APP_ERR_APL_CRC_ERROR	2019	APPLICATION CRC ERROR,
APP_ERR_APL_VERS_ERROR	2020	APPLICATION VERSION ERROR,
APP_ERR_APL_COMPLETE_PAYMENT	2021	APPLICATION COMPLETE PAYMENT,
APP_ERR_APL_CREDIT_CANNOT_BIGGER_THAN_REMAIN_AMO UNT	2022	APPLICATION CREDIT CANNOT BIGGER THAN REMAIN AMOUNT,
APP_ERR_APL_CREDIT_PAID_AMOUNT_MISSING	2023	APPLICATION CREDIT PAID AMOUNT MISSING,
APP_ERR_APL_CREDIT_PAID_ONLY_ONE_TIME_ALLOWED	2024	APPLICATION CREDIT PAID ONLY ONE TIME ALLOWED,
APP_ERR_RECEIPT (FIS)CAL_INVALID_DATE_TIME	2025	INVALID DATE TIME,
APP_ERR_MAX_RECEIPT_COUNTER_REACHED	2026	REACHED TO MAXIMUM RECEIPT COUNTER,
APP_ERR_APL_NO_PAPER	2027	NOT USED,
APP_ERR_APL_PARAMETRE_NOT_FOUND	2028	APPLICATION PARAMETER NOT FOUND,
APP_ERR_EKU_SOFTWARE_VERSION	2029	EKU SOFTWARE VERSION,
APP_ERR_RECEIPT (FIS)CAL_SOFTWARE_VERSION	2030	RECEIPT (FIS)CAL SOFTWARE VERSION,
APP_ERR_CIHAI_GECICI_OLARAK_KAPATILDI	2031	DEVICE CLOSED,
APP_ERR_CIHAI_ACIL_MODA_ALINMIS	2032	DEVICE ACIL MODA ALINMIS,
APP_ERR_APL_DAILY_MEMORY_DELETED	2033	APPLICATION DAILY MEMORY DELETED,
APP_ERR_APL_CUMULATIVE_COUNTER_FULL	2034	APPLICATION CUMULATIVE COUNTER FULL,
APP_ERR_APL_OUT_OF_SERVICE	2035	APPLICATION OUT OF SERVICE,
APP_ERR_APL_OUT_OF_SERVICE_WARNING	2036	APPLICATION OUT OF SERVICE WARNING,
APP_ERR_APL_FONT_FILE_LOAD	2037	APPLICATION FONT FILE LOAD,
APP_ERR_APL_FONT_POLICY_NOT_FOUND	2038	APPLICATION FONT POLICY NOT FOUND,
APP_ERR_APL_CREDIT_PAY_AMOUNT_ZERO	2039	APPLICATION CREDIT PAY AMOUNT ZERO,
APP_ERR_APL_DB_ERROR	2040	APPLICATION DB ERROR,
APP_ERR_TICKET_SALE_ITEM_LIMIT_EXCEED	2041	TICKET SALE ITEM LIMIT EXCEED,
APP_ERR_APL_PIN_ERROR	2042	APPLICATION PIN ERROR,
APP_ERR_PAYMENT_APPL_MESSAGE	2043	PAYMENT APPLICATION MESSAGE,
APP_ERR_APL_DB_DUPLICATE_RECORD	2044	APPLICATION DB DUPLICATE RECORD,

APP_ERR_APL_TRANSACTION_CORRUPTED	2045	APPLICATION TRANSACTION CORRUPTED,
APP_ERR_APL_PAYMENT_MUST_COMPLETE	2046	APPLICATION PAYMENT MUST COMPLETE,
APP_ERR_INVOICE_LIMIT_EXCEED	2047	INVOICE LIMIT EXCEED,
APP_ERR_TRANSACTION_NOT_SAVED	2048	TRANSACTION NOT SAVED,
APP_ERR_INVOICE_NO_ENTRY	2049	INVOICE NO ENTRY,
APP_ERR_INACTIVE_PERIPHERAL	2050	INACTIVE PERIPHERAL,
APP_ERR_DEVICE_CLOSED	2051	DEVICE CLOSED,
APP_ERR_TEST_REQUIRED	2052	TEST REQUIRED,
APP_ERR_CASHIER_ENTRY_REQUIRED	2053	CASHIER ENTRY REQUIRED,
APP_ERR_FLIGHT_MODE	2054	FLIGHT MODE,
APP_ERR_NO_PARAM	2055	ECR PARAMETER IS NOT AVAILABLE,
APP_ERR_NOT_RECEIPT (FIS)CAL	2056	ECR MUST BE RECEIPT (FIS)CALIZED,
APP_ERR_RECEIPT (FIS)CAL_RECORD_NOT_AVAILABLE	2057	RECEIPT (FIS)CAL RECORD NOT AVAILABLE,
APP_ERR_CITIZEN_NUMBER_ENTRY	2058	CITIZEN NUMBER ENTRY,
APP_ERR_INVALID_USER_MODE	2059	INVALID USER MODE,
APP_ERR_PAYMENT_NOT_ALLOWED	2060	PAYMENT NOT ALLOWED,
APP_ERR_NO_CASH	2061	CASH IN THE ECR IS NOT AVAILABLE,
APP_ERR_PARAMETER_DW_ERROR_FOR_TEST	2062	PARAMETER DW ERROR FOR TEST,
APP_ERR_MAX_TRY_COUNTER_EXCEED	2063	MAX TRY COUNTER EXCEED,
APP_ERR_NOT_ALLOWED	2064	NOT ALLOWED,
APP_ERR_RESOURCE_PROBLEM	2065	RESOURCE PROBLEM,
APP_ERR_ITEM_PRICE_NOT_EXISTS	2066	ITEM PRICE NOT EXISTS,
APP_ERR_RECEIPT (FIS)_LIMITI_ASILAMAZ	2067	RECEIPT (FIS) LIMITI ASILAMAZ,
APP_ERR_NO_ITEM	2068	NO ITEM,
APP_ERR_PAYMENT_FOUND	2069	PAYMENT FOUND,
APP_ERR_MATRAHSIZ_FOUND	2070	NOT TAXABLE (MATRAHSIZ) FOUND,
APP_ERR_NOT_APPROVED	2071	NOT APPROVED,
APP_ERR_ONLY_COUPON	2072	ONLY COUPON,
APP_ERR_NO_AMOUNT	2073	NO AMOUNT,
APP_ERR_CURRENCY_NOT_SUPPORTED	2074	CURRENCY NOT SUPPORTED,
APP_ERR_TICKET_TOTAL_CANNOT_BE_ZERO	2075	TICKET TOTAL CANNOT BE ZERO,
APP_ERR_INVOICE_NOT_ALLOWED	2076	INVOICE TRANSACTION NOT ALLOWED,
APP_ERR_TICKET_HEADER_NOT_PRINTED	2077	TICKET HEADER NOT PRINTED,
APP_ERR_TICKET_HEADER_ALREADY_PRINTED	2078	TICKET HEADER ALREADY PRINTED,
APP_ERR_PLU_NOT_FOUND	2079	PLU NOT FOUND,
APP_ERR_ALREADY_DONE	2080	ALREADY DONE,
APP_ERR_INVALID_PARAMETER_TAXINDEX	2081	INVALID PARAMETER TAXINDEX,
APP_ERR_INVALID_PARAMETER_TAXRATE	2082	INVALID PARAMETER TAXRATE,
APP_ERR_MISSING_PARAMETER	2083	MISSING PARAMETER,
APP_ERR_MISMATCH_PARAMETER	2084	MISMATCH PARAMETER,
APP_ERR_PAYMENT_NOT_SUCCESSFUL_AND_NO_MORE_ERROR_CODE	2085	PAYMENT NOT SUCCESSFUL AND NO MORE ERROR CODE,
APP_ERR_PAYMENT_NOT_SUCCESSFUL_AND_MORE_ERROR_CODE	2086	PAYMENT NOT SUCCESSFUL AND MORE ERROR CODE,
APP_ERR_INVALID_PAYMENT_TYPE	2087	INVALID PAYMENT TYPE,
APP_ERR_VAS_NOT_AVAILABLE	2088	VAS NOT AVAILABLE,
APP_ERR_INVALID_CAR_IDENTIFICATION	2089	INVALID CARD IDENTIFICATION ,
APP_ERR_ALLOCATION	2090	NOT USED,
APP_ERR_MISSING_TAXFREE_PARAMETER	2091	MISSING TAXFREE PARAMETER,

APP_ERR_INVALID_TAXREFUND_VALUE	2092	INVALID TAXREFUND VALUE,
APP_ERR_TAXFREE_NOT_STARTED	2093	TAXFREE NOT STARTED,
APP_ERR_TAXFREE_NOT_COMPLETED	2094	TAXFREE NOT COMPLETED,
APP_ERR_TAXLESS_NOT_SUPPORTED	2095	TAXLESS NOT SUPPORTED,
APP_ERR_INVALID_TICKET_TYPE	2096	INVALID TICKET TYPE,
APP_ERR_NOT_ALLOWED_AS_GMP3_TRANSACTION_IS_PENDIN G	2097	NOT ALLOWED AS GMP3 TRANSACTION IS PENDING,
APP_ERR_DEVICE_REASSIGNED	2098	DEVICE REASSIGNED,
APP_ERR_DEPT_FOOD_RECEIPT_ITEM_NOT_ENABLE	2099	DEPT_FOOD_RECEIPT_ITEM_NOT_ENABLE,
APP_ERR_CHECK_IP_ADDRES_FORMAT	2100	CHECK IP ADDRES FORMAT,
APP_ERR_WRONG_SERVICE_CARD_PIN	2101	WRONG SERVICE CARD PIN,
APP_ERR_UNDEFINED_COM_INTERFACE	2102	UNDEFINED COM INTERFACE,
APP_ERR_BASE_NOT_FOUND	2103	BASE NOT FOUND,
APP_ERR_ETHERNET_MODUL_CALISMIYOR	2104	ETHERNET MODUL NOT WORKING,
APP_ERR_ETHERNET_NOT_PLUGGED	2105	ETHERNET NOT PLUGGED,
APP_ERR_ETHERNET_BASE_NOT_READY	2106	ETHERNET BASE NOT READY,
APP_ERR_ETHERNET_OUT_OF_BASE	2107	ETHERNET OUT OF BASE,
APP_ERR_GPRS_ERROR_NO_SIM	2108	GPRS ERROR NO SIM,
APP_ERR_GPRS_APN	2109	GPRS APN,
APP_ERR_GPRS_BAD_PIN	2110	GPRS BAD PIN,
APP_ERR_GPRS_SIM_LOCK	2111	GPRS SIM LOCK,
APP_ERR_GPRS_NO_SIGNAL	2112	GPRS NO SIGNAL,
APP_ERR_GPRS_DISCONNECTED	2113	GPRS DISCONNECTED,
APP_ERR_GPRS_UNDEFINED	2114	GPRS UNDEFINED,
APP_ERR_SSL_PROFILE	2115	SSL PROFILE,
APP_ERR_WRONG_SERVICE_CARD	2116	WRONG SERVICE CARD,
APP_ERR_COMMUNICATION_CHANNEL_NOT_DEFINED	2117	COMMUNICATION CHANNEL NOT DEFINED,
APP_ERR_EXCEPTION_CODE_NOT_FOUND	2118	EXCEPTION_CODE_NOT_FOUND
APP_ERR_YEMEKCEKI_PAYMENT	2119	ERR_YEMEKCEKI_PAYMENT
APP_ERR_VAS_VERSION_MISMATCH	2120	VAS_VERSION_MISMATCH
APP_ERR_VAS_SERVICE_NOT_AVAILABLE	2121	VAS_SERVICE_NOT_AVAILABLE
APP_ERR_VAS_CUSTOMER_IDENTIFICATION	2122	CUSTOMER_IDENTIFICATION
APP_ERR_INVALID_EXCEPTION_CODE_TYPE	2123	INVALID_EXCEPTION_CODE_TYPE
APP_ERR_INVALID_EXCEPTION_CODE	2124	INVALID_EXCEPTION_CODE
APP_ERR_VAS_LOYALTY_PROCESS	2125	VAS_LOYALTY_PROCESS
APP_ERR_INVALID_INVOICE_ENTRY	2126	INVALID_INVOICE_ENTRY
APP_ERR_GIB_ROOT_CERT_NOT_FOUND	2200	GIB ROOT CERT NOT FOUND,
APP_ERR_GIB_SUB_ROOT_CERT_NOT_FOUND	2201	GIB SUB ROOT CERT NOT FOUND,
APP_ERR_TSM_ROOT_CERT_NOT_FOUND	2202	TSM ROOT CERT NOT FOUND,
APP_ERR_TSM_SUB_ROOT_CERT_NOT_FOUND	2203	TSM SUB ROOT CERT NOT FOUND,
APP_ERR_OKTEM_CERT_NOT_FOUND	2204	OKTEM CERT NOT FOUND,
APP_ERR_GIB_SIGN_CERT_NOT_FOUND	2205	GIB SIGN CERT NOT FOUND,
APP_ERR_GIB_SIFRE_CERT_NOT_FOUND	2206	GIB SIFRE CERT NOT FOUND,
APP_ERR_TSM_SIGN_CERT_NOT_FOUND	2207	TSM SIGN CERT NOT FOUND,
APP_ERR_PACK_GIB	2208	PACK GIB,
APP_ERR_GIB_CERT	2209	GIB CERT,
APP_ERR_DATABASE_SILINMIS	2210	DATABASE REMOVED,
APP_ERR_OKTEM_CERT_VERIFICATION	2211	
APP_ERR_FILE_COPY	2220	FILE COPY,

APP_ERR_FILE_OPEN	2221	FILE OPEN,
APP_ERR_FILE_READ	2222	FILE READ,
APP_ERR_FILE_WRITE	2223	FILE WRITE,
APP_ERR_BUFFER_MALLOC	2224	BUFFER MALLOC,
APP_ERR_FILE_SEEK	2225	FILE SEEK,
APP_ERR_FILE_EOF	2226	FILE EOF,
APP_ERR_FILE_MOUNT	2227	FILE MOUNT,
APP_ERR_GMP_PROTOCOL	2230	GMP PROTOCOL,
APP_ERR_GMP_VERIFICATION	2231	GMP VERIFICATION,
APP_ERR_GMP_LRC	2232	GMP LRC,
APP_ERR_COMPONENT_ABSENT	2233	COMPONENT ABSENT,
APP_ERR_SQL	2234	SQL,
APP_ERR_GMP_RESPONSE_CODE_ERROR	2235	GMP RESPONSE CODE ERROR,
APP_ERR_FRAM_Z_REPORT_UPDATE	2236	FRAM Z REPORT UPDATE,
APP_ERR_DAILY_WRITE	2237	DAILY WRITE,
APP_ERR_RECEIPT (FIS)CAL_CUMULATIVE_WRITE	2238	RECEIPT (FIS)CAL CUMULATIVE WRITE,
APP_ERR_GMP_PROTOCOL_CANNOT_RECEIVED_FULL_MESSAGE	2239	GMP PROTOCOL CANNOT RECEIVED FULL MESSAGE,
APP_ERR_GMP_PROTOCOL_TPDU_NOT_RECEIVED	2240	GMP PROTOCOL TPDU NOT RECEIVED,
APP_ERR_GMP_AUTO_FICAL_FAILED	2241	GMP AUTO FICAL FAILED,
APP_ERR_GMP_EFTPOS_BAGLANTISI_BULUNAMADI	2242	GMP EFTPOS BAGLANTISI BULUNAMADI,
APP_ERR_GMP_GIB_KEY_IMZA_VERIFICATION	2243	GMP GIB KEY SIGNATURE VERIFICATION,
APP_ERR_GMP_GIB_SHA2_IDENTIFIER	2244	GMP GIB SHA2 IDENTIFIER,
APP_ERR_GMP_MISSING_GROUP	2245	GMP MISSING GROUP,
APP_ERR_GMP_BUFFER_ERROR	2246	GMP BUFFER ERROR,
APP_ERR_GMP_PADDING_ERROR	2247	GMP PADDING ERROR,
APP_ERR_GMP_TREK_TRAK_MISMATCH	2248	GMP TREK TRAK MISMATCH,
APP_ERR_GMP_TSM_KEY_IMZA_VERIFICATION	2249	GMP TSM KEY SIGNATURE VERIFICATION,
APP_ERR_GMP_NEGATIVE_TRANSACTION	2250	GMP NEGATIVE TRANSACTION,
APP_ERR_GMP_UNEXPECTED_RESULT	2251	GMP UNEXPECTED RESULT,
APP_ERR_GMP3_UNDEFINED_OKC_MODEL	2301	GMP3 UNDEFINED OKC MODEL,
APP_ERR_GMP3_UNDEFINED_OKC_VENDOR	2302	GMP3 UNDEFINED OKC VENDOR,
APP_ERR_GMP3_UNDEFINED_OKC_SICILNO	2303	GMP3 UNDEFINED OKC SICILNO,
APP_ERR_GMP3_UNDEFINED_CHAZ_MODEL	2304	GMP3 UNDEFINED DEVICE MODEL,
APP_ERR_GMP3_UNDEFINED_CHAZ_VENDOR	2305	GMP3 UNDEFINED DEVICE VENDOR,
APP_ERR_GMP3_UNDEFINED_CHAZ_SERINO	2306	GMP3 UNDEFINED DEVICE SERINO,
APP_ERR_GMP3_UNDEFINED_MSG_TYPE	2307	GMP3 UNDEFINED MSG TYPE,
APP_ERR_GMP3_UNDEFINED_MISSING_PARAMETER	2308	GMP3 UNDEFINED MISSING PARAMETER,
APP_ERR_GMP3_INVALID_SEQUENCE_NUMBER	2309	GMP3 INVALID SEQUENCE NUMBER,
APP_ERR_GMP3_INVALID_DATE_TIME	2310	GMP3 INVALID DATE TIME,
APP_ERR_GMP3_MISSING_TDES_KEY	2311	GMP3 MISSING TDES KEY,
APP_ERR_GMP3_INVALID_KCV	2312	GMP3 INVALID KCV,
APP_ERR_GMP3_UNDEFINED_STATUS	2313	GMP3 UNDEFINED STATUS,
APP_ERR_GMP3_TIMEOUT	2314	GMP3 TIMEOUT,
APP_ERR_GMP3_CERTIFICATE	2315	GMP3 CERTIFICATE,
APP_ERR_GMP3_VERIFY	2316	GMP3 VERIFY,
APP_ERR_GMP3_INVALID_HANDLE	2317	GMP3 INVALID HANDLE,
APP_ERR_GMP3_CRC	2318	GMP3 CRC,
APP_ERR_GMP3_LEN	2319	GMP3 LEN,

APP_ERR_GMP3_STX	2320	GMP3 STX,
APP_ERR_GMP3_ETX	2321	GMP3 ETX,
APP_ERR_GMP3_NACK	2322	GMP3 NACK,
APP_ERR_GMP3_ACK	2323	GMP3 ACK,
APP_ERR_GMP3_RECEIVE	2324	GMP3 RECEIVE,
APP_ERR_GMP3_SEND	2325	GMP3 SEND,
APP_ERR_GMP3_PARSE	2326	GMP3 PARSE,
APP_ERR_GMP3_USER_BREAK	2327	GMP3 USER BREAK,
APP_ERR_GMP3_PROTOCOL	2328	GMP3 PROTOCOL,
APP_ERR_GMP3_PAIRING_REQUIRED	2329	GMP3 PAIRING REQUIRED,
APP_ERR_GMP3_UNKNOWN_DEVICE	2330	GMP3 UNKNOWN DEVICE,
APP_ERR_GMP3_VERSION_MISMATCH	2331	GMP3 VERSION MISMATCH,
APP_ERR_GMP3_NO_PRIME_NUMBER	2332	GMP3 NO PRIME NUMBER,
APP_ERR_GMP3_PERMISSION	2333	GMP3 PERMISSION,
APP_ERR_GMP3_INCORRECT_DEVICE	2334	GMP3 INCORRECT DEVICE,
APP_ERR_GMP3_MEMORY_READ_ERROR	2335	GMP3 MEMORY READ ERROR,
APP_ERR_GMP3_MEMORY_WRITE_ERROR	2336	GMP3 MEMORY WRITE ERROR,
APP_ERR_GMP3_MEMORY_ERASE_ERROR	2337	GMP3 MEMORY ERASE ERROR,
APP_ERR_GMP3_APP_CHECKSUM_MISMATCH	2338	GMP3 APP CHECKSUM MISMATCH,
APP_ERR_GMP3_APP_DATE_EXPIRED	2339	GMP3 APP DATE EXPIRED,
APP_ERR_GMP3_TCP_RECEIVE_ERROR	2340	GMP3 TCP RECEIVE ERROR,
APP_ERR_GMP3_NO_HANDLE	2341	GMP3 NO HANDLE,
APP_ERR_GMP3_PING	2342	GMP3 PING,
APP_ERR_GMP3_TCP_NO_DATA	2343	GMP3 TCP NO DATA,
APP_ERR_GMP3_EOT	2344	GMP3 EOT,
APP_ERR_GMP3_TAG_NOT_AVAILABLE	2345	GMP3 TAG NOT AVAILABLE,
APP_ERR_GMP3_ENCRYPT_DECRYPT	2346	GMP3 ENCRYPT DECRYPT,
APP_ERR_GMP3_MESSAGE_TYPE	2347	GMP3 MESSAGE TYPE,
APP_ERR_GMP3_DECOMPRESS	2348	GMP3 DECOMPRESS,
APP_ERR_GMP3_INVALID_MERCHANT_UNIQUE_ID	2349	GMP3 INVALID MERCHANT UNIQUE ID,
APP_ERR_GMP3_INVALID_BRANCH_UNIQUE_ID	2350	GMP3 INVALID BRANCH UNIQUE ID,
APP_ERR_GMP3_DATA_GROUP_NOT_FOUND	2351	GMP3 DATA GROUP NOT FOUND,
APP_ERR_GMP3_VERIFY_GROUP_NOT_FOUND	2352	GMP3 VERIFY GROUP NOT FOUND,
APP_ERR_GMP3_VERIFY_GROUP_MISMATCH	2353	GMP3 VERIFY GROUP MISMATCH
APP_ERR_GMP3_RECEIPT_CAN_NOT_VOID	2354	GMP3 RECEIPT CAN NOT VOID
APP_ERR_GMP3_BITMAP_NOT_EXIST	2355	GMP3 BITMAP NOT EXIST
APP_ERR_GMP3_LRC	2356	GMP3 LRC
APP_ERR_GMP3_PAYMENT_CAN_NOT_VOID	2357	GMP3 PAYMENT CANNOT VOID
APP_ERR_GMP3_NO_TICKET_TO_VOID	2358	GMP3 NO TICKET TO VOID
APP_ERR_GMP3_PAYMENT_KDV_NOT_AVAILABLE	2400	GMP3 PAYMENT KDV NOT AVAILABLE,
APP_ERR_GMP3_ITEM_NOT_AVAILABLE	2401	GMP3 ITEM NOT AVAILABLE,
APP_ERR_GMP3_OKC_KISIM_UNKNOWN	2402	GMP3 OKC KISIM UNKNOWN,
APP_ERR_GMP3_DOVIZ_UNKNOWN	2403	GMP3 CURRENCY UNKNOWN,
APP_ERR_GMP3_KASIYER_UNKNOWN	2404	GMP3 CASHIER UNKNOWN,
APP_ERR_GMP3_FUNC_OKC_PARAMETRE	2405	PARAMETER DOWNLOAD FAILED,
APP_ERR_GMP3_FATURA_ODEME_PARAMETRE	2406	GMP3 INVOICE (FATURA) ODEME PARAMETER,
APP_ERR_GMP3_Z_RAPOR	2407	GMP3 Z REPORT,
APP_ERR_GMP3_X_RAPOR	2408	GMP3 X REPORT,

APP_ERR_GMP3_BATTERY_LEVEL	2409	GMP3 BATTERY LEVEL,
APP_ERR_GMP3_EKU_RAPOR_TYPE	2410	GMP3 EKU REPORT TYPE,
APP_ERR_GMP3_EKU_STATE	2411	GMP3 EKU STATE,
APP_ERR_GMP3_ECR_NOT_RECEIPT (FIS)CAL	2412	GMP3 ECR NOT RECEIPT (FIS)CAL,
APP_ERR_GMP3_NO_FUNCTION	2413	GMP3 NO FUNCTION,
APP_ERR_GMP3_MALI_KUMULATIF_RAPOR	2414	GMP3 FINANCIAL CUMULATIVE REPORT,
APP_ERR_GMP3_SINEMA_BILET_DESTEGI	2415	GMP3 CINEMA TICKET DESTEGI,
APP_ERR_GMP3_PARAMETRE_HATASI	2416	GMP3 PARAMETER ERROR,
APP_ERR_GMP3_Z_REQUIRED	2417	GMP3 Z REQUIRED,
APP_ERR_GMP3_UNSUPPORTED	2418	GMP3 UNSUPPORTED,
APP_ERR_GMP3_SINEMA_URUN_SAYISI	2419	GMP3 CINEMA NUMBER OF PRODUCTS,
APP_ERR_GMP3_PAYMENT_CANCELLED	2420	GMP3 PAYMENT CANCELLED,
APP_ERR_GMP3_ITEM_URUN_SAYISI	2421	GMP3 ITEM NUMBER OF PRODUCTS,
APP_ERR_GMP3_RESOURCE_PROBLEM	2422	GMP3 RESOURCE PROBLEM,
APP_ERR_GMP3_PLU_NOT_FOUND	2423	GMP3 PLU NOT FOUND,
APP_ERR_GMP3_NOT_PROPER_DISCOUNT	2424	GMP3 NOT PROPER DISCOUNT,
APP_ERR_GMP3_BIRIM_NOT_FOUND	2425	GMP3 UNIT NOT FOUND,
APP_ERR_GMP3_MIKTAR_NOT_FOUND	2426	GMP3 QUANTITY NOT FOUND,
APP_ERR_GMP3_URUN_ADI_NOT_FOUND	2427	GMP3 PRODUCT NAME (URUN ADI) NOT FOUND,
APP_ERR_GMP3_TUTAR_NOT_FOUND	2428	GMP3 AMOUNT (TUTAR) NOT FOUND,
APP_ERR_GMP3_UNKNOWN_ISLEM_TIPI	2429	GMP3 UNKNOWN TRANSACTION TYPE,
APP_ERR_GMP3_VERGI_NOT_AVAILABLE	2430	GMP3 TAX NOT AVAILABLE,
APP_ERR_GMP3_ISLEM_NOT_ALLOWED	2431	GMP3 ISLEM NOT ALLOWED,
APP_ERR_GMP3_CURRENCY_NOT_AVAILABLE	2432	GMP3 CURRENCY NOT AVAILABLE,
APP_ERR_GMP3_RECEIPT (FIS)_ITEM_NOT_ALLOWED	2433	GMP3 RECEIPT (FIS) ITEM NOT ALLOWED,
APP_ERR_GMP3_FATURA_PARAMETER_MISSING	2434	GMP3 INVOICE (FATURA) PARAMETER MISSING,
APP_ERR_GMP3_FATURA_TARIHI_MISSING	2435	GMP3 INVOICE (FATURA) DATE MISSING,
APP_ERR_GMP3_TUTAR_NOT_REQUIRED	2436	GMP3 AMOUNT (TUTAR) NOT REQUIRED,
APP_ERR_GMP3_TUTAR_REQUIRED	2437	GMP3 AMOUNT (TUTAR) REQUIRED,
APP_ERR_GMP3_INCORRECT_PASSWORD	2438	GMP3 INCORRECT PASSWORD,
APP_ERR_GMP3_INACTIVE_CASHIER_INDEX	2439	GMP3 INACTIVE CASHIER INDEX,
APP_ERR_GMP3_APPLICATION_NOT_FOUND	2440	GMP3 APPLICATION NOT FOUND,
APP_ERR_GMP3_UNDEFINED_TAG	2441	GMP3 UNDEFINED TAG,
APP_ERR_GMP3_SAVE_ECR_HEADER	2442	GMP3 SAVE ECR HEADER,
APP_ERR_GMP3_INVALID_RECIPIT_LIMIT	2443	GMP3 INVALID RECIPIT LIMIT,
APP_ERR_GMP3_ADVANCE	2444	GMP3 ADVANCE,
APP_ERR_GMP3_PAYMENT	2445	GMP3 PAYMENT,
APP_ERR_GMP3_TRANSACTION_IS_ACTIVE	2446	GMP3 TRANSACTION IS ACTIVE,
APP_ERR_GMP3_FATURA_TYPE_NOT_AVAILABLE	2447	GMP3 INVOICE (FATURA) TYPE NOT AVAILABLE,
APP_ERR_GMP3_FATURA_INVALID_TCK	2448	GMP3 INVOICE (FATURA) INVALID TCK,
APP_ERR_INVALID_UNIQUE_ID	2449	INVALID UNIQUE ID,
APP_ERR_GMP3_TICKET_TYPE	2450	GMP3 TICKET TYPE,
APP_ERR_GMP3_SETTINGS	2451	GMP3 SETTINGS,
APP_ERR_GMP3_BITMAP	2452	GMP3 BITMAP,
APP_ERR_GMP3_FILE	2453	GMP3 FILE,
APP_ERR_GMP3_ADMIN_PASSWORD_BLOCKED	2454	GMP3 ADMIN PASSWORD BLOCKED,
APP_ERR_GMP3_CASHIER_PASSWORD_BLOCKED	2455	GMP3 CASHIER PASSWORD BLOCKED,
APP_ERR_GMP3_CASHIER_PASSWORD_INCORRECT	2456	GMP3 CASHIER PASSWORD INCORRECT,

APP_ERR_GMP3_VAS_SUPPORT	2457	GMP3 VAS SUPPORT,
APP_ERR_GMP3_OTOPARK_PARAMETER	2458	GMP3 OTOPARK PARAMETER,
APP_ERR_GMP3_CUSTOMER_INFO	2459	GMP3 CUSTOMER INFO,
APP_ERR_GMP3_RECEIPT (FIS)EKU_PING_ERROR	2460	GMP3 RECEIPT (FIS)EKU PING ERROR,
APP_ERR_GMP3_TICKET_SALE_NOT_ALLOWED	2461	GMP3 TICKET SALE NOT ALLOWED,
APP_ERR_GMP3_INVALID_CASHIER_PASSWORD	2462	GMP3 INVALID CASHIER PASSWORD,
APP_ERR_GMP3_INVALID_CASHIER_NAME	2463	GMP3 INVALID CASHIER NAME,
APP_ERR_GMP3_ITEM_NOT_ALLOWED_TO_INC_DEC	2464	GMP3 ITEM NOT ALLOWED TO INCREASE DECREASE,
APP_ERR_GMP3_NO_ZERO_TAXRATE_ON_TAX_TABLE	2465	APP_ERR_GMP3_NO_ZERO_TAXRATE_ON_TAX_TAB LE
APP_ERR_GMP3_NO_PERMISSION_OF_USER	2466	APP_ERR_GMP3_NO_PERMISSION_OF_USER
APP_ERR_GMP3_NO_PERMISSION_OF_TSM	2467	APP_ERR_GMP3_NO_PERMISSION_OF_TSM
APP_ERR_GMP3_MISSING_INVOICE_PARAMETER	2468	APP_ERR_GMP3_MISSING_INVOICE_PARAMETER
APP_ERR_GMP3_ETTN_COULD_NOT_DOWNLOAD	2469	APP_ERR_GMP3_ETTN_COULD_NOT_DOWNLOAD
APP_ERR_GMP3_INTEGRATOR_COMMUNICATION	2470	APP_ERR_GMP3_INTEGRATOR_COMMUNICATION
APP_ERR_GMP3_INVALID_ONLINE_INVOICE_TYPE	2471	APP_ERR_GMP3_INVALID_ONLINE_INVOICE_TYPE
APP_ERR_GMP3_NOT_ALLOWED_PROCESS_IN_ONLINE_INVOI CE	2472	GMP3_NOT_ALLOWED_PROCESS_IN_ONLINE_INVOI CE
APP_ERR_GMP3_EXCEPTION_CODE_REQUIRED	2473	GMP3_EXCEPTION_CODE_REQUIRED
APP_ERR_GMP3_FORCED_VOID	2474	GMP3_FORCED_VOID
APP_ERR_GMP3_PROFILE_INDEX_CANNOT_SELECTED	2475	GMP3_PROFILE_INDEX_CANNOT_SELECTED
APP_ERR_GMP3_PROFILE_INDEX_NOT_SELECTED	2476	GMP3_PROFILE_INDEX_NOT_SELECTED
APP_ERR_GMP3_PDM_APP_ACTIVE	2477	GMP3_PDM_APP_ACTIVE
APP_ERR_GMP3_PDM_INVALID_MODE	2478	GMP3_PDM_INVALID_MODE
APP_ERR_GMP3_INVALID_FILE_NAME	2479	GMP3_INVALID_FILE_NAME
APP_ERR_GMP3_MONTHLY_REPORT	2480	APP_ERR_GMP3_MONTHLY_REPORT
APP_ERR_GMP3_MONTHLY_REPORT_WRITE_ERROR	2481	APP_ERR_GMP3_MONTHLY_REPORT_WRITE_ERRO R
APP_ERR_GMP3_MONTHLY_REPORT_READ_ERROR	2482	APP_ERR_GMP3_MONTHLY_REPORT_READ_ERROR
APP_ERR_GMP3_MONTHLY_REPORT_SEND_ERROR	2483	APP_ERR_GMP3_MONTHLY_REPORT_SEND_ERROR
APP_ERR_GMP3_MONTHLY_REPORT_FRAM_READ_ERROR	2484	APP_ERR_GMP3_MONTHLY_REPORT_FRAM_READ_ ERROR
APP_ERR_GMP3_MONTHLY_REPORT_FRAM_WRITE_ERROR	2485	APP_ERR_GMP3_MONTHLY_REPORT_FRAM_WRITE_ ERROR
APP_ERR_GMP3_DAILY_REPORT_FRAM_READ_ERROR	2486	APP_ERR_GMP3_DAILY_REPORT_FRAM_READ_ERR OR
APP_ERR_GMP3_DAILY_REPORT_FRAM_WRITE_ERROR	2487	APP_ERR_GMP3_DAILY_REPORT_FRAM_WRITE_ERR OR
APP_ERR_GMP3_MONTHLY_REPORT_RECEIPT (FIS)CAL_CUMULATIVE	2488	APP_ERR_GMP3_MONTHLY_REPORT_RECEIPT (FIS)CAL_CUMULATIVE
APP_ERR_GMP3_MONTHLY_NO_RECORD	2489	APP_ERR_GMP3_MONTHLY_NO_RECORD
APP_ERR_GMP3_TERMINAL_NOT_RECEIPT (FIS)CAL	2490	APP_ERR_GMP3_TERMINAL_NOT_RECEIPT (FIS)CAL
APP_ERR_GMP3_PAYMENT_STATUS_NO_AVAILABLE	2491	PAYMENT_STATUS_NO_AVAILABLE
APP_ERR_GMP3_INVALID_TCKN_VKN	2492	INVALID_TCKN_VKN
APP_ERR_GMP3_FOODCARD_VATRATE_MISMATCH	2493	FOODCARD_VATRATE_MISMATCH
APP_ERR_GMP3_FOODCARD_BIGGER_THAN_PAYABLE_AMOUN T	2494	FOODCARD_BIGGER_THAN_PAYABLE_AMOUNT
APP_ERR_GMP3_STOPPAGE_RATE_NOT_FOUND	2495	STOPPAGE_RATE_NOT_FOUND
APP_ERR_GMP3_CAN_NOT_ACCEPT_AFTER_HEADER_PRINTED	2496	CAN_NOT_ACCEPT_AFTER_HEADER_PRINTED
APP_ERR_GMP3_INVALID_PAYMENT_INDEX	2497	INVALID_PAYMENT_INDEX
APP_ERR_GMP3_GET_SLIP_REQUEST_FAILED	2498	GET_SLIP_REQUEST_FAILED
APP_ERR_GMP3_GET_SLIP_RESPONSE_FAILED	2499	GET_SLIP_RESPONSE_FAILED
APP_ERR_GMP3_GET_SLIP_INVALID_INDEX	2500	GET_SLIP_INVALID_INDEX
APP_ERR_ECR_RECEIPT_NOT_ALLOWED	2501	RECEIPT_NOT_ALLOWED

APP_ERR_INVOICE_INFO_RECEIPT_NOT_ALLOWED	2502	INVOICE_INFO_RECEIPT_NOT_ALLOWED
APP_ERR_FOODCARD_INFO_RECEIPT_NOT_ALLOWED	2503	FOODCARD_INFO_RECEIPT_NOT_ALLOWED
APP_ERR_PAY_AMOUNT_MISMATCH	2504	PAY_AMOUNT_MISMATCH
APP_ERR_INVALID_FOODCARD_VATRATE	2505	INVALID_FOODCARD_VATRATE
APP_ERR_GMP3_TOTAL_FOODCARD_BIGGER_THAN_PAYABLE_AMOUNT	2506	TOTAL_FOODCARD_BIGGER_THAN_PAYABLE_AMOUNT
APP_ERR_E_BILET_NO_PARAMETER_ENTRY	2507	E_BILET_NO_PARAMETER_ENTRY
APP_ERR_E_BILET_INVALID_BILET_TYPE	2508	E_BILET_INVALID_BILET_TYPE
APP_ERR_SMM_INVALID_RATE	2509	SMM_INVALID_RATE
APP_ERR_E_BILET_ITEM_CAN_NOT_HAVE_A_NAME	2510	E_BILET_ITEM_CAN_NOT_HAVE_A_NAME
APP_ERR_INVALID_UNIT_TYPE	2511	INVALID_UNIT_TYPE
APP_ERR_PAYMENT_TYPE_MISMATCH	2512	PAYMENT_TYPE_MISMATCH
APP_ERR_REVERSE_PAYMENT_CANNOT_VOID	2513	REVERSE_PAYMENT_CANNOT_VOID
APP_ERR_TERMINAL_DATE_INVALID	2520	TERMINAL_DATE_INVALID
APP_ERR_24H_RESET_TIME_DELAY_MUST_BIGGER	2521	24H_RESET_TIME_DELAY_MUST_BIGGER
APP_ERR_24H_RESET_TIME_DELAY_MUST_SMALLER	2522	24H_RESET_TIME_DELAY_MUST_SMALLER
APP_ERR_24H_RESET_INVALID_BEFORE_MANAGER_PARAMETER	2523	24H_RESET_INVALID_BEFORE_MANAGER_PARAMETER
APP_ERR_24H_RESET_INVALID_TIME_SET	2524	24H_RESET_INVALID_TIME_SET
APP_ERR_PAYMENT_APP_NOT_SELECTED	2530	PAYMENT_APP_NOT_SELECTED
APP_ERR_GMP3_NOT_ALLOWED_BEFORE_HEADER	2531	GMP3_NOT_ALLOWED_BEFORE_HEADER

Gmp3SmartDll Return Codes	Error Codes	Description
DLL_RETCODE_PORT_NOT_OPEN	0xF000	PORT NOT OPEN,
DLL_RETCODE_ECR_DATA_ERR	0xF001	ECR DATA ERR,
DLL_RETCODE_POS_DATA_ERR	0xF002	POS DATA ERR,
DLL_RETCODE_TIMEOUT	0xF003	TIMEOUT,
DLL_RETCODE_DATA_SEND_ERR	0xF004	DATA SEND ERR,
DLL_RETCODE LENGHT_ERR	0xF005	LENGHT ERR,
DLL_RETCODE_SESSIONID_ERR	0xF006	SESSIONID ERR,
DLL_RETCODE_DATA_RECV_ERR	0xF007	DATA RECV ERR,
DLL_RETCODE_RETRY_ERR	0xF008	RETRY ERR,
DLL_RETCODE_RECV_EOT	0xF009	RECV EOT,
DLL_RETCODE_LEN_TOO_LONG	0xF00A	LEN TOO LONG,
DLL_RETCODE_FAIL	0xF00B	FAIL,
12467893487654wq1	812357p*0*2470*9	ERROR STX,
DLL_RETCODE_ERROR_ETX	0xF00D	ERROR ETX,
DLL_RETCODE_ERROR_CRC	0xF00E	ERROR CRC,
DLL_RETCODE_ERROR_MSGTYPE	0xF00F	ERROR MSGTYPE,
DLL_RETCODE_ERROR_ERR	0xF010	ERROR ERR,
DLL_RETCODE_ERROR_SID	0xF011	ERROR SID,
DLL_RETCODE_ERROR_EOT	0xF012	ERROR EOT,
DLL_RETCODE_STATUS_CODE_ERR	0xF013	STATUS CODE ERR,
DLL_RETCODE_LEN_TOO_SHORT	0xF014	LEN TOO SHORT,
DLL_RETCODE_DEMO_VERSION	0xF015	DEMO VERSION,
DLL_RETCODE_FILE_OPEN_ERR	0xF016	FILE OPEN ERR,
DLL_RETCODE_KEY_ERROR	0xF017	KEY ERROR,
DLL_RETCODE_TERMSN_ERROR	0xF018	TERMSN ERROR,

DLL_RETCODE_ERROR_LRC	0xF019	ERROR LRC,
DLL_RETCODE_REC_EOT_OK	0xF01A	REC EOT OK,
DLL_RETCODE_ACK_NOT_RECEIVED	0xF01B	ACK NOT RECEIVED,
DLL_RETCODE_RECV_BUSY	0xF01C	RECV BUSY,
DLL_RETCODE_RECV_ACK	0xF01D	RECV ACK,
DLL_RETCODE_RECV_NAK	0xF01E	RECV NAK,
DLL_RETCODE_INTERCHAR_TIMEOUT	0xF01F	INTERCHAR TIMEOUT,
DLL_RETCODE_PAIRING_REQUIRED	0xF020	PAIRING REQUIRED,
DLL_RETCODE_WORG_PING_RETURN	0xF021	WORG PING RETURN,
DLL_RETCODE_MEMORY_ERROR	0xF022	MEMORY ERROR,
DLL_RETCODE_ECR_VALUE_ERROR	0xF023	ECR VALUE ERROR,
DLL_RETCODE_PORT_OPEN_ERROR	0xF024	PORT OPEN ERROR,
DLL_RETCODE_JSON_ERROR	0xF025	JSON ERROR,
DLL_RETCODE_ECR_VERSION_TOO_OLD	0xF026	ECR VERSION TOO OLD,
DLL_RETCODE_PROCESSING_NUMBER_MISMATCH	0xF027	PROCESSING NUMBER MISMATCH,
DLL_RETCODE_DECRYPTION_ERR	0xF028	DECRYPTION ERR,
DLL_RETCODE_HASH_CALCULATE_ERROR	0xF029	HASH CALCULATE ERROR,
DLL_RETCODE_INVALID_STRUCT_SIZE	0xF02A	INVALID STRUCT SIZE,
DLL_RETCODE_COMPRESSION_ERR	0xF02B	COMPRESSION ERR,
DLL_RETCODE_DECOMPRESSION_ERR	0xF02C	DECOMPRESSION ERR,
DLL_RETCODE_PRIME_NUMBER_ERR	0xF02D	PRIME NUMBER ERR,
DLL_RETCODE_ENCRYPTION_ERR	0xF02E	ENCRYPTION ERR,
DLL_RETCODE_PADDING_ERROR	0xF02F	PADDING ERROR,
DLL_RETCODE_ENC_DEC_MISMATCH_ERROR	0xF030	ENC DEC MISMATCH ERROR,
DLL_RETCODE_INVALID_AMOUNT	0xF031	INVALID AMOUNT,
DLL_RETCODE_OLD_STYLE_FUNCTION_CALL_ERROR	0xF032	OLD STYLE FUNCTION CALL ERROR
DLL_RETCODE_NEW_STYLE_FUNCTION_CALL_ERROR	0xF033	NEW STYLE FUNCTION CALL ERROR
DLL_RETCODE_INTERFACE_HANDLE_ERROR	0xF034	INTERFACE HANDLE
DLL_RETCODE_INVALID_INTERFACE_HANDLE	0xF035	INVALID INTERFACE HANDLE
DLL_RETCODE_XML_FILE_ERROR	0xF036	XML FILE ERROR
DLL_RETCODE_INTERFACE_ID_ALREADY_EXIST	0xF037	INTERFACE ID ALREADY EXIST
DLL_RETCODE_PRINTER_FRONT_STATION_BUFFER_ERROR	0xF038	PRINTER FRONT STATION BUFFER ERROR
DLL_RETCODE_INVALID_CERTIFICATE_DLL	0xF039	INVALID_CERTIFICATE_DLL
DLL_RETCODE_INVALID_CERTIFICATE_ECR	0xF03A	INVALID_CERTIFICATE_ECR
DLL_RETCODE_INVALID_SIGN_DLL	0xF03B	INVALID_SIGN_DLL
DLL_RETCODE_INVALID_KCV	0xF03C	INVALID_KCV
DLL_RETCODE_INVALID_SIGN_ECR	0xF03D	INVALID_SIGN_ECR
DLL_RETCODE_MAX_BUFFER_OVERFLOW	0xF03E	MAX_BUFFER_OVERFLOW

8 GDP_v15r32

Name	Date modified	Type
CSharpClass	5.07.2022 15:23	File folder
DLL	15.03.2022 14:29	File folder
DLL_DEBUG	15.03.2022 15:59	File folder
Doc	12.08.2022 09:07	File folder
Inc	15.03.2022 14:29	File folder
LinuxClass	15.03.2022 14:29	File folder
Simulator	15.03.2022 14:29	File folder
TeliumLib	15.03.2022 14:29	File folder

CSharpClass:Header for C lib files for corresponding platforms

DLL:Contains DLL and packages of GMP3.

Doc: Intagration help document.

Inc:Header file for C.

Linux:Header file for Linux OS

Simulator:Simulator source codes

8.1 GMP3 Simulator

GMP3 Simulator is included in GDP_v15r32 file.For downloading in GDP_v15r32 click the link below.

<https://www.redmine.ingenico.com/attachments/50025>

Name	Date modified	Type
CSharpClass	5.07.2022 15:23	File folder
DLL	15.03.2022 14:29	File folder
DLL_DEBUG	15.03.2022 15:59	File folder
Doc	12.08.2022 09:07	File folder
Inc	15.03.2022 14:29	File folder
LinuxClass	15.03.2022 14:29	File folder
Simulator	15.03.2022 14:29	File folder
TeliumLib	15.03.2022 14:29	File folder

EntityFramework.dll	15.11.2019 10:23	Application extens...	4.887 KB
EntityFramework.SqlServer.dll	15.11.2019 10:23	Application extens...	603 KB
EntityFramework.SqlServer	15.11.2019 10:23	XML File	146 KB
EntityFramework	15.11.2019 10:23	XML File	3.181 KB
ePos.iobj	15.11.2019 10:25	IOBJ File	9.809 KB
ePos.ipdb	15.11.2019 10:25	IPDB File	2.922 KB
GMP	15.08.2022 14:32	XML File	3 KB
GMPDLL_2022_08_19_143240	19.08.2022 14:38	TXT File	638 KB
GmpSampleSim	30.06.2021 14:41	Application	645 KB
GmpSampleSim.exe.config	15.11.2019 08:58	XML Configuration...	4 KB
GMPSimulatorDb_v15r32.sqlite	15.08.2022 08:06	SQLITE File	16 KB
GMPSmartDLL.dll	30.06.2021 14:35	Application extens...	2.547 KB
GMPSmartDLL.lib	30.06.2021 14:35	Object File Library	119 KB
Newtonsoft.Json.dll	15.11.2019 10:23	Application extens...	499 KB
Newtonsoft.Json	15.11.2019 10:23	XML File	490 KB
SQLite.Interop.dll	15.11.2019 10:23	Application extens...	1.629 KB
System.Data.SQLite.dll	15.11.2019 10:23	Application extens...	352 KB
System.Data.SQLite.dll.config	15.11.2019 10:23	XML Configuration...	1 KB
System.Data.SQLite.EF6.dll	15.11.2019 10:23	Application extens...	183 KB
System.Data.SQLite.Linq.dll	15.11.2019 10:23	Application extens...	183 KB
System.Data.SQLite	15.11.2019 10:23	XML File	1.064 KB

8.2 GMP.XML

It is an important file that is required to be able to make configurations Of GMP3 DLL.

File content is attached below with some explanations which are needed to make changes.

```
1 <?xml version="1.0" encoding="iso-8859-9" ?>
2 <GMP>
3   <DLL>
4     <IsCheckStructVersion>TRUE</IsCheckStructVersion>
5     <LogPath></LogPath>
6     <LogFileSize>5000000</LogFileSize>
7   </DLL>
8   <LOG>
9     <LogPrintToFileOpen>TRUE</LogPrintToFileOpen>
10    <LogGeneralOpen>TRUE</LogGeneralOpen>
11    <LogFunctionOpen>TRUE</LogFunctionOpen>
12    <LogSecurityOpen>FALSE</LogSecurityOpen>
13    <LogCommOpen>TRUE</LogCommOpen>
14    <LogExtDevOpen>TRUE</LogExtDevOpen>
15    <LogJsonOpen>TRUE</LogJsonOpen>
16    <LogJsonDataOpen>TRUE</LogJsonDataOpen>
17    <LogGmp3TagsOpen>TRUE</LogGmp3TagsOpen>
18    <LogThreadOpen>FALSE</LogThreadOpen>
19    <LogPrintSerialNumOpen>FALSE</LogPrintSerialNumOpen>
20    <LogPrintDateOpen>FALSE</LogPrintDateOpen>
21    <LogPrintTimeOpen>TRUE</LogPrintTimeOpen>
22    <LogPrintTypeOpen>TRUE</LogPrintTypeOpen>
23    <LogPrintVersionOpen>TRUE</LogPrintVersionOpen>
24    <LogPrintSourceFileOpen>FALSE</LogPrintSourceFileOpen>
25    <LogPrintSourceLineOpen>FALSE</LogPrintSourceLineOpen>
26  </LOG>
27  <INTERFACE ID="COM1" DEFAULT="TRUE">
28    <RetryCounter>3</RetryCounter>
29    <IpRetryCount>3</IpRetryCount>
30    <AckTimeOut>3000</AckTimeOut>
31    <CommTimeOut>10000</CommTimeOut>
32    <InterCharacterTimeOut>1500</InterCharacterTimeOut>
33    <PortName>\\.\COM3</PortName>
34    <BaudRate>115200</BaudRate>
35    <ByteSize>8</ByteSize>
36    <fParity>0</fParity>
37    <Parity>0</Parity>
38    <StopBit>0</StopBit>
39    <IsTcpConnection>FALSE</IsTcpConnection>
40    <IP>5.26.113.219</IP>
41    <Port>7500</Port>
42  </INTERFACE>
43  <LOG>
44    <LogPrintToFileOpen>TRUE</LogPrintToFileOpen>
45    <LogGeneralOpen>TRUE</LogGeneralOpen>
46    <LogFunctionOpen>TRUE</LogFunctionOpen>
47    <LogSecurityOpen>FALSE</LogSecurityOpen>
48    <LogCommOpen>TRUE</LogCommOpen>
49    <LogExtDevOpen>TRUE</LogExtDevOpen>
50    <LogJsonOpen>FALSE</LogJsonOpen>
51    <LogJsonDataOpen>FALSE</LogJsonDataOpen>
52    <LogGmp3TagsOpen>TRUE</LogGmp3TagsOpen>
53    <LogThreadOpen>FALSE</LogThreadOpen>
54    <LogPrintSerialNumOpen>FALSE</LogPrintSerialNumOpen>
55    <LogPrintDateOpen>FALSE</LogPrintDateOpen>
56    <LogPrintTimeOpen>TRUE</LogPrintTimeOpen>
57    <LogPrintTypeOpen>TRUE</LogPrintTypeOpen>
58    <LogPrintVersionOpen>TRUE</LogPrintVersionOpen>
59    <LogPrintSourceFileOpen>FALSE</LogPrintSourceFileOpen>
60    <LogPrintSourceLineOpen>FALSE</LogPrintSourceLineOpen>
61  </LOG>
62 </GMP>
63
```

LogPath : Determines in which directory the log file should be created on the computer.If the file path is not selected, it will be created in the directory where the dll file is located.

LogFileSize:Determines the size of the log file.Creates a new log file when the log file reaches the specified size.

```
<LOG>
  <LogPrintToFileOpen>TRUE</LogPrintToFileOpen>
  <LogGeneralOpen>TRUE</LogGeneralOpen>
  <LogFunctionOpen>TRUE</LogFunctionOpen>
  <LogSecurityOpen>FALSE</LogSecurityOpen>
  <LogCommOpen>TRUE</LogCommOpen>
  <LogExtDevOpen>TRUE</LogExtDevOpen>
  <LogJsonOpen>TRUE</LogJsonOpen>
  <LogJsonDataOpen>TRUE</LogJsonDataOpen>
  <LogGmp3TagsOpen>TRUE</LogGmp3TagsOpen>
  <LogThreadOpen>FALSE</LogThreadOpen>
  <LogPrintSerialNumOpen>FALSE</LogPrintSerialNumOpen>
  <LogPrintDateOpen>FALSE</LogPrintDateOpen>
  <LogPrintTimeOpen>TRUE</LogPrintTimeOpen>
  <LogPrintTypeOpen>TRUE</LogPrintTypeOpen>
  <LogPrintVersionOpen>TRUE</LogPrintVersionOpen>
  <LogPrintSourceFileOpen>FALSE</LogPrintSourceFileOpen>
  <LogPrintSourceLineOpen>FALSE</LogPrintSourceLineOpen>
</LOG>
```

These are the log levels configurations above.We ask you to set the marked logs to true in order to examine the problems you encounter in detail.

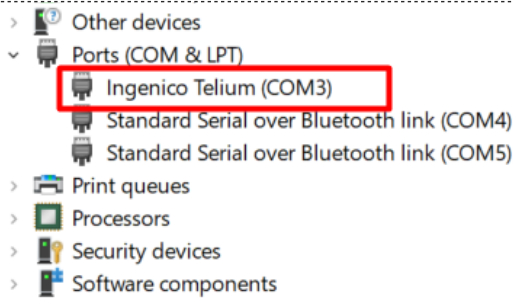
```

<INTERFACE ID="COM1" DEFAULT="TRUE">
  <RetryCounter>3</RetryCounter>
  <IpRetryCount>3</IpRetryCount>
  <AckTimeOut>3000</AckTimeOut>
  <CommTimeOut>10000</CommTimeOut>
  <InterCharacterTimeOut>1500</InterCharacterTimeOut>
  <PortName>\\.\COM3</PortName>
  <BaudRate>115200</BaudRate>
  <ByteSize>8</ByteSize>
  <fParity>0</fParity>
  <Parity>0</Parity>
  <StopBit>0</StopBit>
  <IsTcpConnection>FALSE</IsTcpConnection>
  <IP>5.26.113.219</IP>
  <Port>7500</Port>

```

These are the connection configurations above. If connection type will be by USB cable set **IsTcpConnection** False and set **Portname** of Ports information the computer is connected to. It can be checked following steps below.

Start>Device Manager >Ports(Com&LPT)

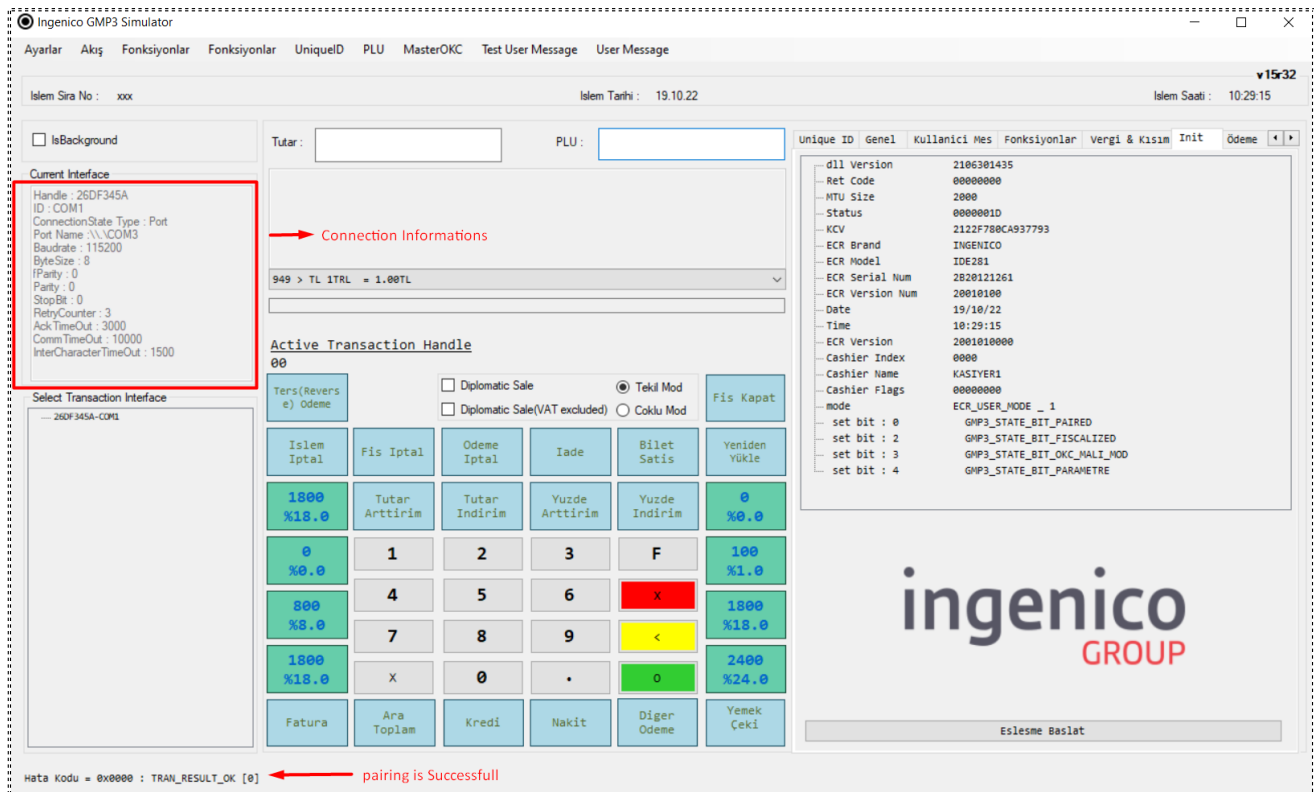
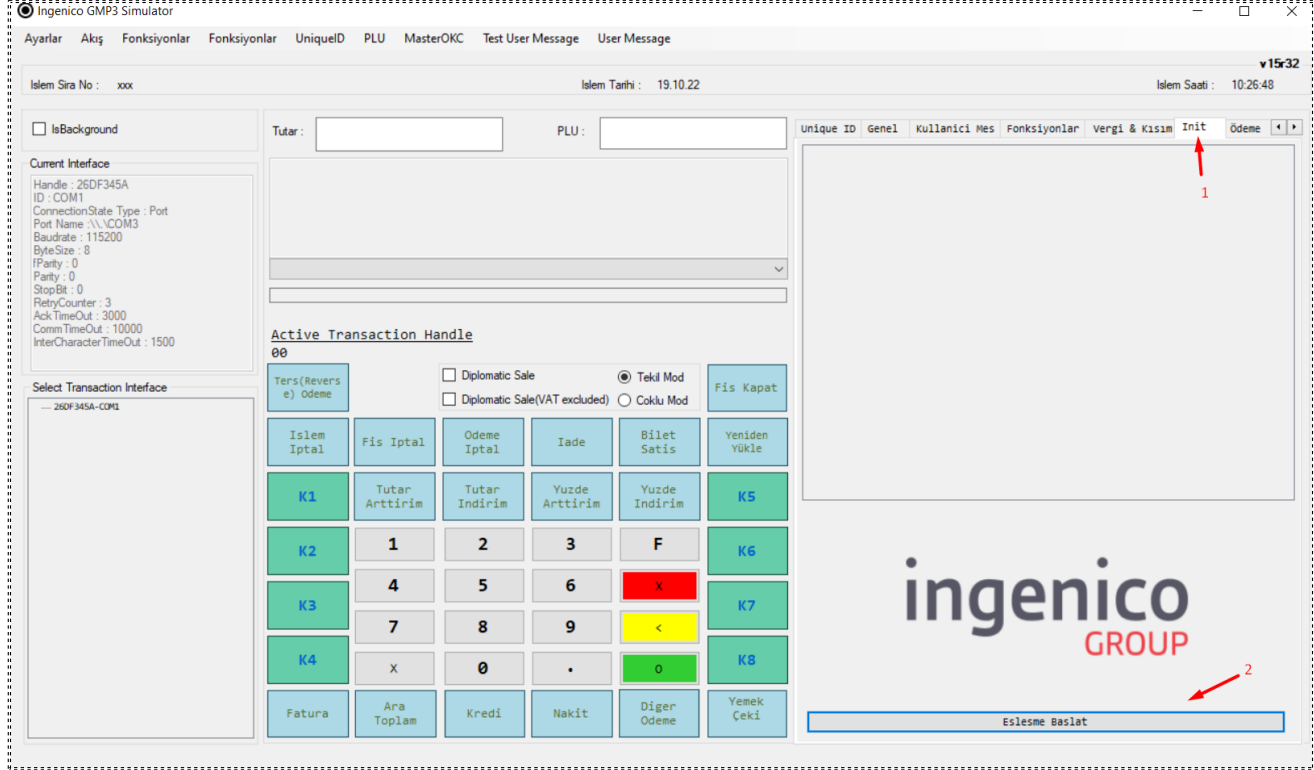


8.3 Using GMP3 Simulator

GDP_v15r32 > DLL > Windows > x64 >

Name	Date modified	Type	Size
en-GB	15.03.2022 14:29	File folder	
x64	15.03.2022 14:29	File folder	
x86	15.03.2022 14:29	File folder	
EntityFramework.dll	15.11.2019 10:23	Application extens...	4.887 KB
EntityFramework.SqlServer.dll	15.11.2019 10:23	Application extens...	603 KB
EntityFramework.SqlServer	15.11.2019 10:23	XML File	146 KB
EntityFramework	15.11.2019 10:23	XML File	3.181 KB
ePos.iobj	15.11.2019 10:25	IOBJ File	9.809 KB
ePos.ipdb	15.11.2019 10:25	IPDB File	2.922 KB
GMP	19.10.2022 09:54	XML File	3 KB
GmpSampleSim	30.06.2021 14:41	Application	645 KB
GmpSampleSim.exe.config	15.11.2019 08:58	XML Configuration...	4 KB
GMPSimulatorDb_v15r32.sqlite	4.10.2022 10:56	SQLITE File	16 KB
GMPSmartDLL.dll	30.06.2021 14:35	Application extens...	2.547 KB
GMPSmartDLL.lib	30.06.2021 14:35	Object File Library	119 KB
Newtonsoft.Json.dll	15.11.2019 10:23	Application extens...	499 KB
Newtonsoft.Json	15.11.2019 10:23	XML File	490 KB
SQLite.Interop.dll	15.11.2019 10:23	Application extens...	1.629 KB
System.Data.SQLite.dll	15.11.2019 10:23	Application extens...	352 KB
System.Data.SQLite.dll.config	15.11.2019 10:23	XML Configuration...	1 KB
System.Data.SQLite.EF6.dll	15.11.2019 10:23	Application extens...	183 KB
System.Data.SQLite.Linq.dll	15.11.2019 10:23	Application extens...	183 KB
System.Data.SQLite	15.11.2019 10:23	XML File	1.064 KB

First of all Pairing should be done.



8.3.1 Sales Ticket

Ingenico GMP3 Simulator

Ayarlar Akış Fonksiyonlar Fonksiyonlar UniqueID PLU MasterOKC Test User Message User Message

İşlem Sıra No : xxx İşlem Tarihi : 19.10.22 İşlem Saati : 10:30:37

IsBackground

Current Interface

Handle : 26DF345A
ID : COM1
ConnectionState Type : Port
Port Name : \\.\COM3
Baudrate : 115200
ByteSize : 8
IF parity : 0
StopBit : 0
RetryCounter : 3
AckTimeOut : 3000
CommTimeOut : 10000
InterCharacterTimeOut : 1500

Select Transaction Interface

26DF345A-COM1

Tutar : 2000 PLU :

1-Amount of the Product
Last 2 digits are kuruş
this is 2 Turkish TL

949 > TL 1TRL = 1.00TL

Active Transaction Handle

00

Ters(Reverse) Ödeme

Diplomatic Sale

Diplomatic Sale(VAT excluded)

Tekil Mod

Coklu Mod

Fis Kapat

Islem Iptal

Fis Iptal

Odeme Iptal

Iade

Bilet Satış

Yeniden Yükle

1800
%18.0

Tutar Arttırım

Tutar İndirim

Yüzde Arttırım

Yüzde İndirim

0
%0.0

1

2

3

F

100
%1.0

0
%0.0

4

5

6

X

1800
%18.0

7

8

9

<

2400
%24.0

800
%8.0

X

0

.

0

2400
%24.0

1800
%18.0

Fatura

Ara Toplam

Kredi

Nakit

Diger Ödeme

Yemek Çeki

Unique ID Genel Kullanici Mes Fonksiyonlar Vergi & Kısım Init Ödeme

dll Version : 2106301435
Ret Code : 00000000
MTU Size : 2000
Status : 00000010
KCV : 2122F780CA937793
ECR Brand : INGENICO
ECR Model : IDE281
ECR Serial Num : 2820121261
ECR Version Num : 20010100
Date : 19/10/22
Time : 10:29:15
ECR Version : 2001010000
Cashier Index : 0000
Cashier Name : KASİYER1
Cashier Flags : 00000000
mode : ECR_USER_MODE _ 1
set bit : 0 : GMP3_STATE_BIT_PAİRED
set bit : 2 : GMP3_STATE_BIT_FISCALIZED
set bit : 3 : GMP3_STATE_BIT_OKC_MALI_MOD
set bit : 4 : GMP3_STATE_BIT_PARAMETRE

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Esleme Başlat

Ingenico GMP3 Simulator

Ayarlar Akış Fonksiyonlar Fonksiyonlar UniqueID PLU MasterOKC Test User Message User Message

İşlem Sıra No : xxx İşlem Tarihi : 19.10.22 İşlem Saati : 10:33:27

IsBackground

Current Interface

Handle : 26DF345A
ID : COM1
ConnectionState Type : Port
Port Name : \\.\COM3
Baudrate : 115200
ByteSize : 8
IF parity : 0
StopBit : 0
RetryCounter : 3
AckTimeOut : 3000
CommTimeOut : 10000
InterCharacterTimeOut : 1500

Select Transaction Interface

26DF345A-COM1

Tutar : 2000 PLU :

1-Amount of the payment

949 > TL 1TRL = 1.00TL

0 %0.0 *20.00

Active Transaction Handle

634FD27B2660D5A4

Ters(Reverse) Ödeme

Diplomatic Sale

Diplomatic Sale(VAT excluded)

Tekil Mod

Coklu Mod

Fis Kapat

Islem Iptal

Fis Iptal

Odeme Iptal

Iade

Bilet Satış

Yeniden Yükle

1800
%18.0

Tutar Arttırım

Tutar İndirim

Yüzde Arttırım

Yüzde İndirim

0
%0.0

1

2

3

F

100
%1.0

0
%0.0

4

5

6

X

1800
%18.0

7

8

9

<

2400
%24.0

800
%8.0

X

0

.

0

2400
%24.0

1800
%18.0

Fatura

Ara Toplam

Kredi

Nakit

Diger Ödeme

Yemek Çeki

İşlem Sonuçları

Unique ID Genel Kullanici Mes Fonksiyonlar Vergi & Kısım

UNIQUE ID : 000000000000000000007B024F630201030000000000
TICKET TYPE : 1
Z NO : 258
F NO : 3
EJNO : 1
TRANSACTION FLAG : 00131074
OPTION FLAG : 00000007
FLG_XTRANS_GMP3 : 19/10/2022
FLG_XTRANS_TICKET_HEADER_PRINTED : 10:33:32
İşlem Tarihi : 20.00
İşlem Saati : 0.00
Toplam Vergi : 1
Urun Tablosu : 0
Kampanya Tablosu : 0
Odeme Tablosu : 0

Hata kodu = 0x0000 : TRAN_RESULT_OK [0]

2-Select type of the payment

The slip and logs of the transaction below


```

FP3_Start(TO : 10000) (hint:01611X) (hTrx:081X)
FP3_Start (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_TicketHeader(TO : 10000) (hInt:01611X) (hTrx:081X)
FP3_TicketHeader (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_OptionFlags(TO : 10000) (hInt:01611X) (hTrx:081X)
FP3_OptionFlags (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_ItemSale(TO : 10000) (hInt:01611X) (hTrx:081X)
FP3_ItemSale (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_Payment(TO : 30000) (hInt:01611X) (hTrx:081X)
FP3_Payment (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_PrintTotalsAndPayments(TO : 10000) (hInt:01611X) (hTrx:081X)
FP3_PrintTotalsAndPayments (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_PrintBeforeMF(TO : 10000) (hInt:01611X) (hTrx:081X)
FP3_PrintBeforeMF (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_PrintUserMessage(TO : 10000) (hInt:01611X) (hTrx:081X)
FP3_PrintUserMessage (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_PrintMF(TO : 100000) (hInt:01611X) (hTrx:081X)
FP3_PrintMF (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])
FP3_Close(TO : 10000) (hint:01611X) (hTrx:081X)
FP3_Close (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

```



8.3.2 Invoice Sales Ticket

Ingenico GMP3 Simulator

Ayarlar Akış Fonksiyonlar Fonksiyonlar UniqueID PLU MasterOKC Test User Message User Message

İşlem Sıra No : xxx İşlem Tarihi : 19.10.22 İşlem Saati : 10:41:00

IsBackground

Current Interface

Handle : 26DF345A
ID : COM1
ConnectionState Type : Port
Port Name : \\.\COM3
Baudrate : 115200
ByteSize : 8
Parity : 0
StopBit : 0
RetryCounter : 3
AckTimeOut : 3000
CommTimeOut : 10000
InterCharacterTimeOut : 1500

Select Transaction Interface

26DF345A-COM1

Tutar : PLU :

TOTAL : 20.00 TL
REMAIN : 0.00 TL

949 > TL 1TRL = 1.00TL

0 %00 *20,00

Active Transaction Handle

00

Ters(Reverse) Odeme

Diplomatic Sale

Diplomatic Sale(VAT excluded)

Tekil Mod

Coklu Mod

Fis Kapat

İşlem İptal

Fis İptal

Odeme İptal

İade

Bilet Satış

Yeniden Yükle

1800 %18.0

Tutar Arttırım

Tutar İndirim

Yüzde Arttırım

Yüzde İndirim

0 %0.0

1

2

3

F

100 %1.0

800 %8.0

4

5

6

X

1800 %18.0

7

8

9

<

2400 %24.0

1800 %18.0

X

0

.

0

2400 %24.0

Fatura

Ara Toplam

Kredi

Nakit

Diğer Odeme

Yemek Çeki

Fatura

2-select type of invoice

Fatura

Fatura Tipi : NAKİT FATURA

Fatura Tipi : NAKİT FATURA

TC Kimlik No : E FATURA

Vergi kimlik No : E ARSIV

Fatura No :

Fatura Tarihi : 19 Ekim 2022 Çarşamba

Fatura Tutarı :

Gonder

Hata Kodu = 0x0000 : TRAN_RESULT_OK [0]

Ingenico GMP3 Simulator

Ayarlar Akış Fonksiyonlar Fonksiyonlar UniqueID PLU MasterOKC Test User Message User Message

İşlem Sıra No : xxx İşlem Tarihi : 19.10.22 İşlem Saati : 10:43:07

IsBackground

Current Interface

Handle : 26DF345A
ID : COM1
ConnectionState Type : Port
Port Name : \\.\COM3
Baudrate : 115200
ByteSize : 8
Parity : 0
StopBit : 0
RetryCounter : 3
AckTimeOut : 3000
CommTimeOut : 10000
InterCharacterTimeOut : 1500

Select Transaction Interface

26DF345A-COM3

Tutar : PLU :

TOTAL : 20.00 TL
REMAIN : 0.00 TL

949 > TL 1TRL = 1.00TL

0 %00 *20,00

Active Transaction Handle

00

Ters(Reverse) Odeme

Islem Iptal Fis Iptal Odeme Iptal Iade Bilet Satış Yeniden Yükle

1800 %18.0 Tutar Arttırım Tutar İndirim Yüzde Arttırım Yüzde İndirim 0 %0.0

0 %0.0 1 2 3 F 100 %1.0

800 %8.0 4 5 6 x 1800 %18.0

1800 %18.0 7 8 9 < 2400 %24.0

X 0 . 0

Fatura Ara Toplam Kredi Nakit Diğer Odeme Yemek Çeki

4-If its empty ECR will write on the slip FSD (Faturası sonradan düzelenecektir -Invoice is going to be created later)

3-Its compulsory field and should be 11 digits

Fatura

Fatura Tipi : KAGIT FATURA ☐ İrsaliye

TC Kimlik No : 11111111111

Vergi Kimlik No :

Fatura No :

Fatura Tarihi : 19 Ekim 2022 Çarşamba

Fatura Tutarı :

Gonder

5-It should be empty.Amount should not be send in this p

6

Hata Kodu = 0x0000 : TRAN_RESULT_OK [0]

Ingenico GMP3 Simulator

Ayarlar Akış Fonksiyonlar Fonksiyonlar UniqueID PLU MasterOKC Test User Message User Message

İşlem Sıra No : xxx İşlem Tarihi : 19.10.22 İşlem Saati : 10:30:37

IsBackground

Current Interface

Handle : 26DF345A
ID : COM1
ConnectionState Type : Port
Port Name : \\.\COM3
Baudrate : 115200
ByteSize : 8
Parity : 0
StopBit : 0
RetryCounter : 3
AckTimeOut : 3000
CommTimeOut : 10000
InterCharacterTimeOut : 1500

Select Transaction Interface

26DF345A-COM3

Tutar : 2000 PLU :

949 > TL 1TRL = 1.00TL

Active Transaction Handle

00

Ters(Reverse) Odeme

Islem Iptal Fis Iptal Odeme Iptal Iade Bilet Satış Yeniden Yükle

1800 %18.0 Tutar Arttırım Tutar İndirim Yüzde Arttırım Yüzde İndirim 0 %0.0

0 %0.0 1 2 3 F 100 %1.0

800 %8.0 4 5 6 x 1800 %18.0

1800 %18.0 7 8 9 < 2400 %24.0

X 0 . 0

Fatura Ara Toplam Kredi Nakit Diğer Odeme Yemek Çeki

1-Amount of the Product
Last 2 digits are kuruş
this is 2 Turkish TL

2-Vat Value of the products

Unique ID Genel Kullanıcı Mes Fonksiyonlar Vergi & Kısım Init Ödeme

dll Version 2106301435
Ret Code 00000000
MTU Size 2000
Status 00000010
KCV 2122F780CA937793
ECR Brand INGENICO
ECR Model IDE281
ECR Serial Num 2820121261
ECR Version Num 20010100
Date 19/10/22
Time 10:29:15
ECR Version 2001010000
Cashier Index 0000
Cashier Name KASIVER1
Cashier Flags 00000000
mode ECR_USER_MODE _ 1
set bit : 0 GMP3_STATE_BIT_PAİRED
set bit : 2 GMP3_STATE_BIT_FISCALIZED
set bit : 3 GMP3_STATE_BIT_OKC_MALI_MOD
set bit : 4 GMP3_STATE_BIT_PARAMETRE

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Esleme Başlat

Ingenico GMP3 Simulator

Ayarlar Akış Fonksiyonlar Fonksiyonlar UniqueID PLU MasterOKC Test User Message User Message

İşlem Sıra No : xxx İşlem Tarihi : 19.10.22 İşlem Saati : 10:50:03

IsBackground

Current Interface

Handle : 26DF345A
ID : COM1
ConnectionState Type : Port
Port Name : \\COM3
Baudrate : 115200
ByteSize : 8
Parity : 0
StopBit : 0
RetryCounter : 3
AckTimeOut : 3000
CommTimeOut : 10000
InterCharacterTimeOut : 1500

Select Transaction Interface

26DF345A-COM1

Tutar : 20000 PLU :

TOTAL : 2000.00 TL
REMAIN : 0.00 TL

Amount of the payment

949 > TL 1TRL => 1.00TL

800 %8 *2.000,00

Active Transaction Handle

00

Ters(Reverse) Odeme

Diplomatic Sale

Diplomatic Sale(VAT excluded)

Tekil Mod

Coklu Mod

Fis Kapat

İşlem iptal

Fis iptal

Odeme iptal

İade

Bilet Satış

Yeniden yükle

1800 %18.0

Tutar Arttırım

Tutar İndirim

Yüzde Arttırım

Yüzde İndirim

0 %0.0

1

2

3

F

100 %1.0

800 %8.0

4

5

6

X

1800 %18.0

7

8

9

<

2400 %24.0

1800 %18.0

X

0

.

0

2400 %24.0

Fatura

Ara Toplam

Kredi

Nakit

Diğer Odeme

Yemek Çeki

Hata Kodu : 000000 : TRAN_RESULT_OK [0]

Select type of the payment

The slip and logs of the transaction below

*** TEST CİHAZIDIR ***

ingenico

ingenico test

Maslak

Sarıyer

İSTANBUL

ŞİŞLİ 1880997800

TARİH : 19/10/2022

SAAT : 10:48:08

FİŞ NO : 0004

BİLGİ FİŞİ

TÜR: FATURA

BELGE NO: FSD

MÜŞTERİ VKN/TKN:1111111111

BELGE TARİHİ : 19/10/2022

800 %8 *2.000,00

TOPKDV *148,15

TOPLAM *2.000,00

NAKİT *2.000,00

T.SİCİL NO: 172839020000001

*** MALİ DEĞERİ YOKTUR ***

İRSALİYE YERİNE GEÇMEZ

EKÜ NO:0001 Z NO:0258

Teşekkür Ederiz

*** TEST CİHAZIDIR ***

OK 2B 20121261

FP3 Start(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 Start (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 OptionFlags(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 OptionFlags (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 SetInvoice(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 SetInvoice (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 TicketHeader(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 TicketHeader (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 ItemSale(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 ItemSale (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 Payment(TO : 30000) (hInt:01611X) (hTrx:081X)

FP3 Payment (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 PrintTotalsAndPayments(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 PrintTotalsAndPayments (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 PrintBeforeMF(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 PrintBeforeMF (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 PrintUserMessage(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 PrintUserMessage (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 PrintMF(TO : 100000) (hInt:01611X) (hTrx:081X)

FP3 PrintMF (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

FP3 Close(TO : 10000) (hInt:01611X) (hTrx:081X)

FP3 Close (return:0-0x00000000:TRAN_RESULT_OK [0]:İşlem Başarılı sonuçlandı. [0])

After completing the integration process, you should create a redmine issue for compatibility testing before deployment. You will have some tests related to your exe. You can access the list of these tests from the link below and be ready before the Hash test.

https://www.redmine.ingenico.com/attachments/25537/Ingenico_GMP3_test_cases_v1.6.xlsx

There are 3 tests that are the most important among these tests.

8.3.3 Use of Dynamic Vat Table

GMP3 integrations, if the VAT rates change on the device or in any other way, there is a possibility of transacting with incorrect VAT information. In order to avoid such situations, there are some points to be considered in GMP3 integrations. The item coded as TR01 in the approval tests tests this integration.

Situations where VAT control is required

- Immediately after each match,
- every time the ZNo value changes,

If VAT rates are requested to be changed on the device, the ECR will force the Z report, otherwise the VAT rate cannot be changed. For this reason, the correctness of the VAT information should be checked, as there is a risk of VAT change when the ZNo increases.

There are 2 functions of using dynamic vat table.FP3_GetDepartments and FP3_SetDepartments.

Departments of the devices which are defined already can be got By FP3_GetDepartments then you can set them to the device from external software as correct index by FP3_SetDepartments.So after each pairing your Vat value can be set to departments as your desired.

8.3.4 Disconnection and Out of Paper Scenarios

In scenarios of cable disconnection or paper run out during the process should be handled on the exe.For handling these scenarios there is a function to handle these scenarios.This function is FP3_GetTicket.With this function, you can check the interrupted ticket on the device in paper-out scenarios or disconnections, and if you wish, you can cancel this ticket or complete the unfinished ticket.There is an array named st_payment in st_Ticket, which is one of the references of this function.All payments are kept in the st_payment array.if the paper has run out or the connection has been lost during the Transaction,It can be check whether there is a payment or not, the FP3_GetTicket Function can be called and the transaction can be continued or canceled.Thera ara some videos about it.Click the link below.

<https://www.redmine.ingenico.com/documents/471>

Also there is a sample code block is below.

```
ReloadTransaction()
{
    UInt32 RetCode = 0;
    ST_TICKET m_stTicket = new ST_TICKET();
    UInt64 activeFlags = 0;

    RetCode = GMPSmartDLL.FP3_OptionFlags(CurrentInterface, GetTransactionHandle(CurrentInterface), ref
activeFlags, Defines.GMP3_OPTION_ECHO_PRINTER | Defines.GMP3_OPTION_ECHO_ITEM_DETAILS |
Defines.GMP3_OPTION_ECHO_PAYMENT_DETAILS, 0, Defines.TIMEOUT_DEFAULT);

    if (RetCode != Defines.TRAN_RESULT_OK)
        return RetCode;

    RetCode = Json_GMPSmartDLL.FP3_GetTicket(CurrentInterface, GetTransactionHandle(CurrentInterface),
ref m_stTicket, Defines.TIMEOUT_DEFAULT);

    if (RetCode != Defines.TRAN_RESULT_OK)
        return RetCode;

    else if (m_stTicket.stPayment[0].payAmount == m_stTicket.stPayment[0].orgAmount)
    {
        GMPSmartDLL.FP3_PrintTotalsAndPayments(CurrentInterface, GetTransactionHandle(CurrentInterface),
3000);
        GMPSmartDLL.FP3_PrintBeforeMF(CurrentInterface, GetTransactionHandle(CurrentInterface), 3000);
        GMPSmartDLL.FP3_PrintMF(CurrentInterface, GetTransactionHandle(CurrentInterface), 3000);
        GMPSmartDLL.FP3_Close(CurrentInterface, GetTransactionHandle(CurrentInterface), 3000);
    }
}
```